



PROCESS-VOLTAGE-TEMPERATURE (PVT) ANALYSIS OF DOUBLE GATE TUNNEL FIELD EFFECT TRANSISTOR

DISSERTATION REPORT

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STUDENT’S DECLARATION

I hereby certify that the work which is being presented in this Dissertation report entitled **“Process-voltage-temperature(PVT) analysis of DGTFET transistor”** in partial fulfilment of the requirements for the award of the Degree of Master of Technology and submitted in the Department of Electronics Engineering of the Zakir Husain College of Engineering & Technology, Aligarh Muslim University, Aligarh is an authentic record of my own work carried out during final year of M.Tech (ECSD) under the guidance of **Dr. M. J. Siddiqui**, Professor, Department of Electronics Engineering, Aligarh Muslim University, Aligarh.

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ABSTRACT

In this thesis, we have analysed the effect of process-voltage and temperature (PVT) on the conventional DGTFET device. In the process, we have analyzed the effect of gate misalignment towards the source and drain, effect of interface fixed charges in the gate oxide region, impact of variation in both gate oxide thickness, impact of variation in upper gate oxide thickness and effect of variation in gate metal work function in the conventional DG-TFET device. The impact of process variation especially affects the electrical parameter like I_{ON} , I_{OFF} , $v_{T,SS}$ and I_{ON}/I_{OFF} ratio etc. We have also analyzed the effect of scaling in drain voltages on the performance of TFET device by using Silvaco TCAD simulator. We have also analyzed the different regions of operation by varying the both gate to source voltage (V_{GS}) and drain to source (V_{DS}) from negative to positive voltages in the TFET device. At last, we have analyzed the effect of temperature with variation of temp. from 298K to 448K on the TFET device.

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TABLE OF CONTENTS

STUDENT'S DECLARATION	i
Abstract.....	ii
ACKNOWLEDGEMENTS.....	iii
Table of Contents	iv
List of Figures.....	vi
List of Tables	x
List of Symbols	xi
List of Abbreviations.....	xii
Chapter 1	1
Introduction to Field-Effect Transistors.....	1
1.1 What is Transistor Action	1
1.2 Metal-Oxide-Semiconductor Field Effect Transistor (MOSFET)	2
1.2.1 TYPES OF MODES IN A MOSFET-	5
1.2.2 Regions of operation of MOSFET-.....	6
1.2.3 Scaling of MOSFET: -	8
1.2.4 Limitations of MOSFET	8
1.3 Short channel effects (SCEs) in MOS transistors	10
1.3.1 Channel Length Modulation (C.L.M.) in MOS transistors	10
1.3.2 Drain induced barrier lowering (DIBL) in MOS transistors	12
1.3.3 Carrier Velocity Saturation effect in MOS transistors	13
1.3.4 Punch-through effect in MOS transistors.....	15
1.3.5 Hot electron effect in MOS transistors.....	16
1.3.6 Gate induced drain leakage (GIDL) in MOS transistors.....	17
1.4 New and Emerging FET Architectures	19
1.4.1 Tunnel FETs.....	19
1.4.2 Two-Dimensional FETs	21
1.4.3 Impact Ionization MOSFET.....	21
1.4.4 Gallium Nitride JLTFETs	23
1.5 Thesis Organization	23
Chapter 2	24
2.1 Introduction to Tunnel field effect transistors (TFETs)	24
2.2 Device structure of TFET:-	24
2.3 Tunneling: -.....	26
2.4 Operation: -	28
2.5 Schematic band diagram of TFET: -	29

2.6	Transfer characteristics of TFET: -	31
2.6.1	OFF-State	32
2.6.2	Subthreshold Region	33
2.6.3	Super-Threshold Region	34
2.7	Subthreshold swing in Tunnel FETs: -	34
2.8	Band-to-band tunneling transmission: -	37
2.9	Output characteristics	40
2.9.1	Tunnel Resistance–Dominated Region	41
2.9.2	Channel Resistance–Dominated Region	42
2.9.3	Saturation Region	42
2.10	Threshold Voltage	43
2.11	Effect of Temperature	44
Chapter 3		46
3.1	ATLAS	46
3.1.1	MESH	47
3.1.2	Regions and materials	48
3.1.3	Specifying electrodes	48
3.1.4	Specifying doping	48
3.1.5	Specifying contact work function and type of electrode	49
3.1.6	Specifying physical model	49
3.1.7	Specifying numerical solution technique	51
3.1.8	Obtaining Solution	51
3.1.9	Run time output	52
3.1.10	LOG File.	53
3.1.11	Solution file	53
Chapter 4		54
4.1	Analysis of Process-voltage-temperature (PVT) in DG TFET	54
4.2	Device structure and simulation parameters	55
4.3	Simulation results and discussion	57
4.3.1	Impact of process variation in DG-TFET	57
4.3.2	<i>Temperature effect on DG-TFET</i>	82
4.3.3	<i>Impact of scaling in supply voltage</i>	87
Chapter 5		94
5.1	Conclusion	94
5.2	Future work and scope	96
References		97

LIST OF FIGURES

Figure 1.1: (a) n-MOS transistor 3D diagram (b) the MOS capacitor action (c) accumulation region in MOS, (d) depletion region in MOS, and (e) inversion regime of MOS capacitor.....	3
Figure 1.2: MOSFET showing its 4 terminals drain (D), body (B), gate (G) and source (S). An insulating layer (white) separates the gate from the body.....	5
Figure 1.3: Schematic cross-sectional view of (a) an NMOS and (b) a PMOS	6
Figure 1.4: I_D - V_{DS} characteristic of an ideal n-channel MOSFET	8
Figure 1.5: A typical I_{DS} - V_{GS} curve for a highly-optimized conventional MOSFET	10
Figure 1.6: Channel length modulation effect in MOSFET.....	11
Figure 1.7: Ideal I_D - V_{DS} characteristic shown by the solid lines and channel length modulation effect shown by dotted lines for an n-channel MOSFET	11
Figure 1.8: Plotted energy band diagram of MOSFET in ON and OFF states	12
Figure 1.9: The electron drift velocity V_d as a function of electric field E along the channel, which shows that first increase linearly with E and then it saturates.	13
Figure 1.10: An n-channel MOSFET under the punch through form of breakdown (a) surface punch through and (b) bulk punch through.....	15
Figure 1.11: Mechanisms of the origination of substrate current and gate current in an ultra-small channel length n-MOSFET.....	16
Figure 1.12: symmetrically doped p–n junctions Energy band profile with (a) p-type and n-type doping concentration of 10^{15} cm^{-3} and (b) p-type and n-type doping concentration of 10^{20} cm^{-3} .18	
Figure 1.13: Shows double-gate tunnel field-effect transistor Energy band profiles in ON-state.19	
Figure 1.14: schematic view of a double gate n-TFET	20
Figure 1.15: Shows the double-gate tunnel field-effect transistor energy band profiles in OFF-state.	20
Figure 1.16: 2D-FET structure.....	21
Figure 1.17: Three-dimensional view of a double-gate impact ionization MOSFET.....	22
Figure 1.18: I-MOS Energy band diagram in ON-state and OFF-state.	23
Figure 2.1: A simple implementation of a TFET and the biasing scheme: (a) n-type TFET.....	25
Figure 2.2: Schematic cross-sectional view of a DGTFET.....	26
Figure 2.3: Classic and quantum tunneling of electron passing through the energy barrier	27
Figure 2.4: Energy band diagram for a basic lateral TFET structure.....	28
Figure 2.5: Band diagram of an n-type TFET in the OFF-state and in the ON-state.....	29
Figure 2.6: Band diagram of an p-type TFET in the OFF-state and in the ON-state.....	30
Figure 2.7: The transfer characteristics of MOSFET and Tunnel FET along with their sub-threshold slope.	31
Figure 2.8: Simulated transfer characteristics of the DGTFET at $V_{DS} = 1.0 \text{ V}$	31

Figure 2.9: Components of the OFF-state current in a TFET: I_{BTBT} due to BTBT and I_{THERM} due to thermionic emission.	32
Figure 2.10: Fermi–Dirac distribution function at three temperatures T1, T2 and T3. The exponential tail becomes prominent as the temperature increases. The probability of occupying higher energy states increases as the temperature increases.	35
Figure 2.11: Visual definitions of point swing, taken at the steepest point of the I_{DS} - V_{GS} curve, and average swing, taken as the average from turn-on to threshold.	37
Figure 2.12: Band-to-band tunnelling can be calculated by approximating the energy barrier width by a triangular potential energy barrier, where the electrons must tunnel through the widest distance at the base of the triangle.	38
Figure 2.13: Energy band cross section of a Tunnel FET showing the triangular barrier approximation within the bands, $\Delta\Phi$, the screening length λ , and the filtering behaviour of the device in the on-state.....	40
Figure 2.14: Simulated output characteristics in the DGTFET.....	41
Figure 2.15: Effect of the gate work function on the threshold voltage in the DGTFET. The threshold voltage is defined as the as the gate voltage at which the drain current reaches 1×10^{-7} A/ μm at $V_{DS}=1\text{V}$	44
Figure 3.1: Atlas inputs and outputs	46
Figure 3.2: mesh creation using Atlas syntax with non-uniform mesh.....	47
Figure 4.1: Schematic view of (a) Conventional DG-TFET	56
Figure 4.2: Simulation of I_{DS} - V_{GS} Transfer characteristics for misalignment of gate at $V_{DS}=1.5\text{V}$ in log scale.	58
Figure 4.3: Energy band profiles for misalignment of gate 5nm, 10nm, 15nm towards source (+) and drain (-) side in (a) ON-state. ($V_{DS}=1.5\text{V}$, $V_{GS}=1.5\text{V}$) (b) OFF-state. ($V_{DS}=1.5\text{V}$, $V_{GS}=0\text{V}$), cutline 1nm below gate oxide-channel interface.....	60
Figure 4.4: Simulation of (a) ON-current ($V_{DS}=1.5\text{V}$, $V_{GS}=1.5\text{V}$) (b) OFF-current ($V_{DS}=1.5\text{V}$, $V_{GS}= 0\text{V}$) (C) I_{ON}/I_{OFF} ratio variation in misalignment of bottom gate varying from (-15nm to 15nm) at $V_{DS}=1.5\text{V}$	61
Figure 4.5: (a) Threshold voltage (V_T) (b) Subthreshold swing (SS) graphs show the variation in misalignment of bottom gate at $V_{DS}=1.5\text{V}$	62
Figure 4.6: Energy band profiles in the presence of positive and negative interface charge for DGTFET structure as shown in Fig. (a) ON-state. ($V_{DS}=1.5\text{V}$, $V_{GS}=1.5\text{V}$) (b) OFF-state. ($V_{DS}=1.5\text{V}$, $V_{GS}= 0\text{V}$), cutline 1nm below gate oxide-channel interface.....	64
Figure 4.7: Simulation of I_{DS} - V_{GS} Transfer characteristics for different interface charge with gate voltage varying from 0V-1.5V at $V_{DS}=1.5\text{V}$, $T_{ox}= 3\text{nm}$ in log scale.	65
Figure 4.8: Simulation of (a) ON-current (b) OFF-current variation with positive and negative interface charges present Si-SiO ₂ surface at $V_{DS}=1.5\text{V}$	66

Figure 4.9: (a) I_{ON}/I_{OFF} (Current-ratio) and (b) Threshold voltage (V_T) variation with positive and negative interface charges present Si-SiO ₂ surface at $V_{DS}=1.5V$	67
Figure 4.10: Energy band profiles under different both gate oxide thickness variation for DGTFT structure as shown in Fig. (a) ON-state. ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$, $V_{GS}= 0V$), cutline 1nm below gate oxide-channel interface.....	68
Figure 4.11: Simulation of $I_{DS}-V_{GS}$ Transfer characteristics for different both gate oxide thickness variation with gate voltage varying from 0V-1.5V at $V_{DS}=1.5V$, plot in log scale.....	70
Figure 4.12: Simulation of (a) ON-current ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-current ($V_{DS}=1.5V$, $V_{GS}= 0V$) (C) I_{ON}/I_{OFF} (Current-ratio) variation with different both gate oxide thickness value varying from (3nm to 5nm) at $V_{DS}=1.5V$	71
Figure 4.13: (a) Threshold voltage (V_T) graph shows the variation with both gate oxide thickness value varying from (3nm to 5nm) at $V_{DS}=1.5V$	72
Figure 4.14: Energy band profiles under different upper gate oxide thickness variation for DGTFT structure as shown in Fig. (a) ON-state. ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$, $V_{GS}= 0V$), cutline 1nm below gate oxide-channel interface.....	73
Figure 4.15: Simulation of $I_{DS}-V_{GS}$ Transfer characteristics for different upper gate oxide thickness variation with gate voltage varying from 0V-1.5V at $V_{DS}=1.5V$, plot in log scale.....	74
Figure 4.16: Simulation of (a) ON-current ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-current ($V_{DS}=1.5V$, $V_{GS}= 0V$) (C) I_{ON}/I_{OFF} (Current-ratio) ratio variation with different upper gate oxide thickness value varying from (3nm to 5nm) at $V_{DS}=1.5V$	76
Figure 4.17: (a)Threshold voltage (V_T) and (b) Subthreshold swing(SS) graph show the variation with upper gate oxide thickness value varying from (3nm to 5nm) at $V_{DS}=1.5V$	77
Figure 4.18: Energy band profiles under different both gate work function variation for DGTFT structure is shown in above Fig. (a) ON-state. ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$, $V_{GS}= 0V$), cutline 1nm below gate oxide-channel interface.....	78
Figure 4.19: Simulation of $I_{DS}-V_{GS}$ Transfer characteristics for different both gate work function with gate voltage varying from 0V-1.5V at $V_{DS}=1.5V$, $T_{ox}= 3nm$ in log scale.....	79
Figure 4.20: Simulation of (a) ON-current ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-current ($V_{DS}=1.5V$, $V_{GS}= 0V$) (C) I_{ON}/I_{OFF} (Current-ratio) variation in both gate work function of metal value varying from (3.9eV to 5.2eV) at $V_{DS}=1.5V$	81
Figure 4.21: (a) Threshold voltage (V_T) (b) Subthreshold swing (SS) graphs show the variation with both gate work function of metal value at $V_{DS}=1.5V$	82
Figure 4.22: Simulation of $I_{DS}-V_{GS}$ Transfer characteristics for temperature range 298K- 448K with gate voltage varying from 0V-1.5V at $V_{DS}=1.5V$, $T_{ox}= 3nm$ in log scale.....	83
Figure 4.23: Energy band profiles at different temperature for DGTFT structure as shown in Fig. (a) ON-state. ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$, $V_{GS}= 0V$), cutline 1nm below gate oxide-channel interface.....	84

Figure 4.24: Simulation of (a) ON-current (b) OFF-current variation with temperature range 298K- 448K at $V_{DS}=1.5V$	85
Figure 4.25: (a) I_{ON}/I_{OFF} (Current-ratio) (b) Threshold voltage (V_T) variation with temperature range 298K- 448K at $V_{DS}=1.5V$	86
Figure 4.26: Subthreshold swing (SS) variation with temperature range 298K- 448K at $V_{DS}=1.5V$	87
Figure 4.27: Energy band profiles at different drain voltages variation for DGTFET structure as shown in Fig. (a) ON-state. ($V_{DS}=1.5V$ to $0.1V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$ to $0.1V$, $V_{GS}= 0V$), cutline 1nm below gate oxide-channel interface.	88
Figure 4.28: Simulation of $I_{DS}-V_{GS}$ Transfer characteristics for different drain voltages with gate voltage varying from $0V$ - $1.5V$, $T_{ox}= 3nm$ in log scale.	90
Figure 4.29: Simulation of $I_{DS}-V_{GS}$ Transfer characteristics in the DGTFET for different doping on the source and the drain region with varying the gate voltage from ($-1.5V$ to $1.5V$) at $V_{DS}=1.5V$, $Tox= 3nm$ in log scale.	91
Figure 4.30: Simulation of $I_{DS}-V_{DS}$ output characteristics in the DGTFET for different doping on the source and the drain region with varying the drain voltage from ($-1.5V$ to $1.5V$) at $V_{GS}=1.5V$ and $-1.5V$, $Tox= 3nm$ in log scale.....	93

List of Tables

Table 3.1: Carrier statistics Models.....	49
Table 3.2: Mobility Models.....	50
Table 3.3: Recombination Models	50
Table 3.4: Tunneling Models and Carrier Injection Models	50
Table 4.1: Device structure parameters of DGTFET	56
Table 4.2: Device performance parameters.....	59
Table 4.3: Device Performance Parameters	66
Table 4.4: Device Performance Parameters	70
Table 4.5: Device Performance Parameters	75
Table 4.6: Device Performance Parameters	80
Table 4.7: Device Performance Parameters	86

LIST OF SYMBOLS

2D	Two Dimensional
3D	Three Dimensional
V_{DD}	Supply Voltage
V_{DS}	Drain to Source Voltage
V_{GS}	Gate to Source Voltage
V_{th}	Threshold Voltage
V_{tn}	Threshold Voltage of NMOS Devices
V_{tp}	Threshold Voltage of PMOS Devices
N_D	Donor Concentration
N_A	Acceptor Concentration
Φ	Work Function
E_F	Fermi Energy
E_i	Intrinsic Fermi energy
C_{ox}	Gate dielectric capacitance
E_g	Band gap energy
F	Electric field

List of Abbreviations

BTBT	Band-to-Band Tunnelling
CMOS	Complementary Metal Oxide Semiconductor
DIBL	Drain Induced Barrier Lowering
DGTFET	Double gate tunnel field effect transistor
GIDL	Gate induced drain leakage
MOS	Metal Oxide Semiconductor
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
NMOS	n-Channel MOSFET
PMOS	p-Channel MOSFET
SOI	Silicon on Insulator
SCE	Short Channel Effect
TCAD	Technology Computer Aided Design
VLSI	Very Large-Scale Integration
CLM	Channel length modulation
SS	Subthreshold swing
C.B.	Conduction band
PVT	Process voltage temperature
V.B	Valance band

Chapter 1

Introduction to Field-Effect Transistors

We are living in a world which is full of advanced technology and Innovation where cell phone, advanced technology has become an important part of human life. All these innovation and advance technology is possible because of huge amount of research work and development in the field of semiconductors and its allied area. When we see the history of semiconductor fabricating industry we can find out the development in this area is almost exponential, for instance normal phones which we were using in around 1990 has just function of calling and sending messages, now it can perform massive majority many of function such as Wi-Fi-connected to internet, touch screen display, calling, email, send text messages, photography, record and play audio and music, clock and temperature information, video editing, gaming, real time video streaming and what not. Have you ever thought on to what is the most fundamental building block of semiconductor industry? The most accurate answer is “TRANSISTOR”. Transistor operates actually like cell in our body which group together to perform special task and the more cell we have more functions we can perform.

1.1 What is Transistor Action

When we see Transistor closely, we discovered that it has three terminals and with current or voltage at the third terminal controlling resistance between the two terminals. Transistors for example metal oxide field effect transistor (MOSFETs) can act as switch, But it could be recognized that the MOSFETs are not perfect switches which means ideal switches are normal put away no consume power when turned-OFF and produce a little amount of current immediately when switched-on. MOSFET previously have a little leakage current associated with it and, hence, when MOSFET is in OFF state then a small amount of power dissipation from the power supply even. In a MOSFET device, there are two types of power consumption occur: static power dissipation and other is the dynamic power consumption. This power consumption in OFF state in MOSFET is called as the static power dissipation (P_s), which is calculated by $P_s = V_{DD} \cdot I_{OFF}$ where supply voltage is denoted by V_{DD} and I_{OFF} is the sum of device's leakage current when MOSFET is turned OFF. The static power consumption is ideally zero in a MOSFETs. The power consumed by the circuit is known as dynamic power dissipation when

switching occurs. MOSFET similarly consume power when it is in ON state. Dynamic power is mostly dependent on the frequency of switching or switching activity of transistor and called as dynamic power P_D given by $P_D = V_{DD}^2 C_L f \alpha$ where f denotes switching frequency, supply voltage is V_{DD} and α is the switching probability, and C_L represents the load capacitance at the transistor's output terminal [1].

Recently the focus of researchers and engineers is to minimize the load capacitance by transforming transistor size, functionality of IC remains unchanged and increasing no. of transistor per chip also called as scaling. Due to continue scaling of transistor furthermore increases the on-state current I_{ON} but the problem is the OFF-state current also increases considerably, this effect is fundamentally because of the short-channel effects (SCEs) which are inherently originated by MOSFET gate length scaling pattern. Due to increment in the OFF-state current significantly contributes to static power dissipation in newer transistor. Until recently, however, the researchers were concerned about dynamic power dissipation. Now, the way in which scaling patterns are going on in recent years, it has been projected that static power consumption would ultimately overcome dynamic power consumption if the MOS transistors are scaled with such a speed. This increment in static power dissipation will eventually interrupt long run the battery life of our portable devices essentially.

1.2 Metal-Oxide-Semiconductor Field Effect Transistor (MOSFET)

The transistor action performed by a MOSFET device is rooted in the formation of MOS capacitor in channel region. From fig. 1.1 we can see that a protecting insulating oxide material is located in between metal and semiconductor which is the things that basically needed for capacitive action. The supply of voltage to the gate terminal of a MOS transistor controls the capacitance. Usually, body terminal is connected to ground. Now, if a negative bias is applied to the gate electrode w.r.t to semiconductor substrate, we notice the bottom plate carries a positive charge, whereas the top plate carries a negative charge and an electric field will be formed in the direction of semiconductor to metal. Negative charges on the gate terminal can be detected as a result of the negative voltage at the gate terminal which attracts the majority carrier hole towards the surface of the oxide-semiconductor near interface region and they will start to accumulate at interface of oxide-semiconductor junction corresponding to positive charge exists on the lower plate of the MOS capacitor. Next, when a positive voltage is supplied to the gate

electrode, an electric field is formed in reversed direction and directed from gate metal to semiconductor. A positive voltage at the gate can be viewed as a positive charge on the top metal plate and negative charge on the bottom plate at the gate and therefore it repels holes away from the interface, these repelled holes will create behind a depletion region near interface. now, if we increase positive bias further increase it will start to attract electrons to the surface and they start to pile up at the surface and the point at which concentration of electron at interface is equivalent to the hole concentration in the bulk then we called this as threshold voltage [2]. This phenomenon is also called inversion.

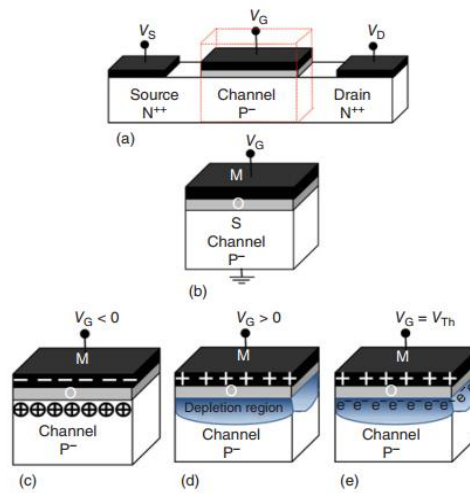


Figure 1.1: (a) n-MOS transistor 3D diagram (b) the MOS capacitor action (c) accumulation region in MOS, (d) depletion region in MOS, and (e) inversion regime of MOS capacitor.

So, for transistor action inversion region is must in the MOSFET devices occur Further if we apply voltage at gate same as threshold voltage and increase drain voltage what we can see is electrons from source finds a low resistance conduction route through inversion layer to drain and we conceptualize as closed switch. However, if we apply gate potential less than threshold voltage, we can find out that channel is depleted and there is no inversion region form for conduction of electron and we can conceptualize this as ON switch. Therefore, MOSFETs work as a switch that can be turn OFF or ON switched. It depending upon on the applied voltage to the gate terminal. Also, the apply voltage at the gate electrode control the flows of current between drain to source in the device. Because the third terminal controls the current flowing through the other two terminals, we can call MOSFETs as Transistor.

When we start nonstop scaling of size of transistor to help boosting its performance like as increased circuit speed, reduced cost per function, reduce the operating voltage in

terms of both fabrication cost and power consumption. then different type of short channel effects occurs in the device. Like as threshold voltage role OFF, drain induced barrier lowering, hot electron effects, channel length modulation, carrier velocity saturation, punch-through effect, gate induced drain leakage, two-dimensional charge sharing, direct source to drain tunnelling, Boltzmann tyranny and ultra-steep doping profile etc. So, in subsequent section we will discuss few short channel effects in MOSFETs.

At the beginning phase of MOSFETs discovered and Lilienfeld and Heil invented it in 1930, but it was first effectively presented in 1960. During at starting phase of innovation there were many technical issues. These issues are particularly connected to the control and decrease of the surface region contact between semiconductor and the oxide layer. At first, just “**depletion mode**” was exist, a negative voltage is first applied to the gate terminal then n-type of channel is formed. Without applying gate voltages, a conducting channel will form in among the source and drain terminal in a depletion-mode device. Fabrication of the most recent device by reducing the surface area, which allowed us to manufacture a device without a beginning conducting channel among the drain and the source. These devices are called as “**enhancement-mode**” devices. The electrons in this device are concentrated in a narrow (10 nm thick) “inversion layer” near the oxide-semiconductor. Currently, most of us used enhancement-mode devices.

Metal oxide field effect transistor is also called as MOSFET, and it is a current-controlled voltage device. MOSFET is a semiconductor device that is often used in electronic devices for electronic signal amplification and switching purpose. Functionality of devices based on the MOS capacitor. The main part of MOSFET is MOS capacitor. It is a device in which the electric field controls the drain and source terminals and electric field is created due to applying voltage at gate terminal. Insulating a very thin oxide layer is located in between metal and semiconductor. Figure 1.2 shows that MOSFETs mainly contains of 4 sections to be specify, which are the namely gate, source, drain and body terminals. The body terminal is ignored because the body is generally connected to the source terminals and it behaves as a three-terminal device [3]. MOSFETs are usually called as transistors and are used in both analog and digital circuits.

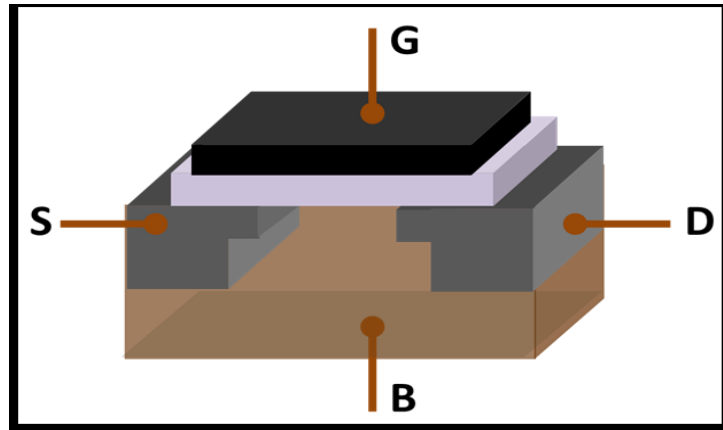


Figure 1.2: MOSFET showing its 4 terminals drain (D), body (B), gate (G) and source (S). An insulating layer (white) separates the gate from the body.

TWO COMPLEMENTARY DEVICES-

- 1. n- channel devices (n-MOSFET) on p-type Substrate**
- 2. P- channel devices (p-MOSFET) on n-type Substrate**

1.2.1 TYPES OF MODES IN A MOSFET-

The apply voltage at the gate terminal in enhancement mode MOSFETs forms a conducting channel between drain and source contacts through the lateral field effect. Enhancement mode means the increase of conductivity along with the oxide field increase. It is also known as the inversion layer because it increases conductivity by increasing gate voltages, which provides additional electrons to the channel. Two types of channels exist in MOSFET. The n-MOSFET or n-MOS is a channel that mainly contains electrons, whereas the p-MOSFET or p-MOS is a channel that contains holes. Figure 1.3 (a) and Figure (b) show cross-sectional views of n-MOS and a p-MOS respectively. However, the type of doping in substrate is completely inverse which means, in p-MOS being formed on an n-type substrate and n-MOS being formed on a p-type substrate [1].

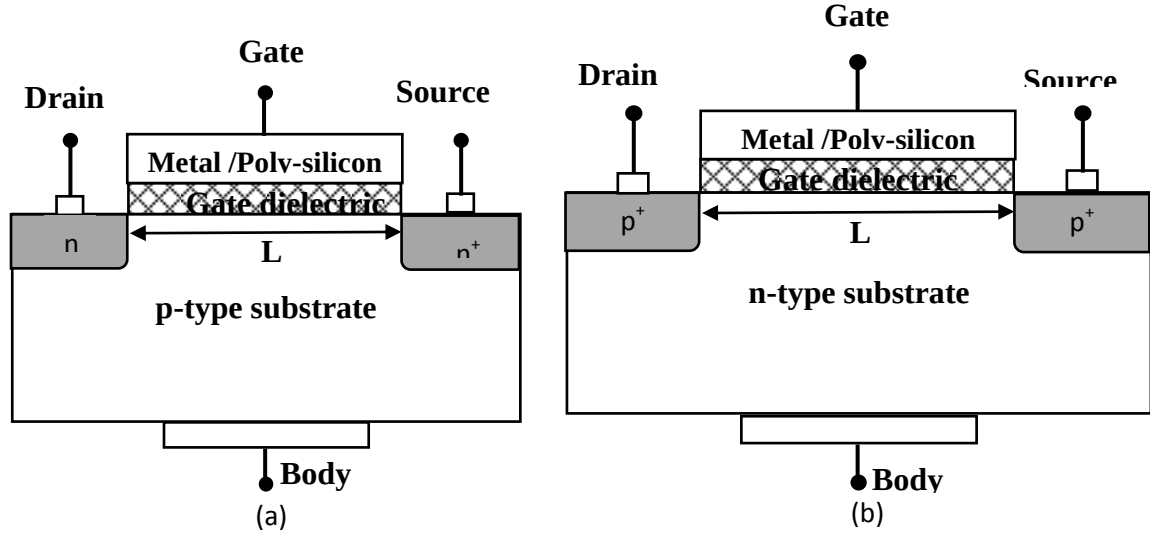


Figure 1.3: Schematic cross-sectional view of (a) an NMOS and (b) a PMOS

If we apply the voltage across gate to source $V_{GS} = 0V$ then no channel will be formed in between drain and source. The channel width increases when the gate voltages will be increased. As the increment in gate voltage, a huge number of holes are pulled to the p-type substrate, while electrons are drawn to the n-region under the drain and source terminals. Because the channel width thickens as the gate voltages rise, a huge amount of current flows from drain to source side under the electric field formed by the oxide and silicon layer.

1.2.2 Regions of operation of MOSFET-

The device's operation is basically divided into three modes, which are as follows:

1.2.2.1 Cut-off Region –

In this mode, device will be always in the OFF condition and no current flow between source and drain $I_D = 0$. $V_{GS} < V_t$, $V_{GD} < V_t$ with $V_{DS} > 0$ is the basic cut-off condition for MOSFETs. Where V_t denotes the device's threshold voltage. However, some electrons with greater energies can pass from the source to drain as resulting in sub-threshold current. Transistor is inactive, in this region of operation [3].

1.2.2.2 Linear or triode Region –

Condition for linear region operation of n-MOSFET is -

$$V_{GS} > V_t, V_{GD} > V_t \text{ and } V_{GD} > 0$$

Linear region transistor works in ON-state. They formed of thin layer of channel between drain and source during ON-state. Because of channel formation, current start to flows in the device. MOSFET behaves same as resistor because gate terminal mainly controls the channel in comparative with both the source and drain voltages [4]. When MOSFET devices works in linear mode then they work as amplifiers. The equation for drain current I_D in this region of operation can be written below-

$$I_D = \frac{1}{2} \mu_n c_{ox} \frac{W}{L} ((V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2}) \quad (1.1)$$

Where gate width is W , L denotes the gate length and C_{ox} is the gate oxide capacitance per unit area.

1.2.2.3 Saturation region-

Transistor follows the following Condition to function in Saturation region -

$V_{GS} > V_T$ (formed the channel), $V_{GD} < V_T$ (channel will pinch- off), $V_{DS} > 0$

In saturation mode, if we apply voltage at drain terminal is higher than that of the gate terminal voltage then MOSFET act as an ideal current source, which is regulated by gate terminal voltage and the drain current is unaffected by the applied drain voltage [4].

The drain current equation of a MOSFET at the saturation region is as follows:

$$I_D = \frac{1}{2} \mu_n c_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 \quad (1.2)$$

In above equation, we neglected the channel length modulation effect. For applied drain to source voltage V_{DS} equal to $V_{DS,sat}$ then pinch-off point is right at the end drain. But if we continuous increase the drain potential, this pinch-off point begins to shift towards the source. This potential drop occurs between the drain region and narrow depletion region between at the channel's end. Due to applying voltage speed up the electron that approach the drain side of channel and swept through the depletion region into the drain. Due to this happen, the actual channel length will reduce. The above equation shows the less dependent on drain voltage and controlled mainly by the gate-source voltage and modified new drain current equation approximately follow as:

$$I_D = \frac{1}{2} \mu_n c_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda(V_{DS} - V_{DS,sat})) \quad (1.3)$$

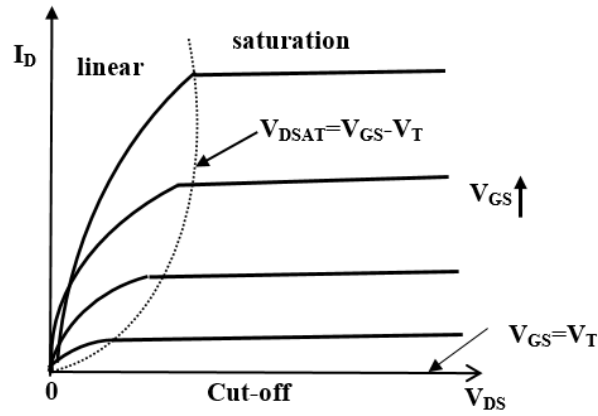


Figure 1.4: I_D - V_{DS} characteristic of an ideal n-channel MOSFET

In the above equation involving additional factor λ is called as channel length modulation and it is inversely proportional to the channel length in general. The above equations represent that drain current is dependent on drain voltages. All above type of region [3] can observed from Figure 1.4

1.2.3 Scaling of MOSFET: -

The MOSFET plays a significant role in integrated circuits (ICs) due to its scalability. Researchers are working to reduce the size of practically all semiconductor devices as technology advances. MOSFETs, which used to have channel lengths of a few micrometres, have been down scaled to tens of nanometres in advanced integrated circuits (ICs) over the last few years. Declination of the length of gate (L_G) raises the circuit's operating frequency, making it faster. The chip's area reduces, allowing for higher transistor density while also lowering the Integrated Circuit costs. Because of the constant switching power density, each function consumes less power, while the same amount of power can be used by a larger number of circuits [3]. The reduction in MOSFET size is challenged by the semiconductor device fabrication process, the necessity to employ relatively low voltages, and decreased the electrical performance, which demanding circuit modification and invention.

1.2.4 Limitations of MOSFET

Conventional MOSFETs have a several drawbacks. A scaling theorem proposed the Robert H. Dennard in 1974, according to which electric field remain same value inside the device. According to this if we reduced V_{DD} around 20% of its original value, V_T

would decrease nearly its half value, in fact, when we reduced V_{DD} around few scaling factors but the value of V_T decreased relatively less. Because of which gate overdrive voltage reduced significantly. Because of that dynamic speed ($C_g V_{DD}/I_{ON}$) decreased and ON- current decreases which influence the device performance [5]. There are two types of suitable solutions to remove this issue with MOSFET.

1.0 Either V_{DD} can remain greater than it should with continuous field scaling or,

2.0 Threshold voltage can be widely downscaled.

V_{DD} has had to dramatically slowdown in order to sustain the gate overdrive value. When V_{DD} does not reduce considerably, there is a more loss of power.

1.0 Dynamic power P_D given by $P_D = V_{DD}^2 C_L f \alpha$

2.0 static power dissipation P_S given by $P_S = V_{DD} \cdot I_{OFF}$

Where f denotes as frequency, C_L represents total switching capacitive load, switching probability is α , the supply voltage is V_{DD} and the sum of leakage current in the devices is I_{OFF} .

The fundamental drawback of conventional MOSFETs is given below-

It features a high-power consumption and a sub-threshold swing of roughly 60 mv/decade at room temperature. However, in the case of conventional MOSFETs, we can't reduce more.

When plotting the $I_{DS}-V_{GS}$ curve on a log-scale in Fig 1.5, the subthreshold slope of conventional MOSFETs remains constant. This indicates that after properly optimising the device to achieve the maximum feasible abrupt transition turn-on with gate voltage and when the subthreshold swing has hit its limit value of 60mV/decade at room temperature in the MOSFET. To reduce the V_T is only way to shifting of graph left [4]. By shifting the V_t by 60mV, the off-current is increased by a decade, resulting an increase in static power. Curve shifting occurs in $I_{DS}-V_{GS}$ graph due to changing the value of gate work function.

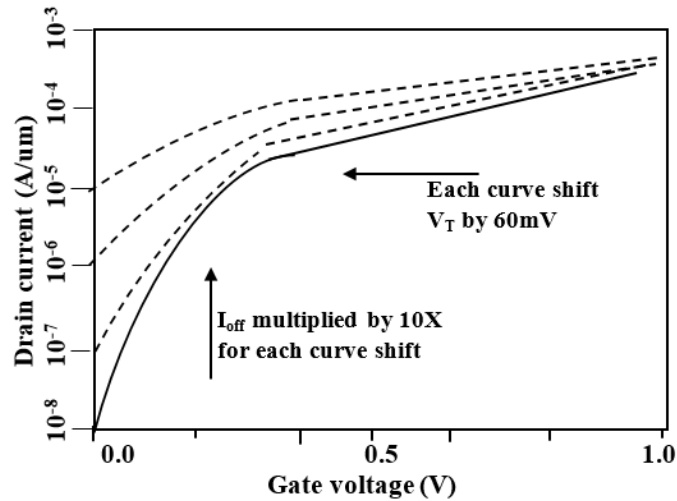


Figure 1.5: A typical I_{DS} - V_{GS} curve for a highly-optimized conventional MOSFET

1.3 Short channel effects (SCEs) in MOS transistors

In this unit we will discuss some of short channel effects which inhibits the performance of the transistor at lower technology node. But first let's know about what short channel MOSFETs are. Short channel MOSFETs are those whose channel length is same order of magnitude of source-channel and channel-drain depletion region length as the depletion layer widths. The so-called short channel effect occurs when the channel length L is decreased to increase both the operating speed and the no. of components per chip. For the large feature size devices, there effect will be negligible. Typically, depletion region is $\sim 350\text{nm}$ therefore, any transistor whose gate length below 350 nm will have short channel effects. Now, we will discuss some of the important short channel effects one by one.

1.3.1 Channel Length Modulation (C.L.M.) in MOS transistors

In a MOSFET, channel length modulation occurs when the drain voltage is increased and length of the reversed channel region is reduced. Channel length modulation causes an increase in current with drain potential and a decrease in output resistance.

This type of effect is similar to early effect in BJTs. We assumed the ideal current-voltage relationship in MOSFET that actual channel length L . When the MOSFET is in saturation mode, the point in the channel which pinch-off must have a drain-to-source potential equal to $V_{DS,sat}$. For applied voltage drain-to-source $V_{DS}=V_{DS,sat}$, the pinch off point is right at the drain end.

For applied V_{DS} voltage greater than $V_{DS,sat}$, however, the pinch off point begins to shift towards the source side, since it is defined as the point where the applied $V_{DS}=V_{DS,sat}$. It means that decreasing the channel length leads to reduces the resistance and more current flow in the device. As the channel length decreases, so called the short channel length. The difference between the applied V_{DS} and $V_{DS,sat}$ ($V_{DS}-V_{DS,sat}$) drops between the pinch off point and drain region, as shown in Fig1.6. The length of this region is called as space charge region, the pinch off region length, drain region length, the saturation region length etc. It is denoted by ΔL . Under this condition, the actual channel length(L) is not equal to effective channel length (L_{eff}), but it is decrease by the amount of ΔL [2].

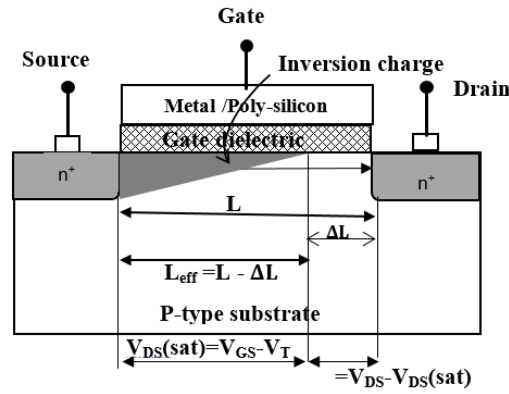


Figure 1.6: Channel length modulation effect in MOSFET

$$L_{eff} = L - \Delta L \quad (1.4)$$

We can observe from the figure 1.7, that the effect of channel length modulation imparts a positive slope in the I_D - V_{DS} characteristics in the saturation region. In mathematically form, the expression for the saturated drain current $I_{D,sat}$ Eq. (1.3) and replace L by L_{eff} .

$$I_{D,sat} = \frac{1}{2} \mu_n c_{ox} \frac{w}{L_{eff}} (V_{GS} - V_{th})(1 + \lambda V_{DS}) \quad (1.5)$$

$$r_o = \frac{1}{\lambda I_{D,sat}} \quad (1.6)$$

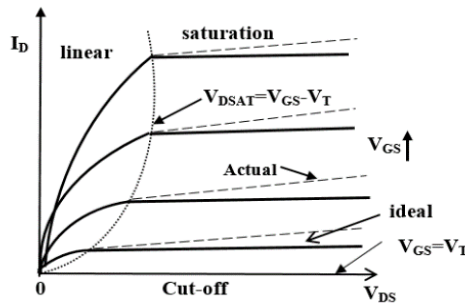


Figure 1.7: Ideal I_D - V_{DS} characteristic shown by the solid lines and channel length modulation effect shown by dotted lines for an n-channel MOSFET

From the above equation, we can define that due to channel length modulation results in a finite output impedance (r_o) and drain current will slightly increase with the factor of $(1+\lambda V_{DS})$ as drain-to-source voltage increase [2]. In the above expression the term λ is known as channel length modulation coefficient, unit is per volt(V^{-1}) and λ is inverse function of channel length(L). The typical value of λ varies between $0.001 V^{-1}$ to $0.1 V^{-1}$.

1.3.2 Drain induced barrier lowering (DIBL) in MOS transistors

This is one of the short channel effects that is called as drain induced barrier lowering, in short form, it is denoted by DIBL. Under the normal mode of operation, in long channel MOSFET, the gate terminal only controls the charges in channel region. It means gate terminal controls the source to channel barrier for amount of carrier injection from source to channel region as shown in fig 1.8. If we would increase the drain positive potential at the higher than source and continue increasing the drain potential. then energy band in drain region would shift downward. After this this PN junction will form between substrate to n+ drain region would get reverse biased more and more the depletion region in this junction would increase into the substrate region. In that case the transition region would increase.

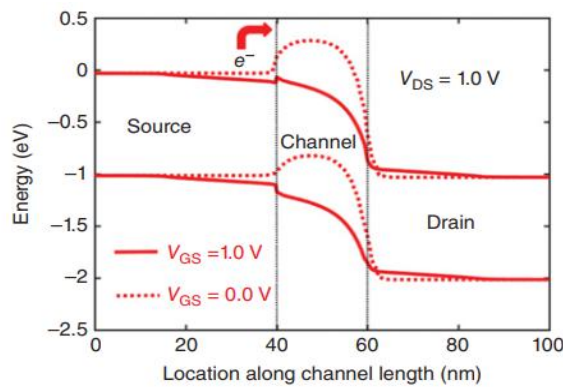


Figure 1.8: Plotted energy band diagram of MOSFET in ON and OFF states

However, due to continuous scaling of the channel length to short channel region. In short channel MOSFET, the drain potential is increased then energy band near the drain would come down along with the depletion region in the substrate near the drain would increase. If we analyse the effect, this bending is more relatively as compared to long channel MOSFET because the available channel length is small. When V_D is increased even beyond this point the depletion region width even increases to start with under equilibrium when source, gate and bulk was grounded. A depletion region extended into the bulk

region and if we increase even further, then barrier height has even reduced which means drain is controlling the barrier height because channel length is small, due to this phenomenon it is referred as DIBL. In this case flow of increasing amount of electron from source to drain would be uncontrolled where the current flow would be really high and in fact, we are not controlling the channel creation from gate here we have not touched the potential at gate and yet there is a current flow in MOSFETs which called as leakage current and this leakage current would be extremely large and the gate would no longer be able to completely switch off the device and losses control of the drain current, due to it sub-threshold current would increase. So, we defined it as if channel length is comparable to the sum of depletion region on source plus the depletion region of drain. The potential applied at drain and barrier was not controlled by gate here it was controlled by drain in this case the gate loses the controlled which was supposed to be controlling the channel here it happens in short channel MOSFET. Reduction in source to channel barrier height reduces due to DIBL will increase leakage current in MOSFETs [2].

1.3.3 Carrier Velocity Saturation effect in MOS transistors

A MOS device is said to be a short channel device, if the length of channel is the same magnitude as depletion region width of the drain and source junctions, it means length of channel is equal to the depletion region thickness of the drain or the source of junction.

We assumed that mobility is constant, In the long channel MOSFET, which means drift velocity will increase with increasing electric field. But in short channel device mobility varies with electric field.

$$E_{lat.} = \frac{V_{DS}}{L} \quad (1.7)$$

$$E_{ver.} = \frac{V_{gs}}{t_{ox}} \quad (1.8)$$

$$V_d = \mu E \quad (1.9)$$

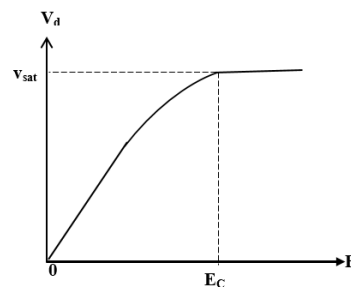


Figure 1.9: The electron drift velocity V_d as a function of electric field E along the channel, which shows that first increase linearly with E and then it saturates.

There exist two types of electric field in MOSFET. One is called lateral field which given by V_{DS} divided by L where V_{DS} is nothing but drain to source voltage and L is the length of channel. Second electric field in MOSFET is called as vertical field which given by V_{gs} divided t_{ox} where the gate to source voltage is denoted by V_{gs} and t_{ox} is gate oxide thickness. We know that drift velocity is given by mobility into the electric field so the drift velocity of carrier is a function of mobility and the electric field lateral and vertical both. If we increase electric field goes on increasing, then drift velocity will also increase. In short channel device, the lateral field is generally larger. Now we understand the effect of lateral field on the drift velocity. if length of channel reduced is reduced and goes on increasing on V_{DS} then transistors enter into the saturation. The channel length pinch-off and after that it keeps on shifting towards the layer. Then lateral electric field increasing constantly so technically by equation so we can say drift velocity has to keep on increasing if lateral electric field will increase. From the Fig 1.9, drift velocity is in the range of E_c , it keeps on increasing linearly but at a certain point of time, it reaches E_c value. After this, the drift velocity of carriers enters into saturation region [2]. In velocity saturation at a particular electric field E_c , mobility start decreasing and over all saturation velocity reduces with apply gate potential due to vertical electric field and surface scattering in the device.

Drift velocity is given by-

$$V_d = \frac{\mu E_{lat}}{1 + \left(\frac{E_{lat}}{E_{cri}}\right)} \quad (1.10)$$

Due to velocity saturation in short channel effect, the saturation current equation will change and saturation current will reduce with a factor of $\left(1 + \frac{V_{gs} - V_{th}}{L E_{cri}}\right)$. if long channel device, then we can ignore this factor in saturation region.

The modified saturation current equation approximately by-

$$I_{D,sat} = W C_{ox} (V_{gs} - V_t) v_{sat} \quad (1.11)$$

Where v_{sat} is the velocity saturation and C_{ox} is denoted as gate oxide capacitance and its unit is per cm^2 . The $I_{D(sat)}$ current is approximately vary linear with V_{gs} , instead of having square law dependency.

1.3.4 Punch-through effect in MOS transistors

In short channel effect, channel length is too short. Punch through is a destructive form of device breakdown, if not design properly then device can completely destroy. Basically, this form of breakdown occurs when the depletion regions associated around the source and drain well, this depletion regions here end up touching each other so let's call this source and drain. Types of two breakdown occur in MOSFET, as shown in Fig 1.10 (a) and Fig 1.10 (b) First breakdown occurs if depletion regions associated with drain-body and source-body junctions touch each other at the surface that is called as surface punch through and other depletion regions touch each other somewhere below the surface means in the bulk, is called as bulk punch through. The surface punch through is an unlikely phenomenon. The bulk punch through is a more likely phenomenon. All the punch through breakdown take place in the device [2].

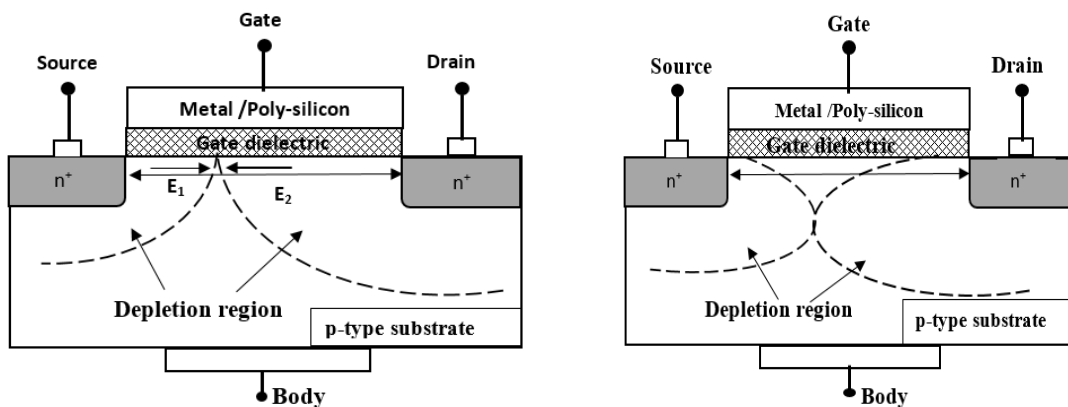


Figure 1.10: An n-channel MOSFET under the punch through form of breakdown (a) surface punch through and (b) bulk punch through

If applying a positive voltage to the drain and source and body will also be grounded. Positive potential to this n-well and lower potential to this P substrate, So PN junction will form between the drain and the substrate and this region device works in reverse bias PN junction then depletion region gets wider. If keep on increasing the drain voltage then depletion region will get wider and wider. Finally, depletion region around the drain is going to contact the depletion region around the source well and they offer very low resistance due to cancellation of directed electric fields across drain to source voltage. Due to this effect, large amount of current flow in the device and total voltage drops across source and drain regions [2]. When the drain potential is raised, the depletion region around the drain might extend more towards the source, resulting in the merging of two

depletion regions. This phenomenon is known as punch-through. This effect is limiting the maximum operating voltage of the device. Punch-through effect increases with decreasing channel length. Due to punch-through, we can't turn off the devices, so heating of the device can meltdown. In order to eliminate the chance of this breakdown, it can be decreased by having higher doping levels either in substrate or near source and drain.

1.3.5 Hot electron effect in MOS transistors

Short channel effects happen in MOSFETs when channel length is approximately equal to depletion layer width of drain and source junction. We know that there are two types of electric field in the MOSFET. First is lateral electric field and other is vertical electric field. Lateral field depends on V_{DS} and channel length and vertical field depends on V_{GS} and oxide thickness. Hot carrier effect is becoming increasingly important in modern MOSFETs having small channel lengths and applied to bias voltage of order 3v or more [2].

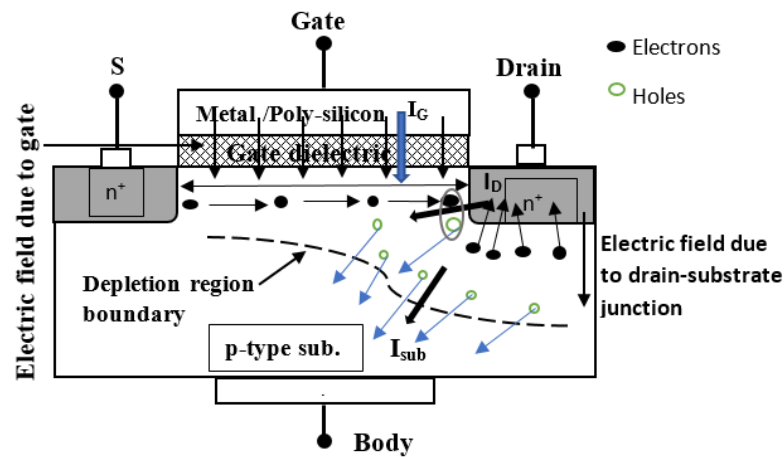


Figure 1.11: Mechanisms of the origination of substrate current and gate current in an ultra-small channel length n-MOSFET

Due to continuous scaling of MOSFETs, the lateral electric field given by V_{DS}/L_g will increase towards the drain. Drift velocity is given by mobility into electric field. If increase the lateral electric field then drift velocity of electron will keep on increasing means the electrons are moving with a very high velocity and we know that kinetic energy is directly proportional to velocity. So kinetic energy of this electrons would increase drastically. They will become hot and hence these electrons are called as hot electrons, as shown in Fig 1.11. Hot electrons are accelerating towards the drain in the channel region with a very high kinetic energy because electric field towards the drain is

high and thus this high energy electron collides with atoms to exchange momentum and energy and they undergo with a phenomenon at the interface called as impact ionization. Due to impact ionization mechanism, this will result in generation of e-h pair if sufficient energy is being transferred to the semiconductor atom. The holes which are going to be p-type will go back to P-substrate will lead to a substrate current which is technically not desirable.

The electron will move towards the semiconductor-insulator region, because of high electric field by the gate voltage, a lot of electrons will move towards the gate. For this transition, a carrier should have high kinetic energy to reach the conduction or valance band in the oxide and there is a possibility that some electrons may will get trapped into oxide region and hit out electrons from atom. Due to hot carrier effect, electrons are entering in the oxide region may damage the oxide's insulating properties and increases to leakage current through gate terminal and therefore, it is bad for good transistor action. This gate current is reducing the isolation properties as well as increase the power dissipation. Also, one important parameter may affect the threshold voltage of the transistor [6].

1.3.6 Gate induced drain leakage (GIDL) in MOS transistors

GIDL or gate induced drain leakage is a quantum tunnelling phenomenon and off state leakage current mechanism. As we all know electrons behave like as both particle as well as wave nature. This phenomenon is called as duality principle. When the gate slightly overlaps the drain, the GIDL current occurs. So, a lot of static power dissipation will consumption and if MOSFETs is not operate in on-state. When a high voltage is supplied to the drain terminal and a low voltage is applied to the gate terminal, this type of impact occurs. The gate-drain overlap region, and therefore the transistor width, are proportional to GIDL current. Because it is a strong function of the electric field, reverse bias and doping concentrations and hence it responds quickly to changes in the drain to gate voltage. The oxide geometry near the gate's edge has a greater impact on this current. By optimising the drain structure and gate to drain overlapping region, the highest electric field in the drain area is enhanced.

Problem arises when applying to positive voltage is at the drain. Under the effect of an x-directional lateral electric field, the n⁺ drain area under the gate is depleted and even inverted when the gate is biased to zero or negative. The surface behaves same as p-

region, due to accumulation of holes. Depletion area develops near the drain due to excessive doping in the drain region. Band bending occurs between the gate and the drain due to the strong electric field. Tunneling from one band to another is only feasible in the presence of a strong electric field in that area. The dopant in the diffusion area, as well as the difference among the drain and gate voltage which affect the Si-SiO₂ interface. While fabricating the modern MOSFETs, an intrinsic gate on drain overlap region is formed. This is because, in the self-aligned gate CMOS process, ion implantation is used to create the source or drain region following gate polysilicon deposition. This gate-on source/drain overlap in MOSFETs will effectively reduce channel length as relative to the defined gate length.

If we plot band diagram at channel-drain interface we can see there is band bending in drain region under the gate as shown in figure 1.12. Also, the band bending in the device is appropriate to align the filled valence band of channel region to the empty conduction band of the drain region. Thus, alignment of band will facilitate tunnelling of electron from channel to drain region if all tunnelling condition meets that is small barrier width, empty density of state in drain region and filled density of electron at channel region. Thus, leakage current in MOSFETs is considerably increased significantly of this band-to-band tunnelling, which is unfavourable to device performance. The GIDL leakage current in MOSFETs is controlled by the gate work-function. Changing the gate work function resulted in a decrease in drain leakage. If the poly-silicon gate's work-function is raised, the leakage current in MOSFET is reduced.

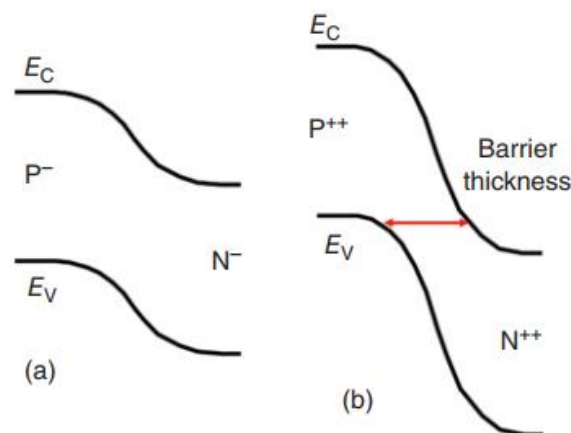


Figure 1.12: symmetrically doped p–n junctions Energy band profile with (a) p-type and n-type doping concentration of 10^{15} cm^{-3} and (b) p-type and n-type doping concentration of 10^{20} cm^{-3} .

Furthermore, we all familiar that subthreshold current depends on gate overdrive voltage exponentially, therefore, if threshold voltage reduces subthreshold current increases exponentially [7] and contribute to very low I_{ON}/I_{OFF} ratio which is not a got indication for low power application, so required new devices which reduce the power.

1.4 New and Emerging FET Architectures

In this segment we should discuss about some new and emerging FET architectures that somehow possible and reliable to overcome some of significant major short channel effects in the Field effect transistors and have capability to replace conventional MOSFET architectures device.

1.4.1 Tunnel FETs

Tunnel FET is basically reversed bias PIN diode with a gate terminal. N-TFETs comprised of P^+ source, intrinsic channel and n^+ drain shown in fig 1.13. Source region in TFETs are heavily doped this is done to guarantee high tunnelling efficiency and effectiveness and enormous band banding for increased tunnelling probability. All TFETs are deals with the principle of interband tunnelling phenomenon. The gate effectively balances the tunnelling at the source to channel barrier height.

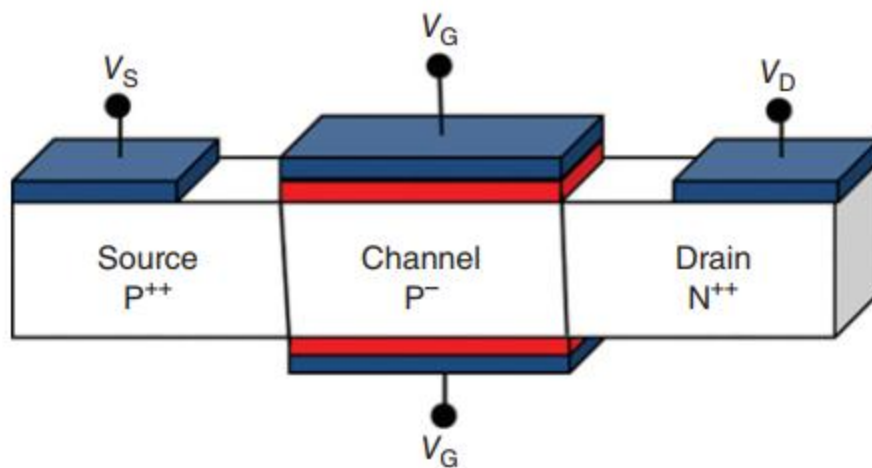


Figure 1.13: Shows double-gate tunnel field-effect transistor Energy band profiles in ON-state.

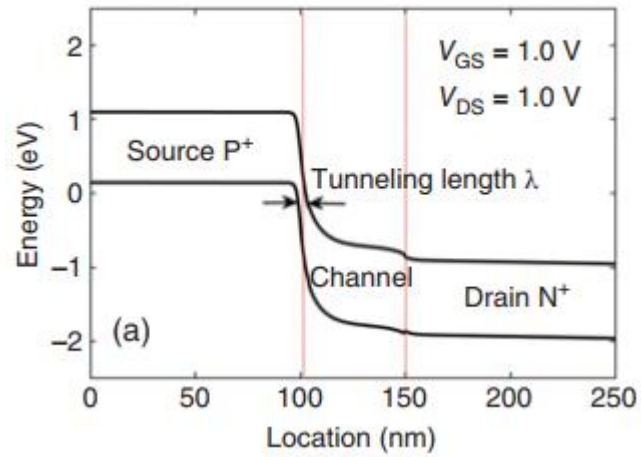


Figure 1.14: schematic view of a double gate n-TFET

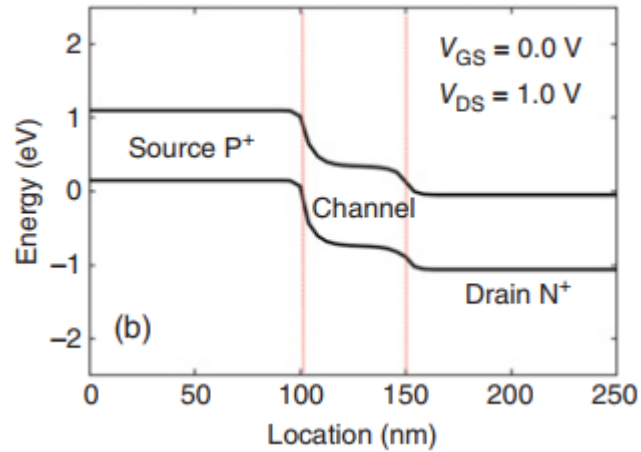


Figure 1.15: Shows the double-gate tunnel field-effect transistor energy band profiles in OFF-state.

If we applied higher gate voltage, the intrinsic channel region moves in to strong inversion region in the ON-state ($V_{GS}=1V$). You can see the band profile in figure 1.14. That there is close area of source valence band to the channel conduction band. This will help Band to band electrons tunnelling from source to the channel region, resulting in a drain current flow in the device. In the OFF-state ($V_{GS}=0.0V$), the valence band overlap with the forbidden energy band gap of channel region shown in figure 1.15. In this manner, there is no drain current flow in the device.

Because of intrinsic channel region there is extremely low leakage current in TFETs devices. TFETs are extremely effective to provide subthreshold slope less than 60mV/decade therefore TFETs as very high potential in low power application. But

efficiency comes at cost of Low ON-state current in the device also due to asymmetric structure it consumes large layout area compared to MOSFETs.

1.4.2 Two-Dimensional FETs

2D materials has open up a new horizon in transistor architecture and their application in optoelectronics and flexible electronics. Some of the 2D materials are MoS_2 , WSe_2 , and WTe_2 . Figure 1.16 shows three-dimensional view of a two-dimensional FET.

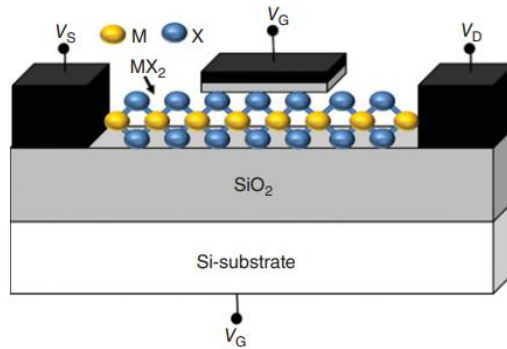


Figure 1.16: 2D-FET structure.

In 2D FET we use 2D-materials as an active layer and a highly doped semiconductor film at the lower part of the active layer which is work as a gate, source and drain contacts are formed at the top of active layer for electrical connection. The utilization of two-dimensional material in the channel region opens the path for ultra-short channel length scaling without compromising the device's electrostatic integrity. We can scale the channel to ultra-thin region even under 5 nm, without any carrier scattering, process-induced thickness variation and trap generation. Likewise, the TMD monolayer 2D materials shows high carrier mobility, improved band gap, a larger effective carrier mass and higher density of state. These in turn provide reduced leakage tunnelling current and significantly lower power dissipation. However, with all these benefits fabrication of 2D-FETs are very challenging. Methods and processes available for realization of atomically thin layers of 2D materials are limited and rudimentary. Also, because of weak bond formation, contact resistance is quite huge and consequently degrades performance of 2D-FET [8].

1.4.3 Impact Ionization MOSFET

Most of the transistors works on thermionic injection or interband tunnelling, aside from this there is one more conduction mechanism that is impact ionization as seen in

avalanche break down p-n junction diode. This impact ionization mechanism prompted positive feedback which in turn leads to very sharp transition from ON-state to OFF-state for small variation in gate voltage. If we somehow control this phenomenon with gate terminal abrupt transition ON-state to OFF-state may be obtained in field effect transistors. This is the fundamental principle behind Impact ionization MOSFETs.

The impact I-MOS has p-i-n diode structure in reverse biased configuration with gate terminal just to some segment of the channel as shown in figure 1.17. A portion of channel is left intentionally. Therefore, we have two different regions in channel. One is with gate and one is ungated intrinsic channel region. But why we choose intrinsic region? This is because for PIN type structure we need little electric field magnitude to induce avalanche multiplication at lower supply voltage.

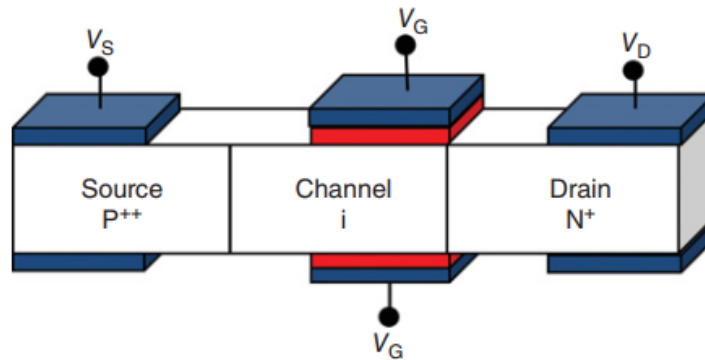


Figure 1.17: Three-dimensional view of a double-gate impact ionization MOSFET.

The gate terminal in I-MOS modulates the carrier in the channel but yet just a segment of it in the ON-state. While, in the OFF-state the total intrinsic area is work like a channel region displayed in figure 1.18. This will expand the effective length of the channel in OFF-state and will suppress major short channel effects. Further, in OFF-state, entire drain voltage is sustained by all intrinsic regions including gated and ungated channel regions. If we increase the gate voltage, the semiconductor below the gated region is inverted and whole drain voltage fall across the little ungated intrinsic region. Once the critical voltage (gate voltage) is reached at which impact ionization happens the current unexpectedly to a large value.

In I-MOS we can get subthreshold swing below 5mV/decade which is most highly desirable. However, In I-MOS output current does not saturate. Also, we need large operating voltages to achieve impact ionization therefore, make them unfit for logic application [9].

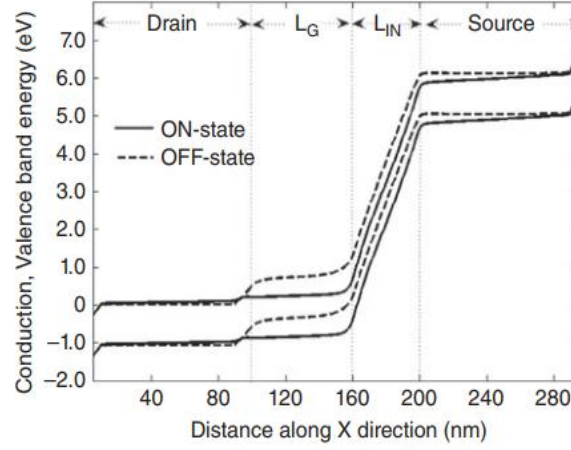


Figure 1.18: I-MOS Energy band diagram in ON-state and OFF-state.

1.4.4 Gallium Nitride JLTFETs

High frequency and high-power devices are becoming hot new topic. GaN is one of those material that is capable to utilize in high frequency and high-power application. This is primarily due to its inherent large critical electric field, high band gap, and high saturation velocity. Previously, GaN has been extensively used in high electron mobility transistors (HEMTs) to get very high bandwidth out of it.

Recently, it has been seen that JLT with Fin architecture made of GaN in the channel region shows a reduced reliance on the fin width differences primarily due to volume conduction mechanism. GaN as a channel material in finFET shows a reduction in OFF-state leakage current and improved breakdown voltage of the device. Therefore, GaN based junctionless transistors are seen as an advantageous alternative to conventional GaN based transistors [10].

1.5 Thesis Organization

After the introduction, we included four more chapters which is as follows:

Chapter 2: this chapter is all about literature review and work done in the past.

Chapter 3: this chapter discuss about method of simulation and different device physics related to the dissertation work.

Chapter 4 this chapter talks about results related to the work performed.

Chapter 5 this chapter discuss conclusion derived from dissertation and future scope of the analysed device.

Chapter 2

2.1 Introduction to Tunnel field effect transistors (TFETs)

In chapter 1 we discussed about various types of MOSFET design that are proposed to solve incessant scaling of conventional MOSFETs. Likewise, we have seen that because of continuous down scaling of MOSFETs causes short channel effects (SCEs) such as drain induced barrier lowering (DIBL), gate induced barrier lowering (GIDL), punch through, tunnelling leakage currents, quantum tunnelling through gate, increased power consumption due to non-scalability of subthreshold slope, impact ionization etc. Of all these short channel effects will increase leakage current, particularly subthreshold leakage due to non-scalability of subthreshold slope is one of the most genuine restraints to further scaling of classical MOSFETs. All these problems degrade the performance of circuit [2]. Therefore, there was a need to develop alternate doping technique to realize ultra-steep doping profile in the device. But what if there will be no P-N junction form in the device? No short channel issues happen due to P-N junctions, requirement of ultra-steep doping profile and no complex thermal budget. Our life should be much simpler.

To overcome this continue scaling limitation, one needs to investigate novel device architectures which use various modes of carrier injection. Tunnel FETs, also known as TFETs, Surface Tunnel Transistors (STTs), or Tunneling FETs, are promising low power devices alternative to the conventional MOSFETs. They have a small subthreshold swing and a very low off-current. Because they are based on a quantum tunneling barrier, Tunnel FETs are fascinating as low-power electronics device. When the device is on-state then the current flows from source to drain and allowing the carriers tunnel through to the barrier. The existence of the barrier keeping the off-current very low when the device is off state, a few orders of magnitude less than off-current of a traditional MOSFET. In T-FETs. Sub 60 mV/decade subthreshold swing is also experimentally found. A semiconductor substrate whose resistance is being controlled by the gate terminal and therefore current flowing through the substrate. Hence, it's also called gated resistor [25].

2.2 Device structure of TFET:-

The basic structure of a TFET is identical to that of a MOSFET, except that the source and drain terminals in a TFET are doped with different kinds of dopants. The drain and source

doping in a conventional MOSFETs, on the other hand, they are doped with the same type of doping. A P-I-N (p-type, intrinsic, n-type) junction is formed in a TFET devices. The intrinsic region's Electrostatic potential is controlled by a gate electrode. The structure of TFET is shown in figure 2.1. TFETs are divided into two categories. (a) n-type TFET (b) p-type TFET. The drain region of an n-type TFET is doped n^+ , while the source side is doped p^+ . The drain region of a p-type TFET is doped p^+ , while the source side is doped n^+ . The channel is a lightly doped or intrinsic n-type or p-type semiconductor. As with conventional MOSFETs, a dielectric isolates the channel from the gate electrode.

For a biasing schemes of n-type TFET, a positive voltage is provided to the drain and the gate terminals and the source terminal is connected to ground [1]. The source terminal of a p-type TFET is connected to ground and a negative potential is provided to the drain and gate terminal as shown in Fig 2.1. When a TFET device is turned on, the dominating carriers in the channel created under the gate terminal determines whether it is a p-type TFET or an n-type TFET. When electrons are the majority carriers in the channel, the TFET is called an n-type TFET, and when holes are the dominating carriers, the TFET is called a p-type TFET. When electrons arrive the channel through the source and exit through the drain, this form of TFET is known as an n-type TFET, and in similar way when holes enter the channel through the source and exit through the drain, this kind of TFET is referred as a p-type TFET.

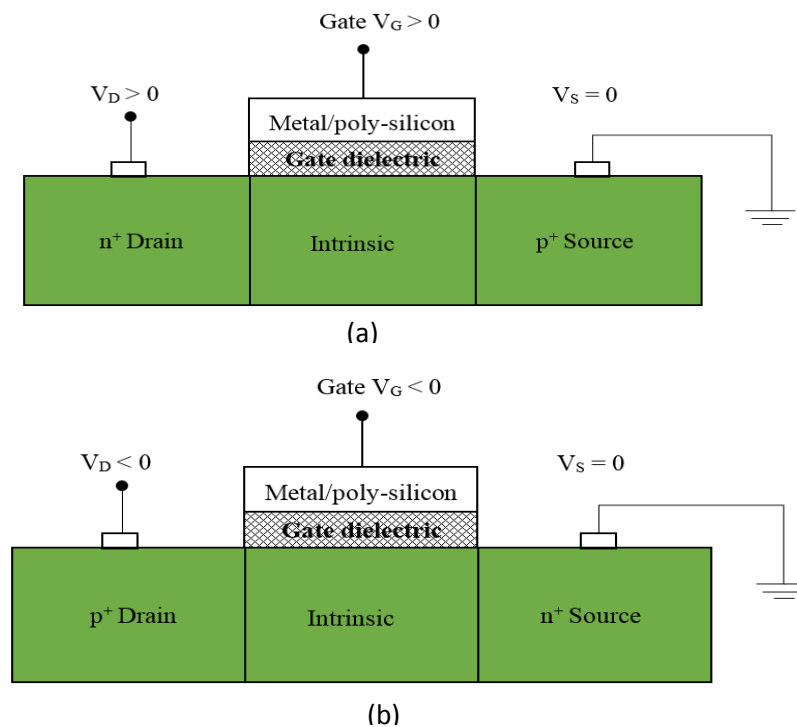


Figure 2.1: A simple implementation of a TFET and the biasing scheme: (a) n-type TFET

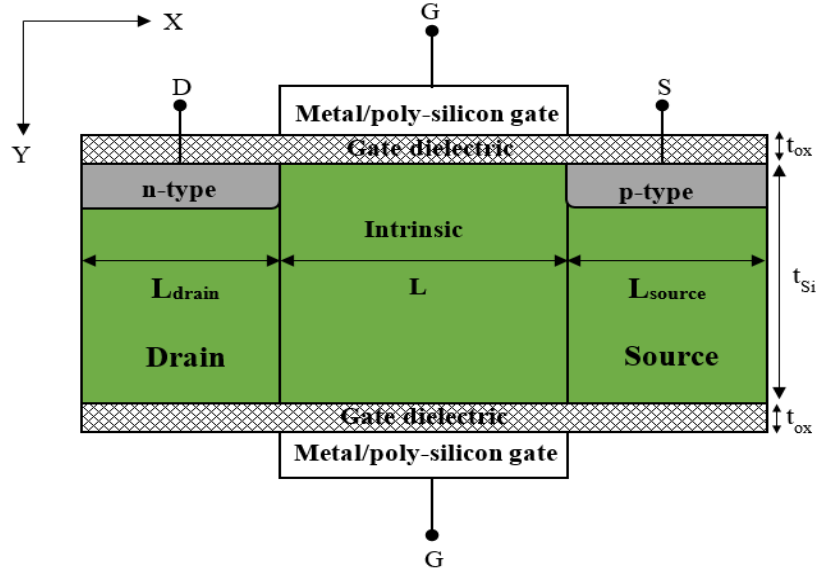


Figure 2.2: Schematic cross-sectional view of a DGTFET

BTBT is the mechanism through which dominating carriers enter the channel. The tunnelling current of TFETs is affected by several device factors such as gate dielectric thickness, body thickness, source-drain doping, channel length, high-k gate dielectric material, and gate work function, among others. When compared to the doping of the drain, the source is kept more doped. This is done to avoid the TFET from acting ambipolarly. A steep doping profile at the source is required to achieve a high ON-current. The ON-current is improved by keeping the silicon body's thickness thin. In a DGTFET construction, the gate control on the channel is increased, and the tunnelling current is estimated to be nearly two times that of a single-gate TFET structure.

2.3 Tunneling: -

TFETs are based on the principal of electronic band to band tunnelling and, in theory, transition between on and off states at lower voltages than MOSFETs. In contrast to conventional MOSFETs, which switch by modulating thermionic emission over a barrier, TFETs switch by modulating quantum tunnelling via a barrier. A quantum mechanical effect in which a wavefunction can pass across a potential barrier is known as tunnelling or quantum tunneling. To pass the carriers junction in classical mechanics, the carrier must have energy at least equivalent to the potential-barrier height. According to **quantum** physics, the possibility of it passing through the barrier for such a narrow width is finite and relies on the height of the barrier. This type of phenomenon is known as tunneling [11]. Quantum physics has far more complex principles than classical mechanics, but it

delivers considerably more precise findings for particles of all sizes, even the smallest ones. The wave-particle duality of atoms is handled by quantum physics.

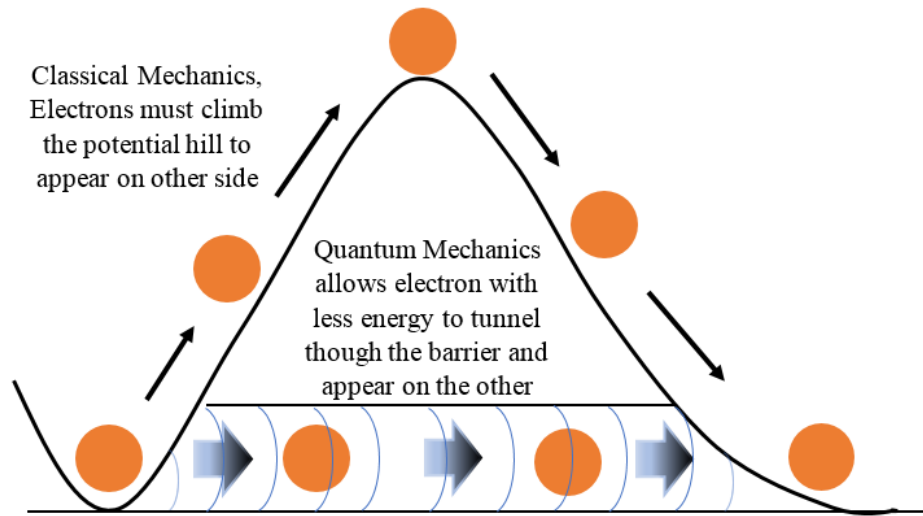


Figure 2.3: Classic and quantum tunneling of electron passing through the energy barrier

The above figure 2.3 shows the passing of electron through the energy barrier in both classic and quantum mechanics. In classical mechanics, an electron [blue] cannot pass through the energy barrier [orange] until and unless its energy is more than the height of the barrier. In fact, there is a finite probability of the electrons for passing through this boundary. For that the barrier should be very thin, such that tunneling event can take place, as shown for quantum mechanics [11].

This property of electrons to be able to tunnel through the thin barriers has been utilized as a principle of operation in the band-to-band tunneling process. From the most recent couple of years, transistor using this principle, referred as Tunnel Field Effect Transistor have caught the attention of the researcher because of its have good sub-threshold swing, very low leakage current, less temperature sensitivity and further improved short channel characteristics. In contrast to the MOSFET, in which the flow of current is controlled by raising or lowering an energy barrier, TFET always keeps its energy barrier high.

Tunnel FET behaves like an ambipolar device. When it is dominant with hole conduction it starts behaving like p-type and as n-type when it is dominant with electron conduction. This device alters the probability of electrons take place one side of the barrier from another side to switch it on or off. When compared to the operational voltage of a metal oxide semiconductor field effect transistor, tunnelling of electrons in tunnel field-effect

transistors can result in exchanging the on and off state at lower voltages. As a result, the devices' power consumption is reduced.

2.4 Operation: -

The BTBT technique is used to operate a TFET. Through the prohibited bandgap, carriers are tunnelled from the valance band to the conduction band, or vice versa, in BTBT. Band to Band tunneling are two types. 1. Direct BTBT 2. Indirect BTBT. Reverse bias is used to operate tunnel field effect transistors, which are really gated p-i-n diode. Not like MOSFETs, which use a thermal injection mechanism for source carrier injection. The band-to-band tunnelling technique is used in the Tunnel FET. Electron must tunnel from a semiconductor's valence band to the conduction band without absorbing or emitting phonons for band-to-band tunnelling to occur. As a result, the particle tunnelling undergoes no change in momentum. It must act as a potential barrier to enable the particle to tunnel over the band gap energy. Unlike indirect tunnelling, an electron does not need to produce or absorb a photon to move from the valence to the conduction band in direct tunnelling. The mechanism of direct tunnelling in silicon, unlike indirect band gap materials, is insignificant. This is happened due to rises the barrier height and the transmission probability significantly falls.

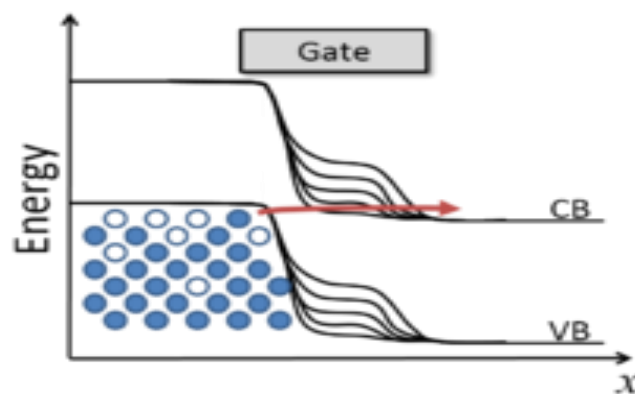


Figure 2.4: Energy band diagram for a basic lateral TFET structure

The energy band diagram for the fundamental TFET structure is shown in fig 2.4. The criterion for perpendicular momentum conservation, i.e., when tunnelling from the valence band to the conduction band, the perpendicular momentum should not change, which can increase tunnelling in direct tunnelling. Current can flow across the device because electrons from the p-type region's valence band tunnel into to the intrinsic region's conduction band. The bands become mismatched as the gate voltage is

decreased, and current could no more flow. When enough gate voltage is provided, electrons can tunnel from the source valence band to the channel conduction band, and the device turns on, bands become aligned and current start flow in the devices.

2.5 Schematic band diagram of TFET: -

1. In OFF state- $V_{GS} = 0V$, $V_{DS} = \text{Positive}$
2. In ON state- $V_{GS} > 0V$, $V_{DS} = \text{Positive}$

The above figure 2.5 shows the OFF and ON conditions energy band diagrams of an n-type TFET. The TFET is in the OFF-state when gate voltage is nearly zero. Because the conduction band in the channel is higher than the valence band in the source, a large potential barrier between the channel and the source prevents tunnelling for Tunnel FET. As a consequence, BTBT is stopped when TFET operates in the OFF-state, the P-I-N diode is reverse biased, and there is only a few fA/um of leakage current flowing. However, when the gate potential exceeds threshold voltage, the TFET enters the ON state. The carrier density under the gate is modulated by the gate voltage, and the conduction band in the channel is pushed down. Band bending occurs at the source, when a sufficiently high voltage is supplied to the gate terminal, causing the valence band in the source to align with conduction band in the channel, as seen in Figure 2.5. As an outcome, electrons in the source's valence band can jump to the channel's conduction band. The positive voltage of the drain pushes the electrons that tunnel through the channel to the drain terminal. The thick potential barrier becomes substantially thinner during the ON-state, allowing a significant current flow to tunnel through the devices [1].

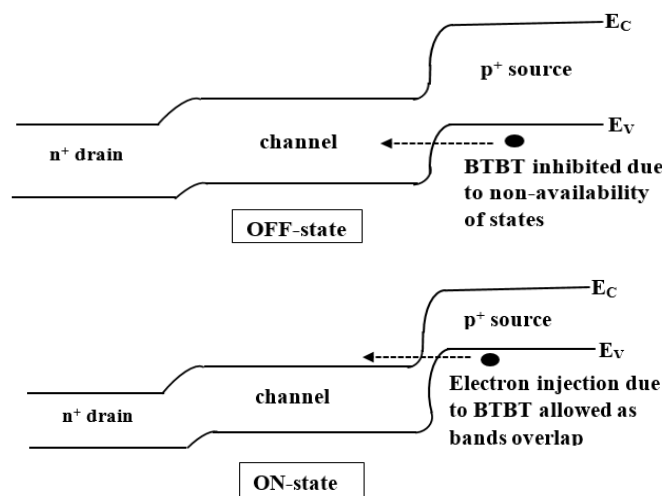


Figure 2.5: Band diagram of an n-type TFET in the OFF-state and in the ON-state.

The width of the energy barrier, as well as the overlap of the valence band with the conduction band and the barrier at the tunnel junction, determines the ON and OFF conditions in tunnel FETs. When the barrier width is reduced, the ON-current rises exponentially. This is the basic operation of an n-type TFET.

In the same way, the energy band diagrams of a p-type TFET in the OFF and ON-state are illustrated in Figure 2.6. The working mechanism of a p-type TFET is identical to that of an n-type TFET. When applying the voltage on the gate terminal is nearly zero, TFET is in the OFF-state. The source's conduction band is higher than the channel's valence band.

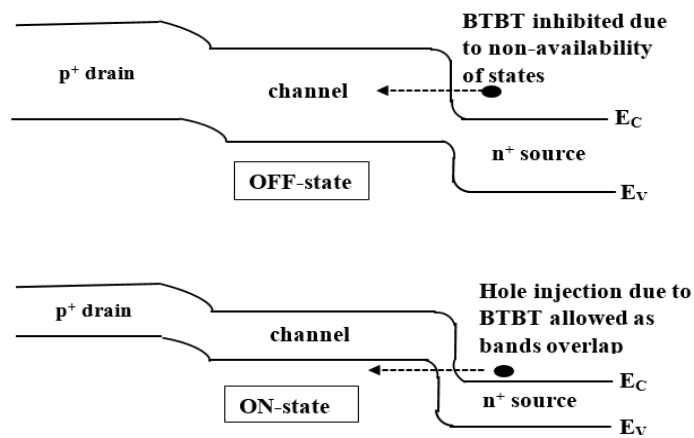


Figure 2.6: Band diagram of an p-type TFET in the OFF-state and in the ON-state.

As a consequence, BTBT is disabled, and the TFET is turned off. When a sufficient enough negative voltage is applied to the gate, the valence band in the channel is pushed over the conduction band in the source and energy barrier becomes narrow sufficiently to allow BTBT tunneling, as seen in Figure 2.6. Consequently, holes are injected into the channel, and the drain's negative bias sweeps the holes to the drain terminal. This is the basic operation of a p-type TFET.

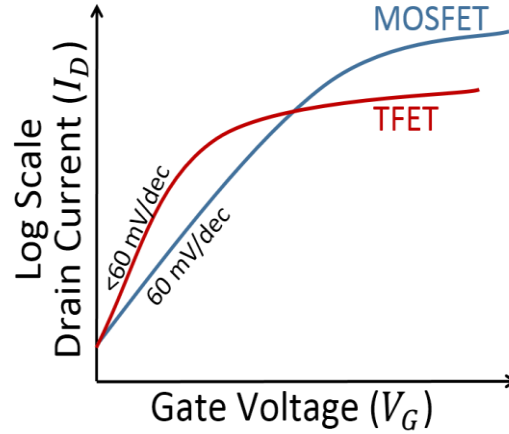


Figure 2.7: The transfer characteristics of MOSFET and Tunnel FET along with their sub-threshold slope.

The transfer properties of MOSFETs and Tunnel FETs are compared in the above Fig 2.7. When compared to MOSFETs, TFETs may be useful for obtaining greater drain currents at lower voltages. In MOSFET current flow due to diffusion phenomena, while TFET is based on tunnel effect. When applied the large voltage on bias gate terminal then the ON current of TFET is low when compared to the current flow through the MOSFET. Since it depends on band to band tunneling in the small area, however not in the entire channel region. This causes issues when tunnel FETs are used in application. Another issue with these transistors is their high threshold voltage.

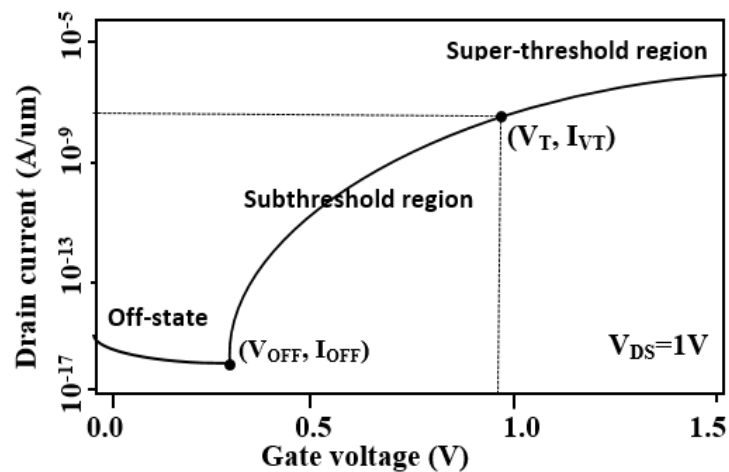


Figure 2.8: Simulated transfer characteristics of the DGTFTFET at $V_{DS} = 1.0$ V

2.6 Transfer characteristics of TFET: -

In general, a TFET's transfer properties can be divided into three categories:

1. OFF-state region ($0 < V_{GS} < V_{OFF}$): The TFET has a very small current in this region, and BTBT is blocked. V_{OFF} is the gate voltage at which the drain current begins to take off.

2. Subthreshold region ($V_{OFF} < V_{GS} < V_T$): In this area, the drain current is quickly increasing. DGTFET's threshold voltage is denoted by V_T .

3. Super-threshold region ($V_{GS} > V_T$): The drain current flows at a slower rate in this region as the gate voltage rises.

2.6.1 OFF-State

When applying the gate voltage is less than V_{OFF} , the source and channel bands are misaligned (equivalent to $V_{GS} = 0V$ in figure 2.5). As a result, BTBT between the source and the channel is hindered, and the TFET's tunnelling current is very low. The drain current does not begin to increase rapidly until the gate voltage $V_{OFF} = 0.3V$ is reached. The OFF-current of the TFET is the drain current (I_{OFF}) at voltage $V_{GS} = V_{OFF}$, the I_{OFF} in a DGTFET is in femtoampere range. It's important to note that a greater gate voltage may be needed to aligning the bands and allow BTBT for low drain voltage. As a result, at lower drain voltages, a greater V_{OFF} could be detected [12].

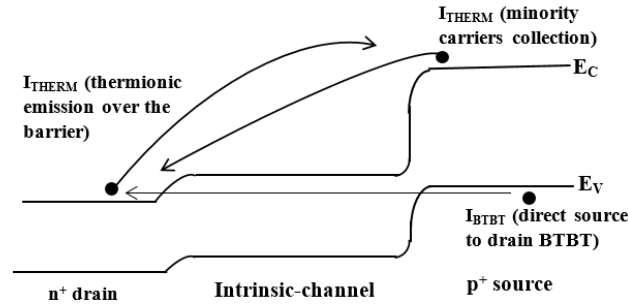


Figure 2.9: Components of the OFF-state current in a TFET: I_{BTBT} due to BTBT and I_{THERM} due to thermionic emission.

The drain current in a TFET has two significant elements in the OFF-state, as shown in Figure 2.9. The first factor, I_{BTBT} , is because of the carriers' direct tunnelling from the source terminal to the drain. The chance of tunnelling the entire length of the channel is quite low, hence this part of the I_{OFF} is typically expected to be very small. However, when the channel length is short. Because of the minority carrier accumulation from the source to the drain terminal and the thermally induced flow of carriers over the drain terminal to source barrier, I_{THERM} is the next main component of TFET in Figure 2.9. The

I_{THERM} is equivalent to $\exp(-E_g/nkT)$, where material's bandgap is denoted by E_g , junction constant is n , the symbol of Boltzmann constant is k , and T represents the absolute temperature in kelvin.

In the TFET, I_{THERM} is frequently found to be a more effective portion of the I_{OFF} [13]. To reduce I_{THERM} , the bandgap of the TFET material might be increased. However, raising the bandgap of the TFET material can result in an unexpected reduction in the TFET's ON-current. When thin gate dielectrics are used in TFETs then direct leakage current flow through the gate dielectric can be considerable, in addition to two parts of the I_{OFF} discussed above.

2.6.2 Subthreshold Region

When applying the gate voltage exceeds V_{OFF} , the drain current begins to quickly increment. The bands in the channel and the source are aligned in this area, and the BTBT is allowed. Accordingly, the tunnelling current begins to flow in this operation area. The drain current increases rapidly near $V_{\text{GS}} = V_{\text{OFF}}$ and the drain current increases at a faster rate as the gate voltage decreases.

The subthreshold slope, or its inverse subthreshold swing, measures the rate at which the drain current increases as the gate voltage increases. The subthreshold swing is measured in mV/decade and is defined as the variation in gate voltage necessary to increase the drain current by a factor of ten. The presence of a little subthreshold swing in a transistor is favourable. When a transistor has a small subthreshold swing, a little gate voltage can be used to boost the drain current from the OFF to the ON-state, making the transistor acceptable for functioning at low power voltage. In comparison to a conventional MOSFET, the sub - threshold swing of a TFET is mostly determined by the gate voltage, as seen by the DGTFTFET's transfer characteristics in Figure 2.8. As a result, there are two types of subthreshold swings for TFETs: the point subthreshold swing and the average subthreshold swing. The point subthreshold swing (SS_{POINT}) is defined as the inverse of the instantaneous gradient of the transfer characteristics at a certain gate voltage V_{GS} , which can be expressed as-

$$(SS_{\text{POINT}})_{V_{\text{GS}}} = \left(\frac{dV_{\text{GS}}}{d \log(I_{\text{DS}})} \right)_{V_{\text{GS}}} \quad (2.1)$$

I_{DS} denotes the drain current. The SS_{POINT} in a TFET can be very small at gate voltages when the drain current begins to rise quickly. Over a few decades of drain current, it is

predicted that the TFETs will have a good subthreshold swing than a conventional MOSFET. The average subthreshold swing (SS_{AVG}) is calculated by using the following [25]:

$$SS_{AVG} = \frac{(V_T - V_{OFF})}{\log(I_{VT}) - \log(I_{OFF})} \quad (2.2)$$

where V_T denotes the TFET's threshold voltage, the gate voltage V_{OFF} is the point at which the drain current starts to increment quickly, I_{VT} is the drain current when $V_{GS} = V_T$, and I_{OFF} is the drain current when $V_{GS} = V_{OFF}$. The gate voltage at which drain current exceeds 1×10^{-7} A/ μ m at $V_{DS} = V_{DD}$, where power supply is V_{DD} , is known as the threshold voltage (V_T). The DGTFET's threshold voltage is estimated to be 0.9V based on the transfer characteristics. The DGTFET is a basic design of a TFET that is not appropriate for low-power CMOS operations due to issues such as low ON-current, high V_T , and non-competitive SS_{AVG} , as shown in Figure 2.2. Several new technologies can be used to increase the performance of the DGTFET

2.6.3 Super-Threshold Region

The drain current continues to increase as the gate voltage exceeds V_T . The ON-current of the TFET is the drain current at $V_{GS} = V_{DS} = V_{DD}$. It is denoted by I_{ON} . Assuming $V_{DD} = 1.0$ V and $I_{ON} \approx 0.2$ μ A/ μ m for the DGTFET in fig 2.2. The DGTFET's ON-current is relatively low. In general, the I_{ON} produced in a TFET is much less than the demand for future low-power uses due to the method of current transportation. This is the most significant disadvantage of TFETs, and researchers are effectively working hard to find ways to boost the I_{ON} in TFETs.

2.7 Subthreshold swing in Tunnel FETs: -

The very good subthreshold swing of a TFET, especially near the gate voltage where the drain current begins to take off, is one of most fundamental electrical features that separates it from conventional MOSFETs.

In a conventional MOSFET, current transport is done by injecting carriers from source into the channel, with only carriers with energies greater than the potential barrier Φ_{max} in the exponential tail of the source Fermi–Dirac distribution contributing to current in the subthreshold region. The drain current in a conventional MOSFET is proportional to

the density of state $D(E)$, the source Fermi–Dirac distribution function $f_s(E)$, and the carrier velocity $v(E)$, where E is the electron energy [15].

$$I_d \propto \int dE \cdot D(E) \cdot v(E) \cdot f_s(E) \quad (2.3)$$

The source Fermi–Dirac distribution function $f_s(E)$ can be approximated by a Boltzmann estimation in the exponential tail region:

$$f_s(E) \approx \exp\left(-\frac{E-E_F^s}{kT}\right) \quad (2.4)$$

Where E_F^s is the Fermi energy level in the source, Boltzmann constant is denoted by k and temperature in Kelvin is T . For the sake of simplicity, the product $D(E) \times v(E)$ can be assumed to be constant when transport occurs in only one particular direction. Furthermore, we can assume that for a perfect MOSFET [16], the change in the conduction band is proportional to the variation in the gate voltage. so, we can write as-

$$\frac{\partial I_d}{\partial V_{GS}} = |e| \frac{\partial I_d}{\partial E} \quad (2.5)$$

where e denotes an electron's charge. The subthreshold swing in a typical MOSFET can be calculated by using Equations (2.4) and (2.5).

$$SS = \left(\frac{\partial \log I_d}{\partial V_{GS}}\right)^{-1} = \ln(10) \left(\frac{\partial I_d}{I_d \partial V_{GS}}\right)^{-1} \approx \ln(10) \frac{kT}{|e|} \quad (2.6)$$

$SS \approx 60$ mV/decade is found at room temperature. In a conventional MOSFET, equation (2.6) relates to the optimum subthreshold swing [15].

Equation (2.6) also indicates that when the temperature rises, the optimum subthreshold swing increases. $f_F(E)$ is the Fermi–Dirac distribution function defined by –

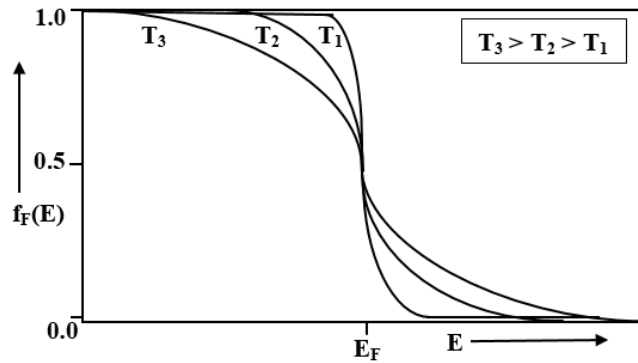


Figure 2.10: Fermi–Dirac distribution function at three temperatures T_1 , T_2 and T_3 . The exponential tail becomes prominent as the temperature increases. The probability of occupying higher energy states increases as the temperature increases.

$$f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)} \quad (2.7)$$

Where Fermi energy is denoted by E_F . As we can see in above equation, temperature's effect on the Fermi–Dirac distribution function is shown in fig 2.10. Around the Fermi energy level, the Fermi–Dirac distribution function rapidly changes, as can be seen. It is obvious that when the temperature rises, the chance of the carriers inhabiting higher energy states rises as well. In a conventional MOSFET, the subthreshold current over the energy barrier is caused by carriers with high energies. As a result, the effect of the temperature-dependent expansion of the Fermi–Dirac distribution function of the carriers is explained in Equation (2.6). TFETs can have a subthreshold swing of less than 60 mV/decade at normal temperature [15]. Subsequently, in order to get a good subthreshold swing and an unexpected current change.

The subthreshold swing in a TFET is influenced by a few different factors, including the the BTBT probability function, control of the gate voltage over the channel potential, the abruptness of the source–channel junction, the interface states, defect-assisted tunnelling, and interface roughness, as well as the arrangement of the bands and Fermi levels.

There are two types of significant swings that are strongly dependent on the gate voltage up to the threshold voltage. The first is that the tunnel FETs' subthreshold swing is not fixed, but it does depend on gate voltage. Second, when gate voltages are low and at room temperature, Tunnel FETs can exhibit subthreshold swings that aren't quite as high as the 60 mV/decade MOSFET limit.

To obtain an equation [17] for the subthreshold swing of a band-to-band tunnelling device, we can start with the expression provided for the tunneling current via a reverse-biased p-n junction:

$$I = aV_{eff}F \exp\left(-\frac{b}{F}\right) \quad (2.8)$$

where

$$a = Aq^3 \sqrt{2m^*/E_g} / 4\pi^2 \hbar^2 \quad (2.9)$$

And

$$b = 4\sqrt{m^*} E_g^{3/2} / 3q\hbar \quad (2.10)$$

The electric field at the tunnel junction is F and the tunnel junction potential is V_{eff} . The result of computing the subthreshold swing is denoted as $S = dV_{GS}/d(\log I_{DS})$ and the expression is written below-

$$S = \ln 10 \left[\frac{1}{V_{eff}} \frac{dV_{eff}}{dV_{GS}} + \frac{F+b}{F^2} \frac{dF}{dV_{GS}} \right]^{-1} \quad (2.11)$$

From above equation some conclusions can be drawn. It should be noted that the subthreshold swing is depend on V_{GS} . This indicates that the subthreshold region doesn't follow a linear curve, when I_{DS} - V_{GS} curve is plotted on a logarithmic scale and the swing doesn't have a single value. Swing is lowest when V_{GS} is lower potential and gradually improves as V_{GS} rises [17].

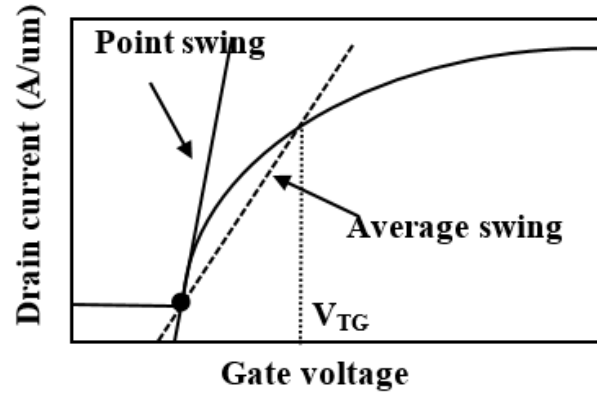


Figure 2.11: Visual definitions of point swing, taken at the steepest point of the I_{DS} - V_{GS} curve, and average swing, taken as the average from turn-on to threshold.

From the above I_{DS} - V_{GS} figure 2.11, It's helpful to differentiate between two kinds of swings: point swing (S_{pt}) and average swing (S_{avg}). The smallest value of the subthreshold swing can be observed anyplace on the I_{DS} - V_{GS} graph, generally precisely when the device exits the off-state and tunnelling current begins to flow.

The average swing is calculated from where the device first turns on until it reaches the threshold, which is commonly calculated using the constant current approach. However, in order to effectively apply the average gradient. For circuit designers, the average swing is the more helpful parameter as shown figure 2.11. In sequence for the turn-on point to occur exactly at $V_{GS} = 0V$, the gate work function would need to be changed.

2.8 Band-to-band tunneling transmission: -

The total transmission function T_t has a strong influence on the drain current. Two types of scattering occur in a TFET: scattering in due to tunneling and scattering within the

channel. However, the BTBT is the most predominant scattering in a TFET, so we can write-

$$T_t \approx T_{BTBT} \quad (2.12)$$

The WKB approximation method with a triangularly formed potential barrier may be used to calculate the BTBT tunnelling current in Tunnel FETs, which is shown in Fig 2.12. The band-to-band tunnelling transmission is obtained by the WKB estimation technique -

$$T_t \approx \exp \left[-2 \int_{-x_1}^{x_2} |k(x)| dx \right] \quad (2.13)$$

The quantum wave vector of the electron is represented by $k(x)$ inside the barrier [18]. The wave vector inside a triangle barrier is -

$$k(x) = \sqrt{\frac{2m^*(PE-E)}{\hbar^2}} \quad (2.14)$$

Here, PE denotes for potential energy, E for incoming electron energy, the reduced plank constant is $\hbar$, and the effective mass of electron is m^* . When a triangle potential energy barrier is drawn at the given coordinates displayed in Fig. 2.12, with the electron reaches at the energy of the extensive portion of the triangle (at $E=0$), then the E component goes away, and PE follows the triangle equation: $E_{g/2}-qF_x$, where E_g denotes the material's band gap and the electric field is F. Then,

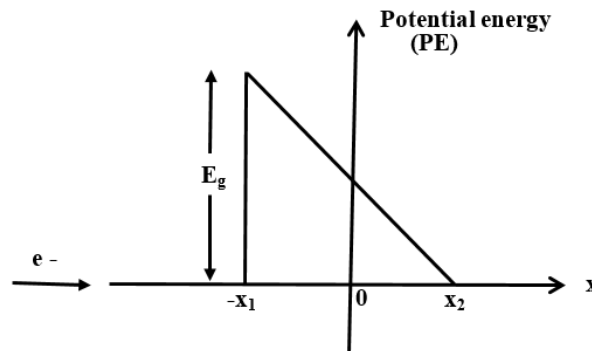


Figure 2.12: Band-to-band tunnelling can be calculated by approximating the energy barrier width by a triangular potential energy barrier, where the electrons must tunnel through the widest distance at the base of the triangle.

$$k(x) = \sqrt{\frac{2m^*}{\hbar^2} \left(\frac{E_g}{2} - qF_x \right)} \quad (2.15)$$

Put the above equation into Eq. (2.13) and it gives -

$$T_t \approx \exp \left[-2 \int_{-x_1}^{x_2} \sqrt{\frac{2m^*}{\hbar^2} \left(\frac{E_g}{2} - qF_x \right)} dx \right] \quad (2.16)$$

The following step is to perform the integration. -

$$T_t \approx \exp \left[\frac{4}{3} \frac{\sqrt{2m^*}}{q\epsilon\hbar} \left(\frac{E_g}{2} - qF_x \right)^{3/2} \right]_{x_1}^{x_2} \quad (2.17)$$

From the graph of triangular potential barrier, we know that when $x = x_2$, $(E_{g/2} - qFx) = 0$, and at $x = -x_1$, $(E_{g/2} - qFx) = E_g$, so after solving, we can get the equation -

$$T_t \approx \exp \left(- \frac{4\sqrt{2m^*} E_g^{3/2}}{3qF\hbar} \right) \quad (2.18)$$

A general expression for band-to-band tunnelling transmission is Equation (2.18). This equation may be modified much more particular to tunnelling transistors by narrowing it down. According to Figure 2.13, the shaded triangle barrier has a height of $\Delta\Phi + E_g$ and width of transition region is λ at the source-channel interface. The electric field F can be replaced with a function that corresponds to the slope of the energy bands then electric field F [19] can replace by $\Delta y/\Delta x = (\Delta\Phi + E_g)/\lambda$. Since electric fields are calculated in V/m, and the new unit is eV/m.

$$I_{BTBT} \propto T_t \approx \exp \left(- \frac{4\lambda\sqrt{2m^*} E_g^{3/2}}{3q(\Delta\Phi + E_g)\hbar} \right) \quad (2.19)$$

From, the drain current or tunneling current is approximately equivalent to T_{BTBT} and the energy overlap window, as shown by the above equation (2.19). $\Delta\Phi$ is the energy range in which tunnelling can occur, E_g is the bandgap at the tunnel junction, λ is a width of tunneling barrier, and m^* is the tunneling mass. For band-to-band tunnelling requires the four main conditions: Available states from which to tunnel and available states to which to tunnel, a thin enough energy barrier to allow tunnelling, and momentum conservation.

The parameter λ has few unique names like as screening length, natural length, and Debye length. It is dependent on gate geometry and can be defined in terms of gate insulating thicknesses, dielectric constants, and semiconductor body of a device [15]. The expression of λ for a double-gate TFET device is denoted by-

$$\lambda = \sqrt{\frac{\epsilon_{Si} t_{Si} t_{ox}}{2\epsilon_{ox}}} \quad (2.20)$$

where ϵ_{Si} and t_{Si} represent the dielectric permittivity and thickness of silicon respectively, the semiconductor material utilized to fabricate the device, ϵ_{ox} and t_{ox} represent the gate insulator of dielectric permittivity and thickness. For a single-gate TFET device, the coefficient $(1/2)^{0.5}$ must be eliminated from the above equation [15].

The Tunnel FET functions as a band pass filter, cutting off the low- and high-energy tails of the n+-type source's Fermi distribution, as shown in Fig. 2.13, The Fermi-Dirac distribution and Fermi level for the source are initially drawn to the left inside the source, and the distribution's low-energy tail is blocked off since carriers cannot exist at energies within the band gap. The high-energy tail is then crossed out on the channel side since those energy levels cannot exist within the bandgap of the channel. Only electrons within the range of energies $\Delta\Phi$ in the source are suitable for tunnelling [15,16], as illustrated in the below figure 2.13.

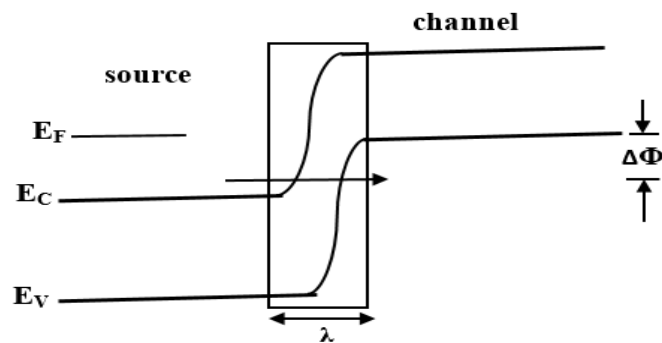


Figure 2.13: Energy band cross section of a Tunnel FET showing the triangular barrier approximation within the bands, $\Delta\Phi$, the screening length λ , and the filtering behaviour of the device in the on-state.

2.9 Output characteristics

The output characteristics of the DGTFTFET at various gate voltages are shown in Figure 2.14. The TFET's output characteristics have a parabolic onset and thereafter saturation [20,12]. The rate of increment in drain current with V_{DS} has been observed, the output properties can be divided into three categories: -

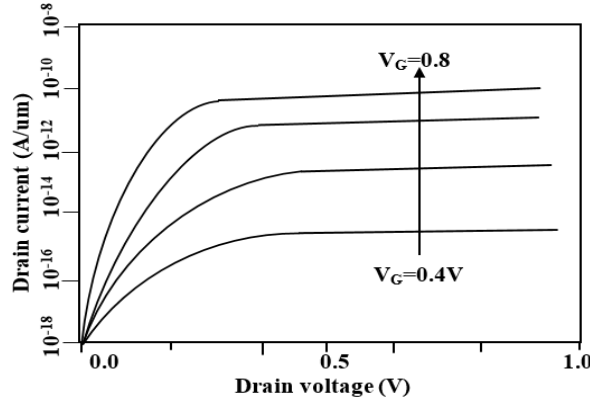


Figure 2.14: Simulated output characteristics in the DGTFTFET

1. Tunnel resistance–dominated region: With increasing V_{DS} , drain current increases quickly, but is restricted by tunnel resistance.

2. Channel resistance–dominated region: The drain current increases at a slower rate with increment V_{DS} in this region, as well as the current is restricted by the channel resistance.

3. Saturation region: The drain current is almost constant flow in this region.

2.9.1 Tunnel Resistance–Dominated Region

When applying the drain potential is very low, the gate voltage is also applied, as a result in the creation of a channel with a high electron concentration. A high concentration of electrons causes the channel to have a small resistance, allowing the drain voltage to keep the conduction band in the channel [20]. As a result, the entire drain voltage occurs across the tunnelling junction, and the drain voltage modulates the tunnelling width at the source-channel junction. The beginning of current is rapidly rising in this area of operation. The change in the available of carriers at the starting of the tunnelling route and the availability of empty states at the ending of the tunnelling path explains the super-linear beginning of current. It should be noted that the availability of the filled states in the source and the empty states in the channel plays a vital role in determining the behaviour of this operation area. The voltage drop due to channel resistance is very small and the source–channel tunnel resistance controls the output characteristic. As a consequently, this type of region is known as the tunnel resistance–dominated region [20].

2.9.2 Channel Resistance–Dominated Region

In a TFET device, most of the electrons from the drain are injected in the channel while a little fraction of the electrons in the channel is provided from the source by the BTBT. As a result, the V_{GD} voltage rather than the V_{GS} voltage is used to create the channel in the TFET [12]. V_{GD} reduces when the drain voltage rises, resulting in a drop in electron concentration in the channel. The electron concentration in the channel sharply falls as the applying voltage V_{DS} is increased more than 0.3V as a larger number of electrons are pushed back to the drain from the channel then channel resistance will be increased. With the supplied drain potential, the drain current increases more slowly due to increased channel resistance [20]. This type of region is called as the channel resistance–dominated region because the channel resistance plays a substantial influence in the behaviour of the output characteristics in this region.

2.9.3 Saturation Region

The output characteristics of the TFET is operated in saturation mode when the drain voltage is applied more than 0.6V. The electron concentration gradually decreases in the channel. The channel begins to disappear away from the source terminal in reality. When applying the drain bias is more than 0.6V, the electron concentration is extremely low and the channel resistance increases [20], due to this reason the horizontal electric field from the drain can't pass through the channel region. In saturation mode, TFET is operated. The extra drain voltage is show up around the drain–channel junction and the gate terminal mainly control the potential at the source–channel junction. Therefore, a high drain voltage is predicted in saturation condition in the output characteristics of a TFET. Since, a TFET is predicted to achieve complete saturation. In a TFET, the optimum voltage equals to the drain voltage when the channel when the channel becomes without carriers. In a TFET, saturation voltage V_{DSAT} can be directly linearly relation with the gate voltage, much like in a normal MOSFET, as drawn below [20,12]:

$$V_{DSAT} = V_{GS} - V_T, \quad (2.21)$$

where V_{GS} denotes the gate voltage and the threshold voltage is represented by V_T in the TFET. From the output characteristics observed in figure 2.14 that in comparison to conventional silicon MOSFETs, the drain current is low. When compared to a normal

MOSFET, the existence of a tunnelling barrier in a TFET limits the drain current because of tunnelling process occurs in small region, not on the whole channel region.

2.10 Threshold Voltage

The threshold voltage of a TFET can be explain in a number of different manners. The constant current technique-based threshold voltage definition is the most widely utilised threshold voltage definition in a TFET. The gate potential at which the drain current exceeds 1×10^{-7} A/um at $V_{DS} = V_{DD}$, where V_{DD} denotes the supply voltage [25], is called as the threshold voltage. Although this description of the threshold voltage is easy to derive from the transfer characteristics, it has no physical importance.

The variation in transconductance is another technique for determining the threshold voltage in a TFET [21]. When the drain voltage V_{DS} is set up to a specific value, the transconductance g_m is described as follows:

$$g_m = \left(\frac{\partial I_D}{\partial V_{GS}} \right)_{V_{DS}} \quad (2.22)$$

In a TFET, the gate voltage V_{GS} affects the g_m value. The gate voltage at which the greatest dg_m/dV_{GS} is obtained is known as the threshold voltage. This technique's threshold voltage defines the transition between the region where the drain current has a quasi-exponential dependency on V_{GS} and the region where it is linearly dependent on V_{GS} . Because the drain current has a complex relationship with the drain voltage, a threshold voltage for a TFET can also be determined with respect to the drain voltage. The value at which the greatest dg_{DS}/dV_{DS} is found is known as the drain threshold voltage, where g_{DS} is the gate voltage V_{GS} at which g_{DS} is defined [21] as–

$$g_{DS} = \left(\frac{\partial I_D}{\partial V_{DS}} \right)_{V_{GS}} \quad (2.23)$$

It should be observed that the transconductance-change technique for determining threshold voltages is sensitive to mathematical errors due to the calculation of higher-order derivatives.

Another definition of the threshold voltage dependent on the band arrangement has been offered in the modelling of TFETs. The gate voltage at which the Fermi energy level in

the source and the channel's edge of the conduction band align is known as the threshold voltage.

The proper work function of the gate material can be used to adjust the threshold voltage in a TFET. Figure 2.15 displays the DGTFET's threshold voltage for various gate material work functions. Even in a typical MOSFET, the threshold voltage has a direct linear relation with the gate's work function. When the gate work function changes, the DGTFET's transfer characteristics move along the X-axis. The output characteristics shift in the same direction along X-axis when the gate's work function changes [20].

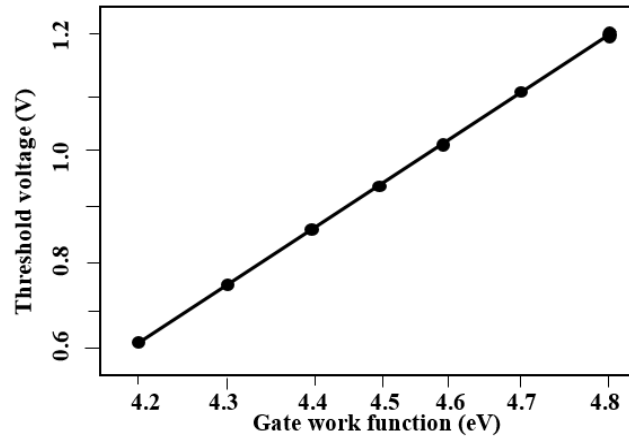


Figure 2.15: Effect of the gate work function on the threshold voltage in the DGTFET. The threshold voltage is defined as the gate voltage at which the drain current reaches 1×10^{-7} A/ μ m at $V_{DS} = 1V$.

2.11 Effect of Temperature

To examine the impact of temperature on the subthreshold swing in a traditional MOSFET versus a TFET is fascinating. From Equation (2.6), we can conclude that the subthreshold swing is predicted to decrease linearly with rising temperature for an ideal MOSFET. As a result, with a normal MOSFET, as the temperature rises, the off current and the power dissipation will also increase [22]. As a consequence, thermal runaway conditions can happen, possibility to destroying the circuit. As a result of the temperature-induced broadening of the Fermi–Dirac distribution around the Fermi energy level, the subthreshold swing in a TFET can vary moderately with rising temperature. The temperature dependency of the subthreshold swing in a TFET is significantly weaker than that of a typical MOSFET. As a result, TFETs can be more resistant to thermal runaway because the leakage currents are predicted to degrade slowly as the temperature rises. The fact that the BTBT in the TFET is inhibited during OFF-state and the drain current is dominated by thermal production of carriers in the

depletion layer should be noted. As a result, even if the OFF-state leakage current flow because of thermal generation is extremely low, it might exhibit a strong temperature dependence.

In a TFET, subthreshold swing is the weak temperature dependence. It's a good way to check if the current transport mechanism in TFETs is working. Engineers frequently use this analysis to analyze whether the current transport process in the fabricated TFET is band-to-band tunnelling. Although BTBT is the major mode of current transport in a TFET, a larger temperature dependence of the subthreshold swing can be seen on occasion. The presence of charge carriers could be the cause for this. The BTBT of the carriers through the tunnelling barrier is the largest component of the drain current in a TFET during ON-state. The material's bandgap E_g is a function of temperature T and is given [23] as-

$$E_g(T) = E_g(0) - \frac{\alpha T^2}{\beta + T} \quad (2.24)$$

where

at $T = 0$, $E_g(0)$ is the bandgap of the material, α and β are constants. As the temperature rises, the material's bandgap reduces. When a result, the drain current rises in the ON-state as the temperature rises due to more BTBT. Yet, at extremely high drain voltages and drain currents, the increased channel resistance caused by rising temperature might result in a drain current reduction [23].

Chapter 3

In this chapter we will discuss method to simulate DGTFET transistor and models that enable us to see true semiconductor device physics behind it. All the simulation has been performed in Atlas Silvaco [24].

3.1 ATLAS

ATLAS provides a way to simulate 2D as well as 3D semiconductor devices. It calculates electrical behaviour of specified semiconductor device and physical mechanism insight about the device. Atlas needs two type of inputs one is text based and other is structure file. Text file contains command that tells atlas what electrical behaviour we are going to simulate in the device whereas structure file contains overall architecture of the device either in 2D form or 3D form. Atlas has 3 output files. First is runtime output second is log file and third one is the solution file. Runtime output gives the advancement and the error and warning about the continuing simulation. All terminal voltages and current from device analysis are saved in Log file. The value of the solution variable within the device at a specific bias point is saved as 2D and 3D data in the solution file. All this is summarised in figure 3.1.

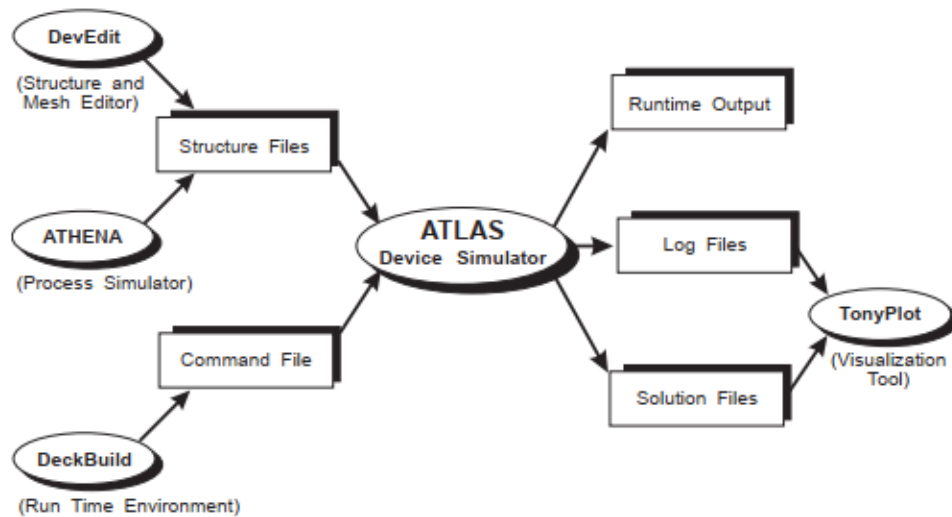


Figure 3.1: Atlas inputs and outputs

In our dissertation we use mesh editor for defining the structure of tunnel field effect transistor. To define structure, we define mesh that contains all the information with respect to its structure.

3.1.1 MESH

To specify a MESH, we use command as: -

The first statement of every mesh initialization must be:

MESH SPACE.MULT=<VALUE>

This is followed by a sequences of X.MESH and Y.MESH commands in the script.

X.MESH LOCATION=<VALUE> SPACING=<VALUE>

.
.
.

Y.MESH LOCATION=<VALUE> SPACING=<VALUE>

.
.
.

The SPACE.MULT statement is a factor for scaling and its default value is one. Vertical and horizontal lines are defined using the X.MESH and Y.MESH expressions, respectively in microns unit, individually, along with the vertical or level dividing related with that line. For each direction, you must define at least two mesh lines. The X.MESH and Y.MESH statements must be defined in such a way that they are in increasing order of x and y. Both negative value and positive values of x and y are permissible in the simulation script. Figure 3.2 shows how some of these statements works on the right we can see mesh with fine with a value of 0.5 and spacing of the vertical lines varies from $1\mu\text{m}$ at $x=0$ and $x=10\mu\text{m}$ to $0.5\mu\text{m}$ at $x=5\mu\text{m}$ where as in the left meshing is coarse with value 1.

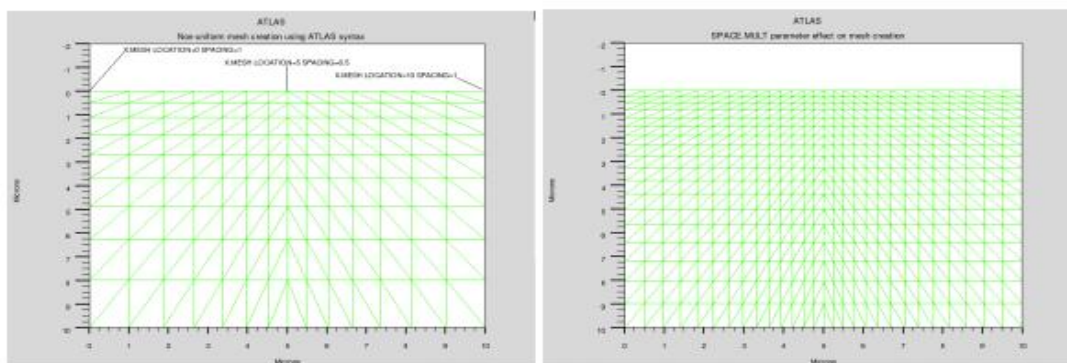


Figure 3.2: mesh creation using Atlas syntax with non-uniform mesh.

3.1.2 Regions and materials

When mesh is defined, we have to define material in every part of it and this is accomplished by region statement in the script. For eg.

```
REGION number=<integer> <material type> <position parameters>
```

Each and every Region number need to begin with a value of 1 and rise with each succeeding region statement. In Silvaco Atlas, we may have up to 15000 distinct types of regions. Atlas Silvaco contains a vast number of material repositories. The x and y composition fractions can also be provided in the REGION declaration if a material with composition type is defined.

The commands X.MIN X.MAX Y.MIN Y.MAX are used to define the position., if there is any overlap area is there than overlapped region is allocated as a type of new material. One more thing is we must specify mesh point if you leave any then atlas won't run and shows error message.

3.1.3 Specifying electrodes

To contact semiconductor device with outside world we need to specify electrodes this is through an ELECTRODE statement as shown below

```
ELECTRODE NAME=<electrode name> <position_parameters>
```

50 electrode contacts can be defined in the device. Position of electrodes are defined with same way as that of materials X.MIN X.MAX Y.MIN Y.MAX. Multiple electrodes can have same name linked with them and they are tied together in simulation.

3.1.4 Specifying doping

Doping can be specified from process simulation file or can be specified in atlas with doping statement. Specifying doping with DOPING statement as:-

```
DOPING <distribution_type> <dopant_type> <position_parameters>
```

Doping can be uniform or Gaussian and can be specified as:-

```
DOPING UNIFORM CONCENTRATION=1E20 N.TYPE REGION=1
```

```
DOPING CONCENTRATION=1E20 CHARACTERISTIC=0.01 N.TYPE X.LEFT=0.0  
X.RIGHT=2.0 PEAK=0.5
```

3.1.5 Specifying contact work function and type of electrode

The CONTACT statement is used to define the function of metal work and type as shown below

CONTACT NAME=GATE WORKFUNCTION=4.8

CONTACT NAME=DRAIN

CONTACT NAME=SOURCE

CONTACT NAME =GATE N.POLYSILICON

3.1.6 Specifying physical model

Semiconductor working physics needs to be specified while simulating the device. It is specified by MODELS and IMPACT commands. Physical model has been categorized in five classes: carrier statistics, mobility, recombination, impact ionization and tunnelling. For example

MODELS CONMOB FLDMOB SRH FERMIDIRAC

IMPACT SELB

Some physical model

Table 3.1: Carrier statistics Models

Model	Syntax	Notes
Boltzmann	BOLTZMANN	Default model
Fermi-Dirac	FERMI	Reduced carrier concentrations in heavily doped regions (statistical approach)
Bandgap Narrowing	BGN	Important in heavily doped regions. Critical for bipolar gain. Use Klaassen Model.
Silicon Ionization model	IONIZ	Accounts for full ionization for heavily doped Si. Use with INCOMPLETE.

Table 3.2: Mobility Models

Model	Syntax	Notes
Concentration Dependent	CONMOB	Lookup table valid at 300K for Si and GaAs only. Uses simple power law temperature dependence.
Lombardi (CVT) Model	CVT	Complete model including N, T, E_{11} , and $E_{\perp}$ effects. Good for non-planer devices.

Table 3.3: Recombination Models

Model	Syntax	Notes
Shockley-Read-Hall	SRH	Uses fixed minority carrier lifetimes. Should be used in most simulations.
Concentration Dependent	CONSRH	Uses concentration dependent lifetimes. Recommended for Si.
Surface	S. N S. P	Recombination at semiconductor to insulator interfaces. This is set in the INTERFACE statement.

Table 3.4: Tunneling Models and Carrier Injection Models

Model	Syntax	Notes
Band- to-Band (standard)	BBT.STD	For direct transitions. Required with very high fields.
Concannon Gate Current Model	N. CONCAN P. CONCAN	Non-local gate model consistent with concannon substrate current model.
Direct Quantum tunneling (Electrons)	QTUNN.EL	Quantum tunneling through conduction band barrier due to an insulator.
Direct Quantum tunneling (Hole)	QTUNN.HO	Quantum tunneling through valance band barrier due to an insulator.
Hot Electron Injection	HEI	Model's energetic carriers tunnelling through insulators. Used for gate current and FLASH EEPROM programming.
Hot Hole Injection	HHI	HHI means hot electron injection.

3.1.7 Specifying numerical solution technique

To calculate solution to device problems various numerical simulation methods can be used. METHOD statement is used to specify numerical solution method.

Atlas can solve up to six equations simultaneously with each numerical method. There are mainly three types of solution technique present in Atlas, namely a) GUMMEL b) NEWTON and c) BLOCK. We can use combination different solution also. GUMMEL Solve equation by finding one unknown at a time and kept other as a constant. NEWTON solves total system of unknown together and BLOCK method will solve some equation fully coupled and other with de coupled. When a system of equations is loosely coupled but exhibits linear convergence, the GUMMEL technique is applied. We use the NEWTON technique when a system of equations is tightly coupled and exhibits quadratic convergence. But it takes extra time solving for quantities. BLOCK method is used when fast convergence is required than newton. Mixed numerical method that utilizes all the combination is given as: - METHOD NEWTON BLOCK GUMMEL

3.1.8 Obtaining Solution

In semiconductor characterization of its electric behaviour, we need DC, AC as well as transient solutions. Atlas supports all of these. Typically, voltages are applied to each electrode or terminal of the device, and Atlas then estimates current flowing through each electrode as well as internal device parameters like as temperature, mobility, electric field, velocity, carrier concentration at each point.

Statement for obtaining solution

For DC

SOLVE VGATE=1.0

SOLVE VGATE=1.5

To sweep

SOLVE VCOLLECTOR=1.5

SOLVE VBASE=0.0 VSTEP=0.01 VFINAL=2.0 NAME=BASE

Here we sweep base voltage from 0 volts to 1 volt in steps of 0.05 volt

For AC

SOLVE VBASE=0.0 VSTEP=0.01 VFINAL=2.0 NAME=base AC FREQ=1.0e6

Here we specify frequency at which we have to perform AC simulation

For transient

SOLVE VGATE=2.0 RAMPTIME=1E-9 TSTOP=10e-9 TSTEP=1e-11

It specifies gate voltage ramped from initial value to 2V with a period of 1 ns.

Now, discuss some of the output files in Atlas

3.1.9 Run time output

While simulating device in the deckBuild window at the bottom we can see run time output. It shows what code it executing and its output correspondingly. In models statement if we add print statement then in run time, we will be able to see all the details related to models in runtime output.

Output will look something like shown below

Obtaining static solution:

```
prev      psi   n   p   psi   n   p
direct     x    x    x   rhs   rhs   rhs

  i   j   m -5.00* -5.00* -5.00* -26.0* -17.3* -17.3*
-----
  1    N -13.5* -14.1* -13.5* -30.9* -18.9* -17.13
```

Electrode	Va(V)	Jn(A/um)	Jp(A/um)	Jc(A/um)	Jt(A/um)
-----------	-------	----------	----------	----------	----------

=====

source	0.000e+000	3.295e-021	-5.743e-035	3.295e-021	3.295e-021
drain	0.000e+000	-7.101e-020	-7.337e-035	-7.101e-020	-7.101e-020
gate	0.000e+001	0.000e+0001	0.000e+001	0.000e+001	0.000e+001

Chapter 4

4.1 Analysis of Process-voltage-temperature (PVT) in DG TFET

In the previous chapters we have learned some of the challenges, we are facing while designing MOS transistors and how newer architectures and technologies tackling these problems. We have also seen what type of benefits TFET architecture can provide and what its inherited problem. So, in this chapter we will focus on the process-voltage-temperature (PVT) analysis of DG-TFET which is one of the main challenges to overcome while designing the device. The TFET works on band-to-band tunneling principle. In the analysis of PVT, we study the misalignment of gates, impact of variation in upper gate oxide thickness (T_{ox1}), effect of interface fixed charges at Si-SiO₂ interface, impact of variation in both gate oxide thickness (T_{ox}), variation of gate work function (Φ_m), impact of variation in temperature (T) and impact of scaling in supply voltage in TFET devices. All these types of parameter variation will affect the TFET device performance. A parametric analysis is carried out, in which effect of one parameter is examined at a time, and the simulation results of the device characteristics are investigated by the Silvaco TCAD software ATLAS [24]. Tunnel FET characteristics are shown to be significantly more sensitive to parameter variation in DG-TFET, when compared to conventional MOSFETs and especially it effects the electrical parameter like I_{ON} , I_{OFF} , threshold voltage, subthreshold swing (SS) and I_{ON}/I_{OFF} current- ratio. Power consumption in OFF state is called as the static power dissipation (P_s), which is calculated by $P_s = V_{DD} \cdot I_{OFF}$ where supply voltage is denoted by V_{DD} and I_{OFF} is the sum of device's leakage current. Dynamic power is mostly dependent on the frequency of switching and is called as dynamic power P_D which is given by $P_D = V_{DD}^2 C_L f \alpha$. From the above equation, we can see that static power consumption is proportional to V_{DD} and dynamic power consumption is proportional to square of supply voltage, there is a more loss of power [1]. So down-scaling of supply voltage (V_{DD}) is one of the most important techniques to decrease the power consumption in the TFET devices. By selecting the appropriate above parameter and excellent gate control over the channel region and charges will lead considerable alignment of channel region valence band to the conduction band of the drain region in n-type TFET transistor. This significant overlap of the bands assists tunnelling of electron from source to channel region which we called as band-to-band tunnelling in short BTBT.

4.2 Device structure and simulation parameters

So, from here onwards, we will discuss the analysis of conventional DG-TFET architecture and simulation parameters. Fig. 4.1(a) shows the schematic view of the conventional double gate TFET (also known as low- k DGTFET). The both gate metal in a conventional DG-TFET comprises of material (M) with work function material M (Φ_m) is taken as 4.5 eV and such that which is the optimum work function in terms of performance [25]. Similarly, the top and bottom gate oxide is comprised of same dielectric material of the device. A low- k gate dielectric silicon dioxide (SiO_2) is employed from middle of metal gate to silicon layer. The body thickness of transistor is manufactured of silicon and gate is composed of silicon dioxide. In this device, dual gate design will be used to better gate electrostatic control at tunnel junction provided by both gates and current will be at least double. The other device structure parameters are mentioned in Table 1. All simulations are performed using SILVACO ATLAS TCAD tool at $V_{GS}=1.5\text{V}$. The DG-TFET structure can be fabricated by using Silicon-on-insulator (SOI) and silicon-on-nothing (SON) technique present used to develop the double-gate devices [25]. Furthermore, dual- k dielectric may be realized using selective etching of SiO_2 followed by deposition of high- k dielectric using ALD process provides as the result of better gate coupling. The device structure and its performance are evaluated using Silvaco atlas 2D device simulator [24] as the device thickness is ≤ 10 nm quantum confinement effect comes into play. Therefore, to incorporate quantum confinement, Hansch quantum confinement model is enabled. To determine tunneling current in OFF-state and Band-to-Band-tunneling (BTBT), non-local Band-to-Band-tunneling model based on Wentzel-Kramer-Brillouin method is enabled. Fermi-Dirac statistics were invoked which is required in Shockley-Read-Hall expression (SRH) for computing intrinsic carrier concentration. For calculation of mobility and carrier lifetime Concentration dependent (CONMOB) mobility, SRH, and CONSRH recombination model were used. Due to high doping of carrier concentration of source, drain and intrinsic region band gap narrowing (BGN) model was also activated. The simulation set up has been calibrated from reference [25] as shown. Also, for tunneling in OFF-state, the parameters were used according to [25]. The tunneling masses of $0.14m_o$ for holes and $0.14m_o$ for electrons are taken as mentioned in [25].

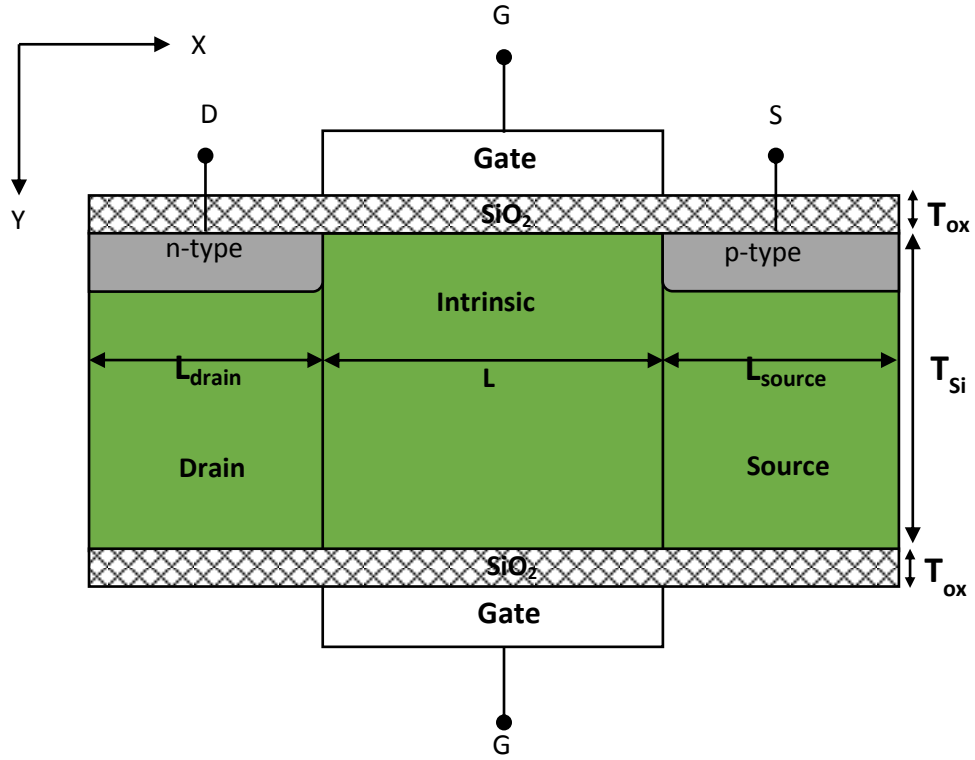


Figure 4.1: Schematic view of (a) Conventional DG-TFET

Table 4.1: Device structure parameters of DGTFET

Parameters	Value
Doping concentration of source (N_A)	$1 \times 10^{20} \text{ atoms/cm}^3$
Doping concentration of drain (N_D)	$5 \times 10^{18} \text{ atoms/cm}^3$
Doping concentration of channel (N_D)	$1 \times 10^{17} \text{ atoms/cm}^3$
Channel length (L)	30 nm
Length of source (L_{source})	100 nm
Length of drain (L_{drain})	100 nm
Gate oxide thickness (T_{OX})	3 nm
Device thickness (T_{Si})	10 nm
Gate work function Φ_m (eV)	4.5
Low k gate dielectric length L_{SiO2}	230 nm

4.3 Simulation results and discussion

4.3.1 Impact of process variation in DG-TFET

If, we are designing any TFET devices during the fabrication process then process is slow means actual parameter value of device changes and if any of parameters change, it will change the device characteristics of electrical parameters of TFET devices. These scaling of devices or transistors can be achieved with the analysis of process variation. If we growth the gate oxide thickness of 3nm but during the fabrication process, oxide thickness growth up to 3.5nm then it changes the device characteristics of DG-TFET device.

We have investigated the effect of process variation on double gate TFET architecture and in the process variation, we have analyzed the misalignment of bottom gates, impact of variation in upper gate oxide thickness (T_{ox1}), effect of interface fixed charges at Si-SiO₂ interface, impact of variation in both gate oxide thickness (T_{ox}) and variation of gate work function (Φ_m) in double gate TFET [26, 27]. The impact of process variation in DG-TFET especially its effect on the electrical parameter like on-current (I_{ON}), off-current (I_{OFF}), threshold voltage (V_T), subthreshold swing (SS) and I_{ON}/I_{OFF} current-ratio etc. Now based on the process variation analyses, the device parameters are optimized in order to operate as performance boosters. After that, all these performance boosters are applied to the device to achieve superior current characteristics. Now, all the various parameter variations will discuss in conventional DG-TFET one by one.

4.3.1.1 *Misalignment of Gates*

Gate alignment is another important booster parameter for DGTFET device. The gate electrode is not guaranteed to be properly aligned on the channel because of fabrication tolerances, especially for short channel devices. We have investigated the impacts of bottom gate misalignment (BGM) towards source and drain side i.e underlap and overlap by using TCAD simulations. A self-aligned process is defined when alignment = 0 which means dopant is properly diffused under the gate. The best performance for a DGTFET device is obtained when the bottom gate is completely aligned and it improves the device speed. Double gate improves the electrostatic control of the gate at the tunnel junction [28]. Due to misalignment of bottom gate towards source side, the control of gate on the channel will be slightly reduced but on the drain side, it has no longer control of the tunnel

junction. When the bottom gate is misaligned towards the source side, a sharp reduction in electron concentration is observed near the p+-region-intrinsic region. The carrier concentration remains constant at the source region when misalignment of bottom gate moves towards drain side, while the electron concentration at the drain side decreases as the bottom gate misalignment increases. For our conventional DG-TFET, it can be observed that when DG-TFET is perfectly aligned means misalignment = 0, the configuration provides the highest current density and the device performance is degraded when the misalignment of the bottom gate increases and drain current consistently decreases. Electric field occurs max near the source-channel interface. Misalignment of gates mainly influences the electrical parameter characteristic of DGTfET [29].

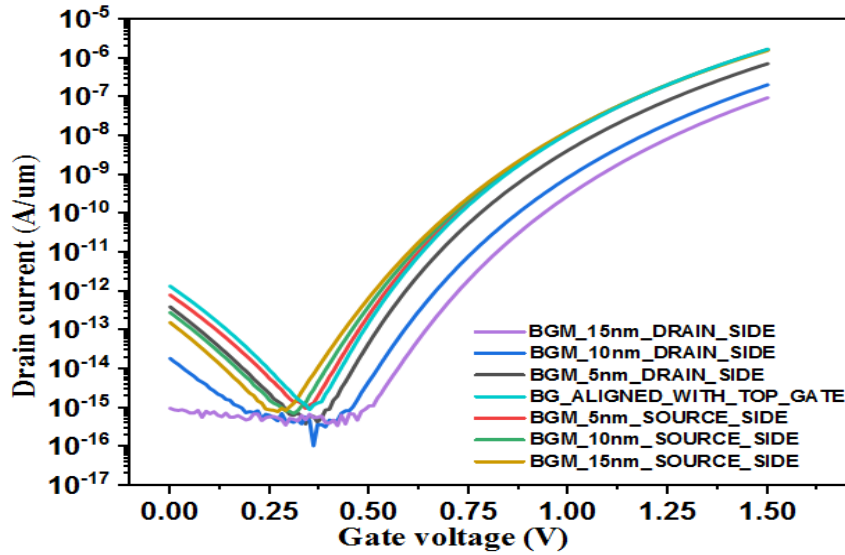


Figure 4.2: Simulation of I_{DS} - V_{GS} Transfer characteristics for misalignment of gate at $V_{DS}=1.5V$ in log scale.

Transfer Characteristics

In this section, we have analyzed the transfer characteristics of conventional DG-TFET due to misalignment of gates towards source and drain side in fig. 4.2. For better switching performance of devices, the I_{ON}/I_{OFF} ratio should be higher. It can be seen at the gate voltage varying from 0V to 1.5V and at supply voltage $V_{DS}=1.5V$. When gate voltage is increased then tunnelling current begins to increase rapidly. This happened due to the alignment of bands in source and the channel and BTBT is enabled then drain current start to flow in DG-TFET device. The drain voltage as well as the gate voltage affects tunneling current. However, around 1V of gate voltage, the drain current starts to saturate. It is due to short channel effect and there is not much improvement on the ON-

current. We observed that the ON-state current of conventional DG-TFET at $V_{DS}=V_{GS}=1.5V$ and OFF-state at $V_{DS}=1.5V$ and $V_{GS}=0V$. In this state, only leakage current flow in the range of pico-ampere and femto- ampere because of the misalignment of the source and channel bands. Band to band tunnelling between source and channel is inhibited and negligible amount of current flows.

The I_D - V_{GS} characteristic for misalignment of bottom gate towards source and drain side configuration at $V_{DS}=1.5V$ are shown in fig. 4.2. When bottom gate is misaligned towards source, the performance of ON-current of the device will decrease. The DG-TFET has produced maximum current 1.73×10^{-6} A/ μm when both gate is perfectly aligned and bottom gate misaligned 5nm, 10nm and 15nm towards source side then drain current will slightly reduce. When bottom gate is shifted towards drain side 5nm, 10nm, 15nm then we observed that drain current decreases slightly fast in comparison of shifting of bottom gate towards source. Table 4.2 shows device performance parameters namely subthreshold slope (SS), threshold voltage V_T , I_{ON} , I_{OFF} , I_{ON}/I_{OFF} ratio under the different misalignment configurations. We have measured threshold voltage V_T using the constant current method with a threshold current of 1×10^{-7} A/ μm at $V_{DS}=V_{DD}$, V_{DD} is the supply voltage. Best performance is shown when both gates will perfectly align. We have analyzed that overall degradation in the device performance is worst when back gate misalignment takes place.

Table 4.2: Device performance parameters

Parameter	SS (mV/dec)	$V_T(V)$	$I_{ON}(A/\mu m)$	$I_{OFF}(A/\mu m)$	I_{ON}/I_{OFF}
Bottom and top gate aligned	50.47	1.176	1.73×10^{-6}	1.37×10^{-12}	1.26×10^6
Bottom Gate misalignment 5nm_Drain side	42.17	1.26	7.34×10^{-7}	4.01×10^{-13}	1.83×10^6
Bottom Gate misalignment 5nm_Source side	54.58	1.175	1.70×10^{-6}	8.10×10^{-13}	2.10×10^6
Bottom Gate misalignment 10nm_Drain side	18.57	1.41	2.10×10^{-7}	1.88×10^{-14}	1.12×10^7
Bottom Gate misalignment 10nm Source side	49.86	1.172	1.69×10^{-6}	2.88×10^{-13}	5.87×10^6
Bottom Gate misalignment 15nm_Drain side	37.12	-	9.76×10^{-8}	9.81×10^{-16}	9.95×10^7
Bottom Gate misalignment 15nm_Source side	56.55	1.174	1.61×10^{-6}	1.59×10^{-13}	1.01×10^7

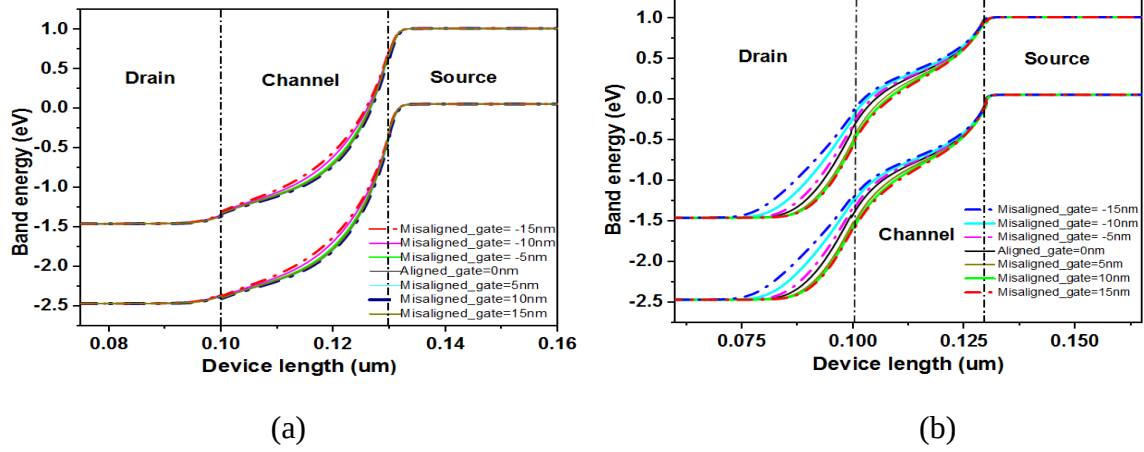


Figure 4.3: Energy band profiles for misalignment of gate 5nm, 10nm, 15nm towards source (+) and drain (-) side in (a) ON-state. ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$, $V_{GS}=0V$), cutline 1nm below gate oxide-channel interface.

Fig. 4.3 shows the energy band profiles (a) ON-state and (b) OFF-state due to misalignment of gate in DGTFET device. In ON-state, the voltage is applied to gate, the potential barrier width between the V.B. of p+-region and C.B. of intrinsic region becomes narrow and leading to a high possibility of BTBT tunneling mechanism because it reduces the screening length and increases the tunneling probability. The high electric field created at the tunnel junction and tunneling current flows in the device. Due to misalignment of bottom gate i.e shifts toward source and drain side then barrier width of tunneling junction slightly increases because of decreasing in gate control ability and electric field at tunnel junction. The energy band becomes slightly less steep curved then BTBT reduces and drain current also decreases. In OFF-state, it can be observed that no potential is applied to gate electrode, the electron from the p+-region cannot pass through the potential barrier between the V.B. of p+-region and C.B. of intrinsic region due to thick tunneling barrier width and hence BTBT of electrons is negligible. As a result, only the leakage current flows in the device. Misalignment of bottom gate causes a decrease in OFF-state current.

Performance analysis of electrical parameters

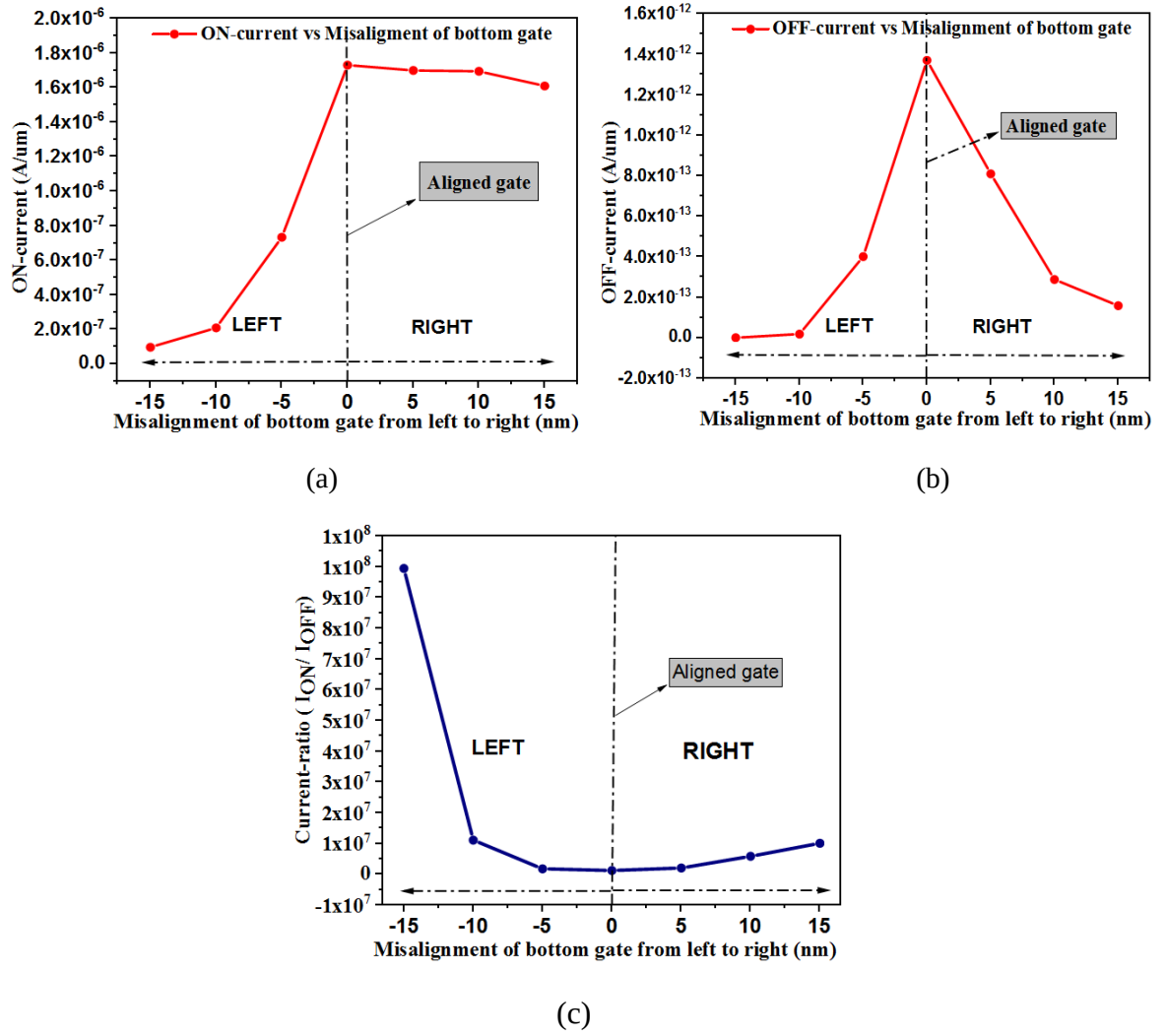


Figure 4.4: Simulation of (a) ON-current ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-current ($V_{DS}=1.5V$, $V_{GS}= 0V$) (C) I_{ON}/I_{OFF} ratio variation in misalignment of bottom gate varying from (-15nm to 15nm) at $V_{DS}=1.5V$.

Fig. 4.4 shows the graph of misalignment of bottom gate move towards source and drain side variation on electrical parameters like (a) ON-current, (b) OFF-current and (c) I_{ON}/I_{OFF} ratio respectively. Table 4.2 shows the electrical performance parameters under the variation in bottom gate of DG-TFET. It is observed from fig.4.4 and device performance table 4.2 that I_{ON} is maximum for aligned gate = 1.73×10^{-6} A/ μ m and is more as compared to misalignment of bottom gate towards source and drain side [29, 30]. But simultaneously, there is significant decrease in I_{OFF} also. For better switching performance of devices, the I_{ON}/I_{OFF} ratio should be high. Best performance of I_{ON}/I_{OFF} is shown when bottom gate moves towards 15nm drain side. I_{ON}/I_{OFF} ratio peaks when misalignment of bottom gate shifts 15nm towards drain side. After this,

current-ratio dropped till 0nm shift of bottom gate and then it starts to increase. Misalignment of bottom gate affects the BTBT in DG-TFET devices. After comparing the parameters, we have found that the best performance occurs when both the gates will be perfectly aligned in DG-TFET device.

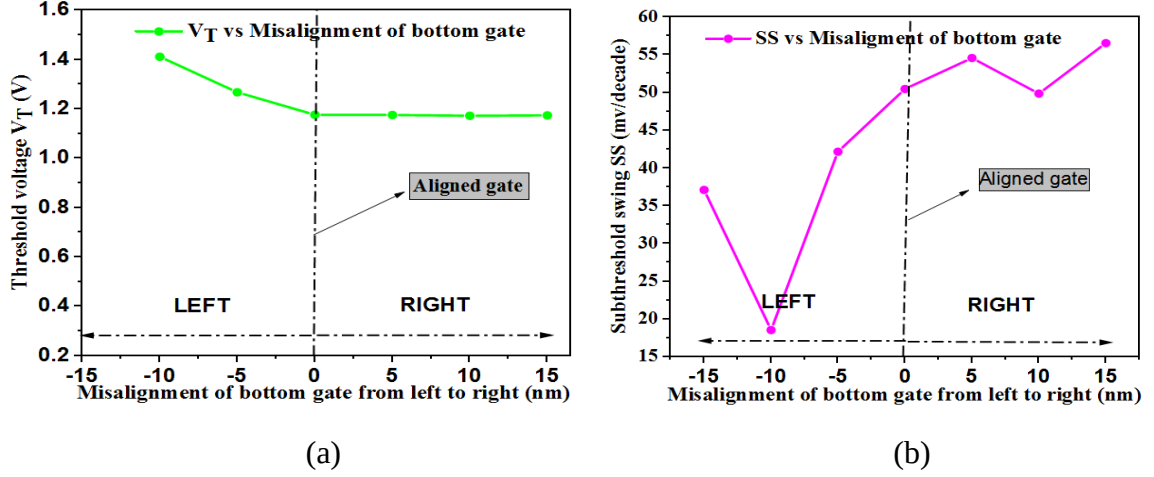


Figure 4.5: (a) Threshold voltage (V_T) (b) Subthreshold swing (SS) graphs show the variation in misalignment of bottom gate at $V_{DS}=1.5V$.

Electrical parameters like subthreshold swing and threshold voltage are most vital parameters to estimate the switching power and exact behaviour of any device or circuits. Subthreshold swing and threshold should be as low as possible for fast switching and low power applications. Due to misalignment of bottom gate towards source side, the control of gate on the channel slightly reduces whereas on the drain side, the gate control of the tunnel junction decreases. It is observed from fig. 4.5 (a) that the threshold voltage is almost constant when bottom gate shift towards source side from (5nm to 15nm) and threshold voltage increases when it moves towards drain side (5nm to 15nm) in DG-TFET device [30]. The subthreshold swing decreases linearly when bottom gate moves 15nm to 10nm towards drain side and then it increases when bottom gate moves towards source side upto 5nm and then it deteriorates after 5nm of misalignment of bottom gate. Thus, from the Table 4.2, we have found that the best performance is observed when both the gates will be perfectly aligned in DG-TFET device.

4.3.1.2 Effect of interface fixed charges at Si-SiO₂ interface

Electron or hole traps, can be negatively or positively charged or be neutral, and will have a net charge density. The interface charge or the interface charge density is the net

charge or charge density of all the electrically active defects located at the interface. There are mainly two types of charges exist at the Si-SiO₂ interface, first charge is fixed and another one is interface trapped charge. The thermal oxidation technique produces fixed oxide charges, which are generally positive charges at the semiconductor-oxide interface. Si-SiO₂ interface is a region of high defect density. The presence of fixed charges at the interface during nanoscale devices fabrication, produced by charge trapping at the Si/SiO₂ interface creates reliability issues on the device performance characteristics. Here, we investigate the effect of donor and acceptor type interface fixed charges at the Si/SiO₂ region [31, 32]. These charges play an important role in affecting the physical and electrical properties of a DG-TFET device. A positive localised charge or a neutral localised charge can exist in a donor interface charges, whereas a negative localised charge or a neutral localised charge can exist in an acceptor interface charge.

In the case of DGTFTs, a larger electric field is needed at the tunnel junction for decreasing the potential barrier width and interface fixed charge are mostly caused by the strong electric field which is present at the tunneling junction. However, the formation of Interface Charges is attributed to raised stress caused by hot carriers and the influence of temperature instability. Furthermore, the accessible Interface Charges at the surface of Si body affect the electric field at the tunnel junction, resulting in a change in the tunneling current and device performance because the BTBT rate is proportional to the electric field. This BTBT rate is extremely susceptible to electric field. Induction of Interface Fixed Charges is caused by conditions such as hot carrier stress and positive bias temperature instability, in addition to a larger electric field at the tunnel junction [33, 34]. In this thesis, we used positive and negative charge density ($\pm 1.0 \times 10^{10}$, $\pm 1.0 \times 10^{11}$, $\pm 1.0 \times 10^{12} \text{ cm}^{-2}$) at the semiconductor/insulator interface. Silvaco TCAD tool has been used to analyzed the impact of interface fixed charge for various relevant DC parameters like I_{ON} , I_{OFF} , V_T , SS , I_{ON}/I_{OFF} etc.

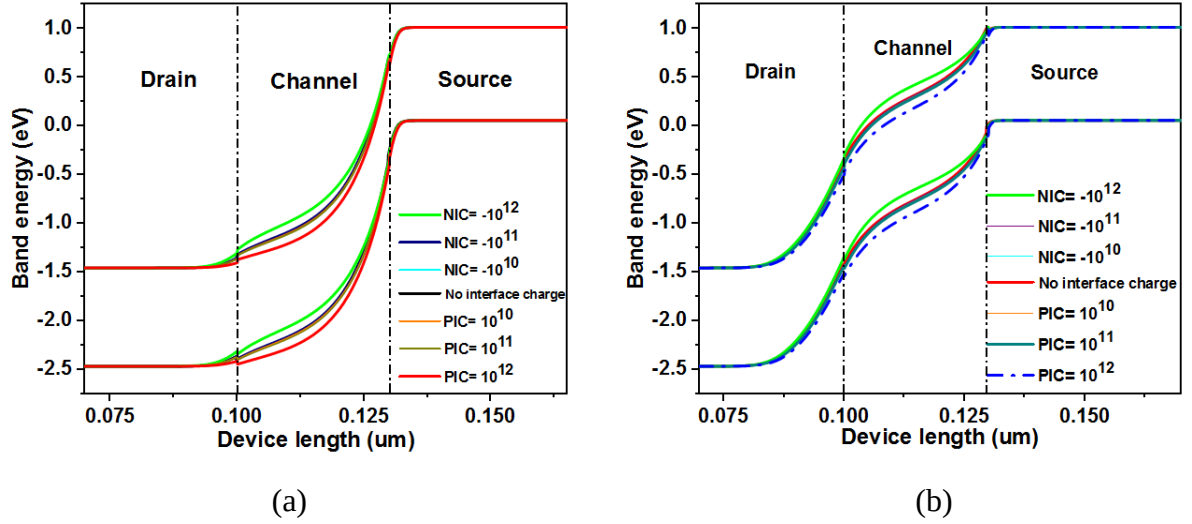


Figure 4.6: Energy band profiles in the presence of positive and negative interface charge for DGTFT structure as shown in Fig. (a) ON-state. ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$, $V_{GS}= 0V$), cutline 1nm below gate oxide-channel interface.

Fig. 4.6 shows the energy band profiles in absence and presence of positive and negative interface charges of DG-TFET in (a) ON-state and (b) OFF-state respectively. In OFF-state, it can be realized that no potential is applied to gate electrode, the electron from the p+-region cannot pass through potential barrier between V.B. of p+-region and C.B. of intrinsic region due to thick tunneling width and hence BTBT of carriers are not feasible. In ON-state, the voltage is applied to gate, C.B. of intrinsic region get proper aligned with the V.B. of p+-region which enables the BTBT carriers at the tunnel junction. In this case, tunneling potential barrier width gets narrow which leads to a high possibility of BTBT mechanism. Tunneling width should be possibly thin for better tunneling. In the above figure 4.6 (a) and (b), we observe the effect of positive, negative and no interface charges at Si-SiO₂ interface. Positive interface charges decrease the tunneling width whereas negative interface charges increase the tunneling width. We have also analyzed the no interface charges. The band bending is higher for positive interface charges and lower for negative interface charges because of high electric field at the tunnel junction (around source side) for positive than negative interface charges [33].

Transfer Characteristics

The figure 4.7 shows the simulation of I_{DS} - V_{GS} transfer characteristic of DG-TFET for variation in different interface charges. From the above figure. 4.7, we have analyzed the impact of positive, negative and zero interface fixed charges on DGTFET. When potential is applied to gate, the accumulation of electron occurs at the silicon surface. Positive interface charges cause a greater accumulation of electron which occurs at the Si surface compared to zero traps, while, in the case of negative interface charges, the opposite phenomenon occurs means accumulation of hole increases at Si-SiO₂ surface, which is combined with the electron concentration in the channel region reduces the electron concentration at the surface of silicon. It is observed that for increasing positive interface charges, drain current increases. But for negative interface charges, drain current decreases. However, no variation takes place in tunneling current and threshold voltage for zero interface charge. From Table 4.3 shows the device performance characteristics. The maximum ON-current is found to be 4.54×10^{-6} A/ μ m at positive interface charge = $+10^{12}$ and OFF-current reduces.

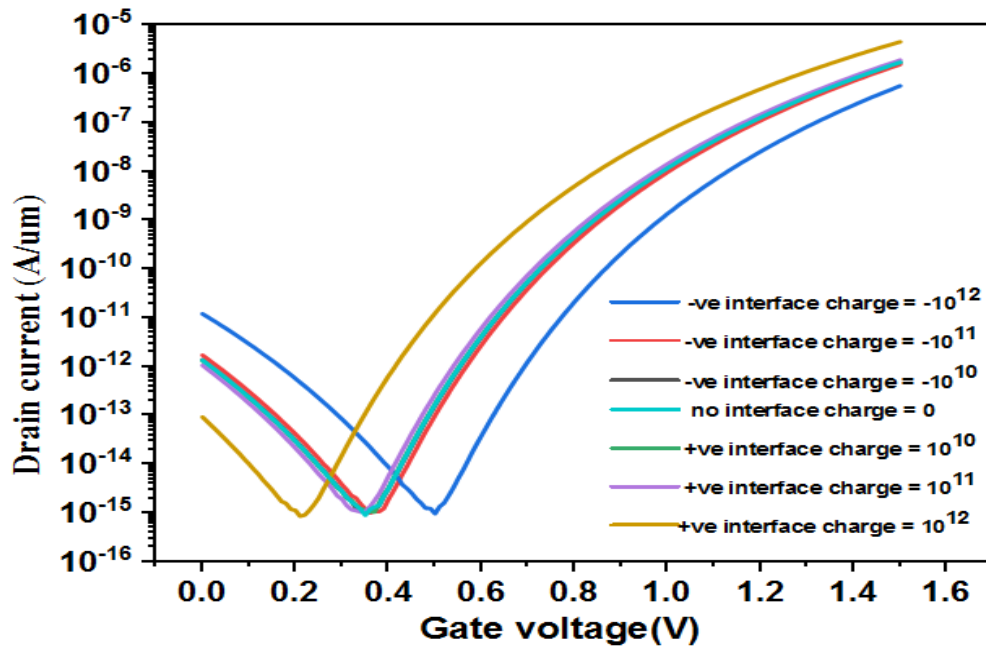


Figure 4.7: Simulation of I_{DS} - V_{GS} Transfer characteristics for different interface charge with gate voltage varying from 0V-1.5V at $V_{DS}=1.5$ V, $T_{ox}= 3$ nm in log scale.

Table 4.3: Device Performance Parameters

Parameter (/cm ²)	SS (mV/dec)	$V_T(V)$	$I_{ON}(A/um)$	$I_{OFF}(A/um)$	I_{ON}/I_{OFF}
Interface charge = -10^{12}	53.01	1.31	5.67×10^{-7}	1.22×10^{-11}	4.64×10^4
Interface charge = -10^{11}	53.37	1.19	1.56×10^{-6}	1.74×10^{-12}	8.96×10^5
Interface charge = -10^{10}	54.76	1.177	1.71×10^{-6}	1.40×10^{-12}	1.22×10^6
No Interface charge =0	50.47	1.176	1.73×10^{-6}	1.37×10^{-12}	1.26×10^6
Interface charge = $+10^{10}$	50.51	1.175	1.75×10^{-6}	1.34×10^{-12}	1.31×10^6
Interface charge = $+10^{11}$	53.71	1.16	1.92×10^{-6}	1.07×10^{-12}	1.78×10^6
Interface charge = $+10^{12}$	55.40	1.03	4.54×10^{-6}	9.38×10^{-14}	4.84×10^7

Performance analysis of electrical parameters

Fig. 4.8 shows the plots of different positive and negative interface charge effect in ON-current and OFF-current. From above fig. 4.8, we have analyzed that ON-current increases when interface charge varies from (-10^{12} to $10^{12}/\text{cm}^2$) because of more BTBT and high electric field at source-channel junction which reduces the energy barrier. But due to negative interface charge in Si-SiO₂ interface, energy barrier will slightly increase in comparison to positive interface charges. The highest ON-current (4.54×10^{-6} A/um) is generated at positive interface charge ($10^{12}/\text{cm}^2$). The OFF-current in DG-TFET decreases because of tunnelling width is too thick when interface charge vary from (-10^{12} to $10^{12}/\text{cm}^2$). In OFF-state, BTBT is negligible. Table 4.3 shows the device performance characteristics of DG-TFET.

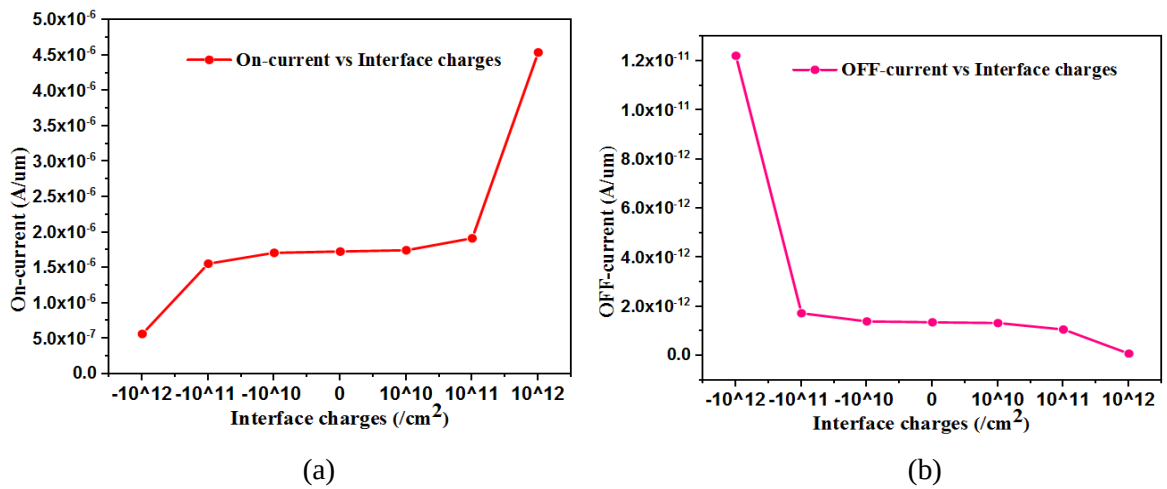


Figure 4.8: Simulation of (a) ON-current (b) OFF-current variation with positive and negative interface charges present Si-SiO₂ surface at $V_{DS}=1.5V$.

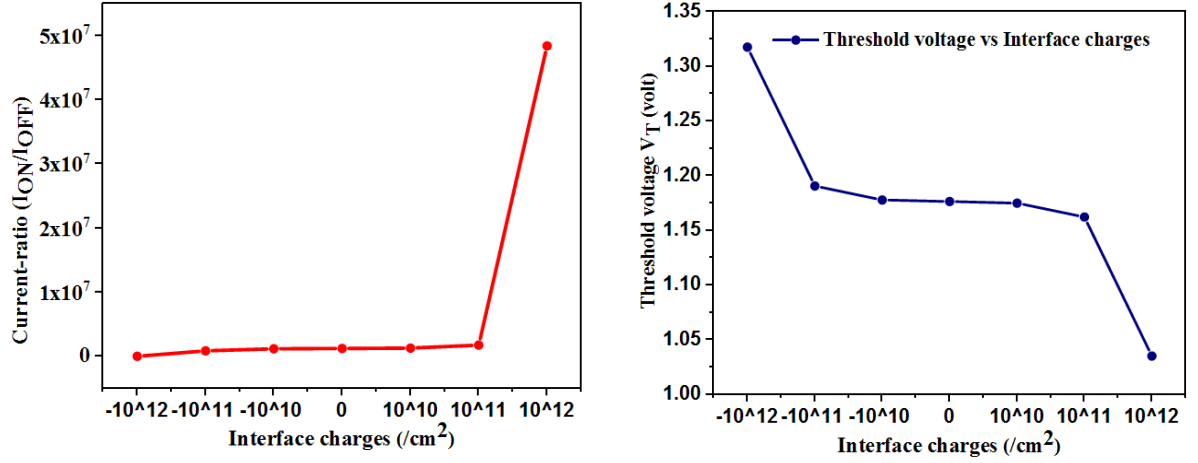


Figure 4.9: (a) I_{ON}/I_{OFF} (Current-ratio) and (b) Threshold voltage (V_T) variation with positive and negative interface charges present Si-SiO₂ surface at $V_{DS}=1.5V$.

Fig. 4.9 shows the plots of different positive and negative interface charge effect in (a) I_{ON}/I_{OFF} (b) Threshold voltage (V_T) respectively. Table 4.3 shows the electrical device performance characteristics of DG-TFET. We have analyzed from the fig 4.9 and device performance Table 4.3 that current-ratio increases in the presence of positive interface charges and is more as compared to negative interface charges. For better switching performance of devices, the I_{ON}/I_{OFF} ratio should be higher means greater than 10^6 which is good. Best performance of I_{ON}/I_{OFF} is shown when positive interface charges = $10^{12}/\text{cm}^2$. Positive and negative interface charges will affect the BTBT in DG-TFET devices.

Fig 4.9 (b) shows the variation of threshold voltage in presence of positive, negative and zero interface trap charges. We observed that the threshold voltage rises with an increase in negative interface fixed charges but it drops with an increase in positive interface charges because of flat band voltage and threshold voltage is directly proportional to flat band voltage.

$$V_{FB} = \Phi_{ms} - \frac{Q}{C_{ox}} \quad (4.1)$$

From the above equation 4.1, we analyzed that flat band voltage changes in presence of interface fixed charges. For positive interface fixed charges, it reduces the flat band voltages. Then threshold voltage also reduces and drain current increases significantly in DG-TFET device. But Flat band voltage increases in presence of negative interface fixed charges and it is also increasing the threshold voltage. We have also analyzed the zero

interface charges at Si-SiO₂ surface then there is no effect on the threshold voltage. This type of variation shown in fig. 4.9.

4.3.1.3 Impact of variation in both gate dielectric thickness

Gate oxide thickness is another important parameter for the performance of DG-TFET and the device is highly sensitive to oxide thickness. Here, we have investigated the influence of variation in both gate oxide thickness range (3nm to 5nm) in DG-TFET device. The impact of both gate thickness variation in DG-TFET especially its effect DC parameters like on-current (I_{ON}), off-current (I_{OFF}), threshold voltage (V_T), subthreshold swing (SS) and I_{ON}/I_{OFF} current ratio etc. The drain current is expected to increase with a decrease in gate oxide thickness and reducing the short channel effects because of higher gate control over the channel potential in DGTFET. Double gate leads to better control, an increase in the effective tunnelling region and enhances On-current. But it should be observed that the gate thickness cannot be reduced by a certain limit because leakage-current will start to increase considerably. OFF-current in TFET is a strong function of gate oxide thickness and is controlled by leakage through gate oxide [35, 36]. All simulations were performed by using ATLAS device simulator.

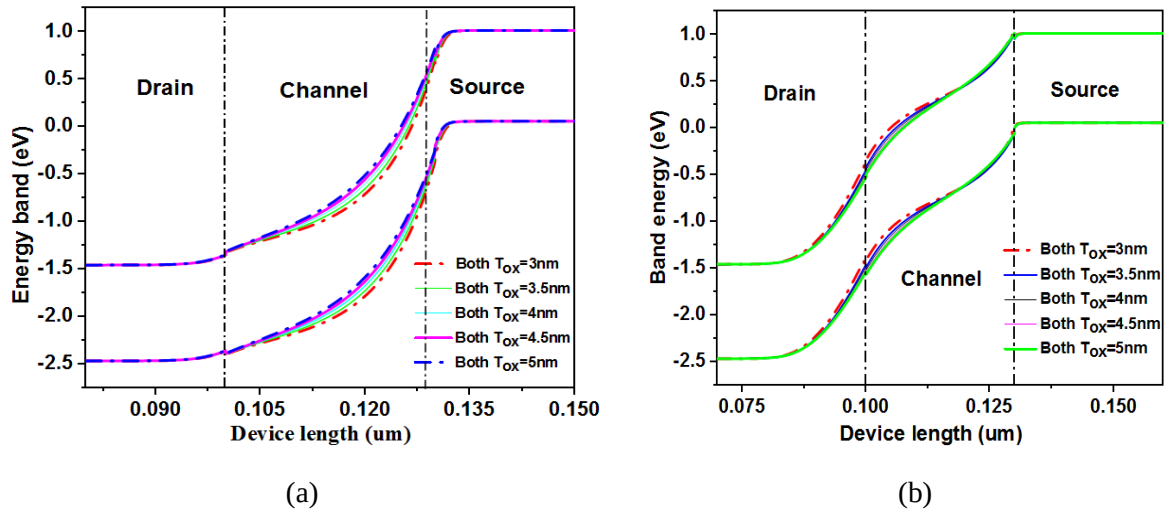


Figure 4.10: Energy band profiles under different both gate oxide thickness variation for DGTFET structure as shown in Fig. (a) ON-state. ($V_{DS}=1.5\text{V}$, $V_{GS}=1.5\text{V}$) (b) OFF-state. ($V_{DS}=1.5\text{V}$, $V_{GS}=0\text{V}$), cutline 1nm below gate oxide-channel interface.

Fig. 4.10 shows the energy band profiles under different both gate oxide thickness variation for DG-TFET in (a) ON-state and (b) OFF-state respectively. Dielectric thickness is important parameter which affects the transmission probability of electron

from the V.B. of p+-region to C.B. of channel. Tunneling current is directly proportional to the transmission probability. The transmission probability can be modelled by WKB approximation technique.

$$T_t \approx \exp\left(-\frac{4\lambda\sqrt{2m^*}}{3q(\Delta\Phi+E_g)\hbar} \frac{E_g^{3/2}}{1}\right) \quad (4.2)$$

$$\lambda = \sqrt{\frac{\epsilon_{Si}t_{Si}t_{ox}}{2\epsilon_{ox}}} \quad (4.3)$$

In the above equation (4.2) and (4.3), m^* is the effective mass, E_g is the material bandgap, λ is the screening tunneling length and it mostly depends on the device geometry, $\Delta\Phi$ is the energy difference between the C.B. in source and the V.B. in the channel. ϵ_{ox} and t_{ox} represent the gate insulator of dielectric permittivity and thickness respectively. In ON-state, we have analyzed that potential barrier width between V.B. of p+-region and C.B. of intrinsic region of device becomes so thin. So, there is a high possibility of BTBT mechanism at the tunnel junction. As a result, a large amount of current flows in the device. In OFF-state, the probability of the tunneling of electrons is very less, since the tunneling barrier width becomes too thick and only leakage current flows in the device.

If we increase the thickness of gate oxide then screening length (λ) will also increase. Due to increasing of screening length, transmission probability starts to decrease. Both On-current and OFF-current will decrease. All these types of effect are observed in fig. 4.10 (a) and (b).

Transfer Characteristics

The figure 4.11 shows the simulation of $I_{DS}-V_{GS}$ transfer characteristic of DG-TFET for variation in both gate thickness. From the above figure. 4.11, we have analyzed the impact of various both gate thickness on DG-TFET. The ON-current of the device is observed to be strongly influenced by gate oxide thickness, since variation in the oxide thickness results changes in the vertical electric field across the tunnel junction. Different oxide thickness parameters considered in the simulation are: 3nm, 3.5nm, 4nm, 4.5nm, 5nm taken. Fig. 4.11 and fig. 4.10 shows the transfer characteristics and the energy band profile impact that is observed at the tunnel junction under the oxide thickness variation. Fig. 4.11 shows that when oxide thickness is 3nm, the band bending is greater when the

oxide thickness is any other parameter investigated in the simulation. This band bending reduces potential barrier width and screening length at the tunnel junction which increases the BTBT rate of the electrons, resulting in an increase in the drain current of the device. Gate controllability decreases over the tunnel junction, due to increase of gate oxide thickness. As a result, at smaller T_{ox} stronger electric field creates at the tunnel junction. Highest value of I_{ON} is achieved for 3nm oxide thickness. Table 4.4 shows device performance parameters under the different gate oxide thickness configurations.

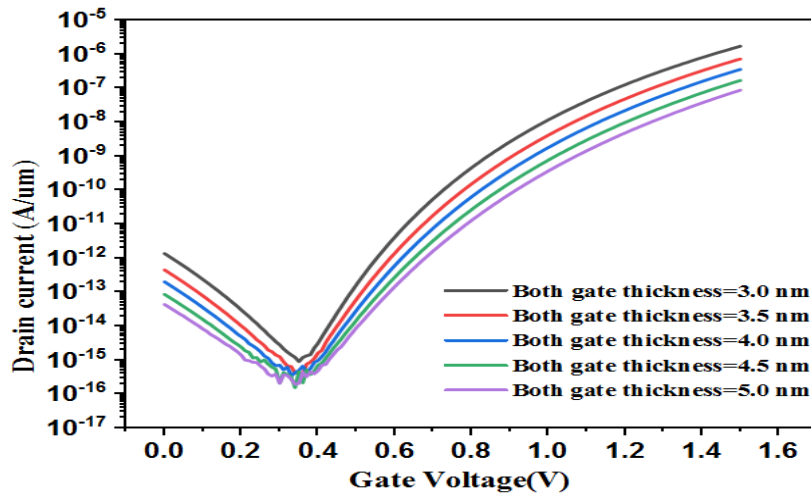


Figure 4.11: Simulation of I_{DS} - V_{GS} Transfer characteristics for different both gate oxide thickness variation with gate voltage varying from 0V-1.5V at $V_{DS}=1.5V$, plot in log scale.

Table 4.4: Device Performance Parameters

Parameter variation in both gate thickness (nm)	$I_{ON}(A/um)$	$I_{OFF}(A/um)$	I_{ON}/I_{OFF}	$V_T(V)$
3.0	1.73×10^{-6}	1.37×10^{-12}	1.26×10^6	1.17
3.5	7.31×10^{-7}	4.52×10^{-13}	1.62×10^6	1.26
4.0	3.62×10^{-7}	2.02×10^{-13}	1.79×10^6	1.34
4.5	1.71×10^{-7}	8.64×10^{-14}	1.98×10^6	1.43
5.0	8.81×10^{-8}	4.37×10^{-14}	2.01×10^6	-

Performance analysis of electrical parameters

Fig. 4.12 shows the graphs of variation in both gate oxide of thickness effect in ON-current, OFF-current and I_{ON}/I_{OFF} current-ratio. From above fig. 4.12, we have analyzed that ON-current decreases when both gate oxide thickness changes from 3nm to 5nm but OFF-current degrades because of increase in oxide thickness while current-ratio increases slightly fast and then slightly start to falls. Hence, I_{ON}/I_{OFF} ratio shows lowest at 3nm oxide thickness and maximum at 5nm. The gate loses control of the channel as the oxide thickness increases, lowering the ON current. Highest value of I_{ON} is achieved for 3nm oxide thickness. So, we can say that in comparison to I_{ON} , I_{OFF} and I_{ON}/I_{OFF} ratio depend much more strongly on the gate dielectric thickness. Without considering gate leakage, very different subthreshold properties are obtained. Hence it is very important to consider gate dielectric leakage, especially for thin oxide thickness [36, 37]. Table 4.4 shows the remaining device performance parameters under the different gate oxide thickness.

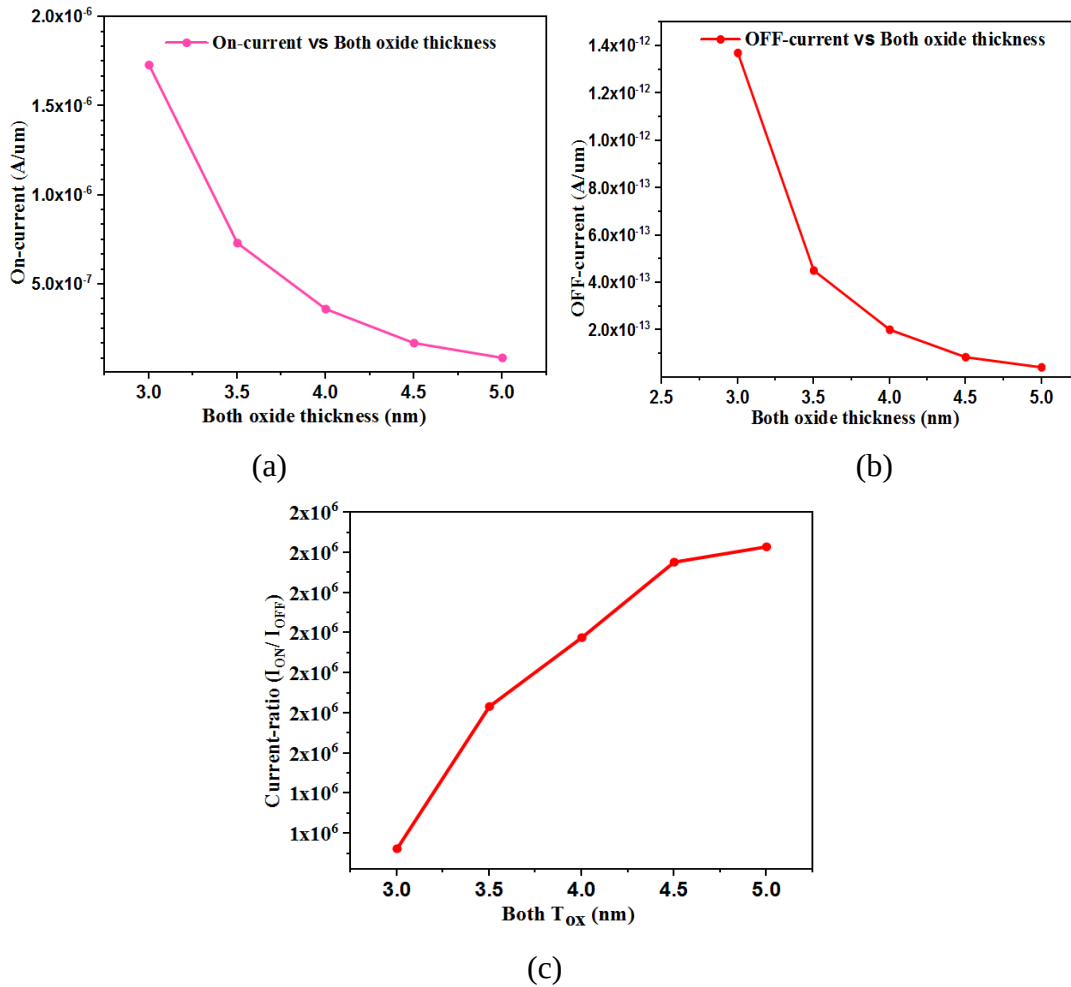


Figure 4.12: Simulation of (a) ON-current ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-current ($V_{DS}=1.5V$, $V_{GS}=0V$) (C) I_{ON}/I_{OFF} (Current-ratio) variation with different both gate oxide thickness value varying from (3nm to 5nm) at $V_{DS}=1.5V$.

We have analyzed the threshold voltage of the devices plotted in fig. 4.13 (a). We know that threshold voltage is a function of gate oxide thickness. Here, we have analyzed that threshold voltage almost increases linearly when both gate oxide thickness changes from 3nm to 5nm. It happens due to decrease in Gate controllability over the tunnel junction, due to increasing of gate oxide thickness. As a result, threshold voltage is obtained lowest value at 3nm T_{ox} and maximum at 5nm T_{ox} . So, we can say that threshold voltage is much more strongly depend on the gate dielectric thickness. Since threshold voltage decides whether a transistor can be turned on or off, it is important to be able to adjust V_T in the designing of devices. Table 4.3 shows the electrical device performance characteristics of DG-TFET under the different gate oxide thickness.

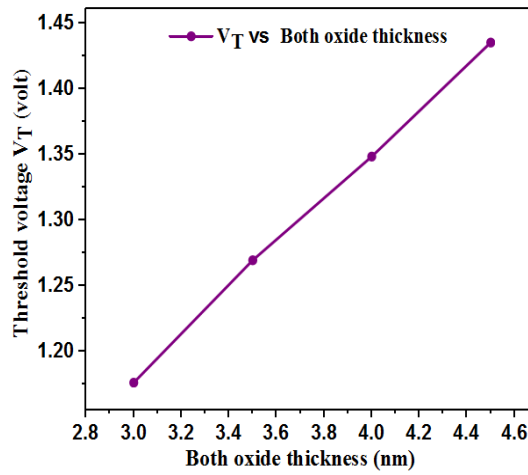


Figure 4.13: (a) Threshold voltage (V_T) graph shows the variation with both gate oxide thickness value varying from (3nm to 5nm) at $V_{DS}=1.5V$.

4.3.1.4 Impact of variation in upper gate oxide thickness

Gate oxide thickness is another important parameter for the performance of DG-TFET and the device is highly sensitive to oxide thickness. Here, we have investigated the impact of variation in upper gate dielectric thickness range (3nm to 5nm) in DG-TFET device. The impact of upper gate thickness variation in DG-TFET especially its effect DC parameters like on-current (I_{ON}), off-current (I_{OFF}), threshold voltage (V_T), subthreshold swing (SS) and I_{ON}/I_{OFF} current ratio etc. The drain current is expected to increases with a decrease in gate oxide thickness and reducing the short channel effects because of higher gate control over the channel potential in DGTFET [36]. Double gate leads to better control, an increase in the effective tunnelling region and improvement in On-current. But it should be noted that the gate oxide thickness cannot be reduced by a

certain limit because leakage-current will start to increase considerably. OFF-current in TFET is a strong function of gate oxide thickness and is controlled by leakage through gate oxide. All simulations were performed by using ATLAS device simulator.

Fig. 4.14 shows the energy band profiles under different upper gate oxide thickness variation for DGTFET in (a) ON-state and (b) OFF-state respectively. Dielectric thickness is important parameter which affects the transmission probability of electron from the V.B. of p+-region to C.B. of channel. Tunneling current is directly proportional to the transmission probability. The transmission probability can be modelled by WKB approximation technique.

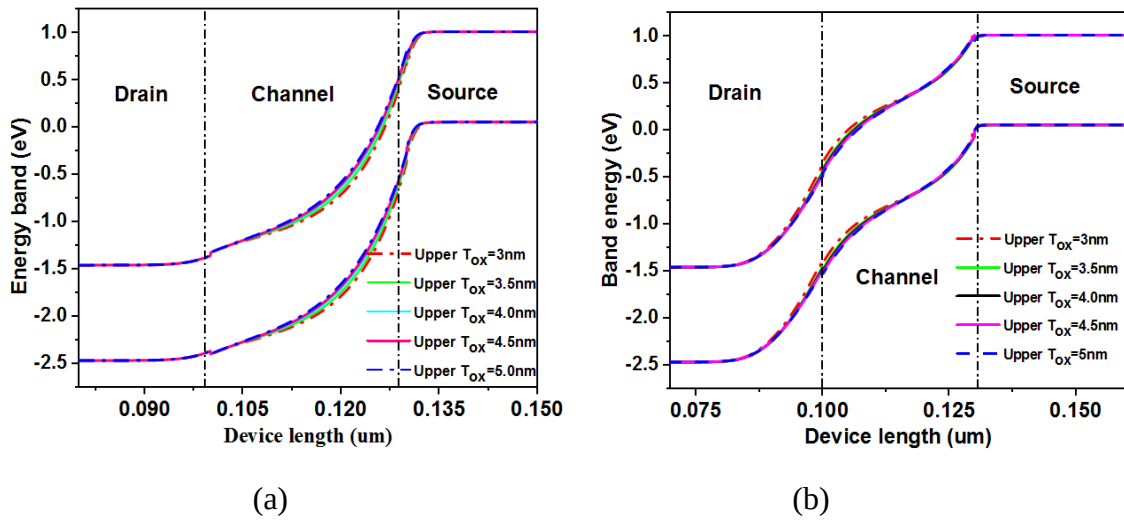


Figure 4.14:Energy band profiles under different upper gate oxide thickness variation for DGTFET structure as shown in Fig. (a) ON-state. ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$, $V_{GS}=0V$), cutline 1nm below gate oxide-channel interface.

$$T_t \approx \exp\left(-\frac{4\lambda\sqrt{2m^*}}{3q(\Delta\Phi+E_g)\hbar} \frac{E_g^{3/2}}{1}\right) \quad (4.4)$$

$$\lambda = \sqrt{\frac{\epsilon_{Si}t_{Si}t_{ox}}{2\epsilon_{ox}}} \quad (4.5)$$

$$\frac{1}{t_{ox}} = \frac{1}{T_{ox1}} + \frac{1}{T_{ox2}} \quad (4.6)$$

In the above equation (4.4), (4.5) and (4.6), m^* is the effective mass, E_g is the bandgap of material, λ is the screening tunneling length and it mostly depends on the device geometry, T_{ox1} and T_{ox2} are the upper and bottom gate oxide thickness, $\Delta\Phi$ is the energy difference between the C.B. in source and V.B. in the channel. ϵ_{ox} and t_{ox} represent the gate insulator of dielectric permittivity and thickness respectively. In ON-state, we have analyzed that potential tunneling barrier width between V.B. of p+-region and C.B. of

intrinsic region of device becomes so thin. So, there is a high probability of BTBT mechanism at the tunneling junction. Hence, a large amount of current flows in the devices. In OFF-state, the probability of the tunneling of carriers is very less, since the tunneling barrier width becomes too wide and only leakage current flows in the device.

If we increase the thickness of upper gate oxide then screening length (λ) will also increase. Due to increasing of screening length, transmission probability starts to decrease. Both On-current and OFF-current will also decrease. All these types of effect are observed in fig. 4.14 (a) and (b).

Transfer Characteristics

The figure 4.15 shows the simulation of $I_{DS}-V_{GS}$ transfer characteristic of DG-TFET for variation in upper gate oxide thickness. From the above fig. 4.15, we have analyzed the impact of various upper gate oxide thickness on DGTFET. The ON-current of the device is observed to be strongly affected by upper gate oxide thickness, since variations in the oxide thickness results as change in the vertical electric field across the tunnel junction.

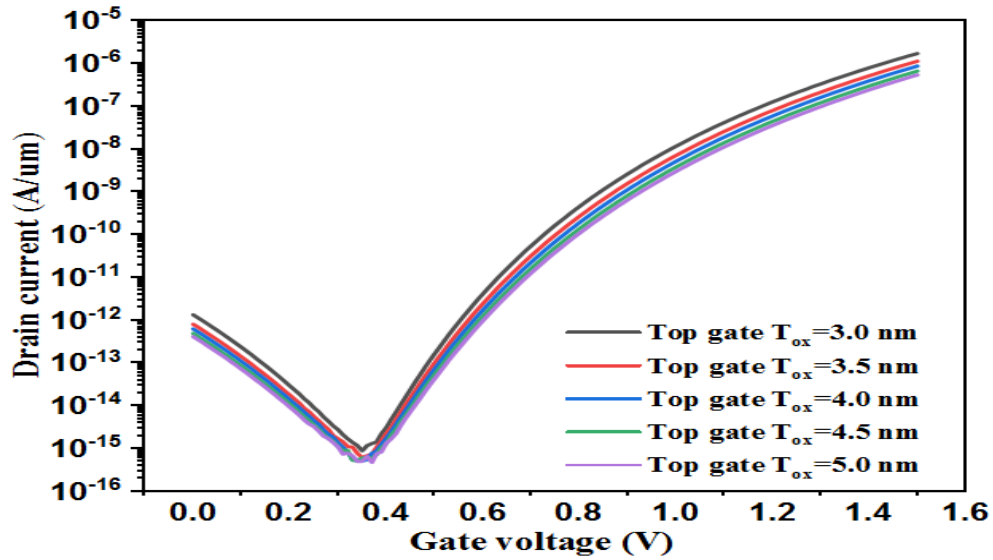


Figure 4.15: Simulation of $I_{DS}-V_{GS}$ Transfer characteristics for different upper gate oxide thickness variation with gate voltage varying from 0V-1.5V at $V_{DS}=1.5$ V, plot in log scale.

Here, Different upper oxide thickness parameters considered in the simulation are: 3nm, 3.5nm, 4nm, 4.5nm, 5nm taken. Fig. 4.15 and fig. 4.14 show the transfer characteristics and impact on energy band profile that is observed at the tunnel junction under the various upper (top) oxide thickness. It is clearly observed from fig. 4.15 that when upper gate oxide thickness is 3nm, the band bending is more than the other oxide thickness

parameters investigated in the simulation. This band bending reduces the potential barrier width and screening length at the tunnel junction which increases the BTBT rate of the electrons, results in increase of the drain current of the device. Gate controllability decreases over the tunnel junction due to increase of upper gate oxide thickness value. As a result, stronger electric field creates at the tunnel junction at smaller T_{ox1} . Highest value of I_{ON} is achieved for 3nm upper gate oxide thickness. Table 4.5 shows device performance parameters under the different upper gate oxide thickness configurations.

Table 4.5: Device Performance Parameters

Parameter variation in upper gate thickness (nm)	SS (mV/dec)	$V_T(V)$	$I_{ON}(A/um)$	$I_{OFF}(A/um)$	I_{ON}/I_{OFF}
3.0	50.47	1.17	1.73×10^{-6}	1.37×10^{-12}	1.26×10^6
3.5	53.00	1.22	1.14×10^{-6}	8.25×10^{-13}	1.38×10^6
4.0	59.55	1.24	8.78×10^{-7}	6.42×10^{-13}	1.36×10^6
4.5	50.85	1.27	6.70×10^{-7}	5.01×10^{-13}	1.33×10^6
5.0	47.50	1.30	5.47×10^{-7}	4.22×10^{-13}	1.29×10^6

Performance analysis of electrical parameters

Fig. 4.16 shows the graphs of variation in upper gate oxide of thickness effect in ON-current, OFF-current and I_{ON}/I_{OFF} current-ratio. From above fig. 4.16, we have analyzed that ON-current decreases when upper gate oxide thickness changes from 3nm to 5nm. But OFF-current degrades because of increase in oxide thickness while current-ratio increases slightly fast and then slightly starts to fall. Hence, I_{ON}/I_{OFF} ratio shows lowest at 3nm oxide thickness and maximum at 5nm. The gate loses control over the channel as the oxide thickness increases, lowering the ON current. Highest value of I_{ON} is achieved for 3nm oxide thickness. The reason behind this fact is the high electric field in the channel region for thin gate oxide that further tends to decrease the tunneling barrier width. So, we can say that in comparison to I_{ON} , I_{OFF} and I_{ON}/I_{OFF} ratio depend much more strongly on the gate dielectric thickness. Without considering gate leakage, very different subthreshold properties are obtained. Hence it is very important to

consider gate dielectric leakage, especially for thin oxide thickness. Table 4.5 shows the remaining device performance parameters under the various upper gate oxide thickness.

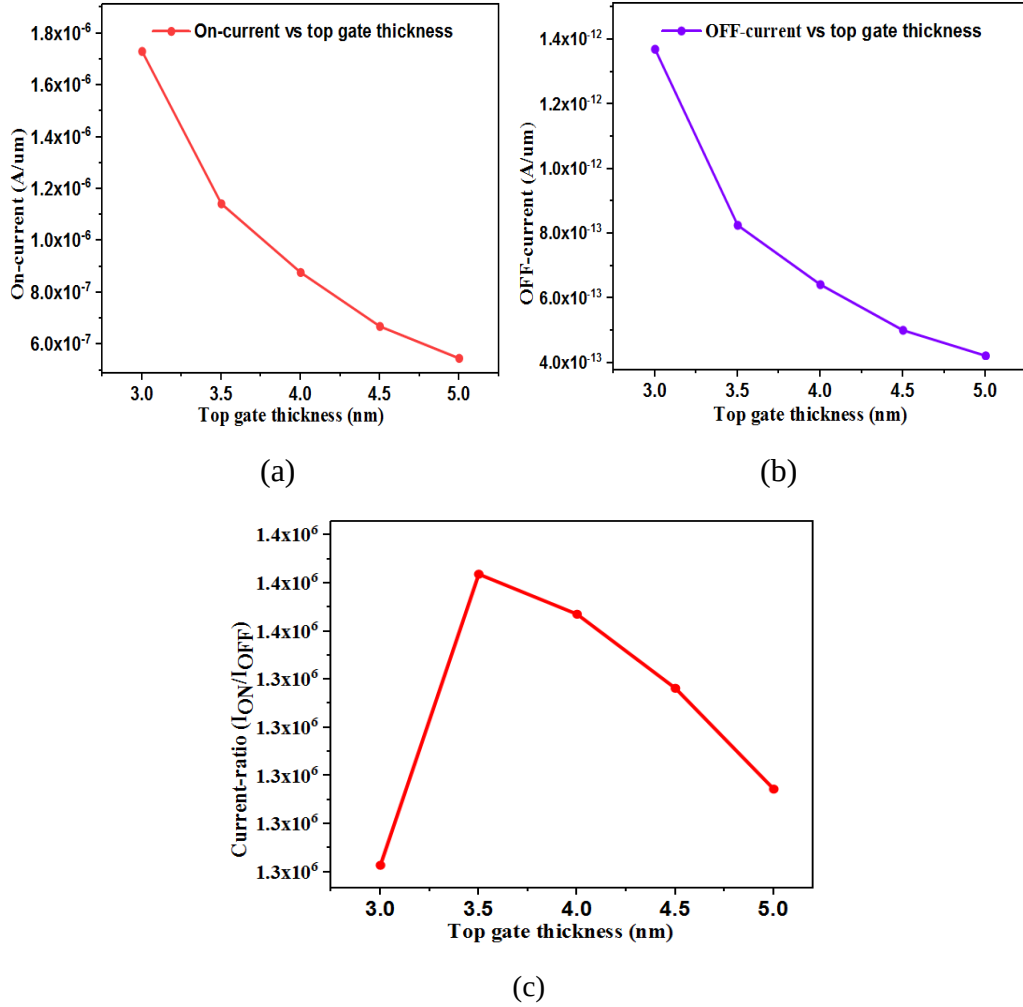


Figure 4.16: Simulation of (a) ON-current ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-current ($V_{DS}=1.5V$, $V_{GS}= 0V$) (C) I_{ON}/I_{OFF} (Current-ratio) ratio variation with different upper gate oxide thickness value varying from (3nm to 5nm) at $V_{DS}=1.5V$.

We analyze the threshold voltage and subthreshold swing of the device plotted in fig. 4.17. We know that threshold voltage is a function of gate oxide thickness. Here, we have analyzed that threshold voltage almost increases linearly when upper gate oxide thickness changes from 3nm to 5nm. It happens due to decrease of Gate controllability over the tunnel junction, due to increase of gate oxide thickness. As a result, threshold voltage shows lowest value at 3nm T_{ox1} and maximum value at 5nm T_{ox1} . So, we can say that threshold voltage is much more strongly depend on the gate dielectric thickness. Since, threshold voltage decides whether a transistor can be turned on or off, it is important to be able to adjust V_T in the designing of devices. It is also observed from fig.4.17 that gate oxide with the thickness of 3nm provides the small swing (50.47

mV/decade) whereas, the increased thickness of gate oxide (4nm) provides the large swing (59.55 mV/decade) and it deteriorates after 4nm of upper gate oxide thickness. Moreover, large electric field helps to improve SS and V_T at oxide thickness of 3nm. Table 4.3 shows the electrical device performance characteristics of DG-TFET under the different gate oxide thickness. Therefore, it is concluded that lower threshold voltage and SS is obtained with 3nm gate oxide thickness.

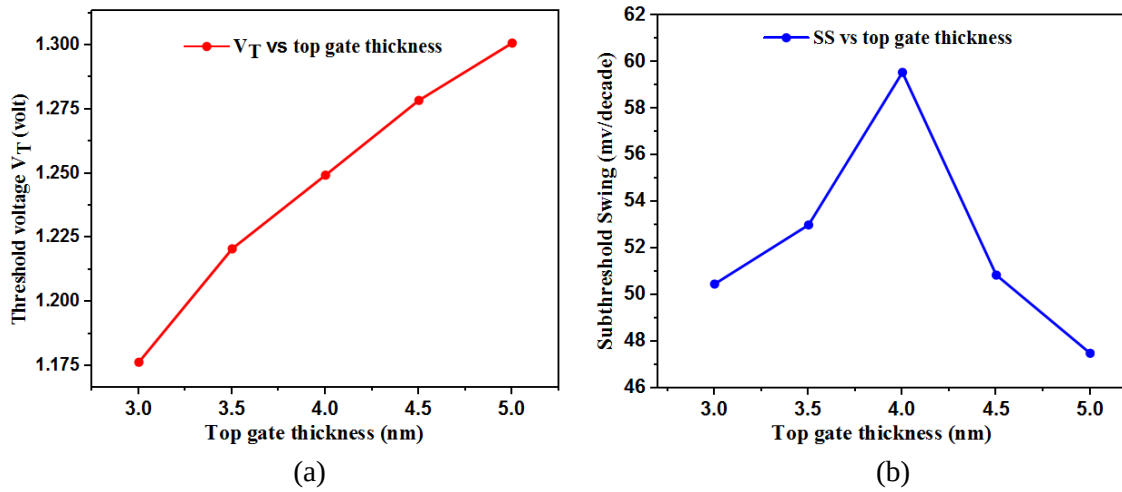


Figure 4.17: (a) Threshold voltage (V_T) and (b) Subthreshold swing (SS) graph show the variation with upper gate oxide thickness value varying from (3nm to 5nm) at $V_{DS}=1.5V$.

4.3.1.5 Effect of gate metal work function variation

Work function of a metal (Φ_m) is defined as the minimum amount of energy needed to pull an electron from the surface of a metal. Different metal has different value of work function because of difference in their electronegativity. Gate work function is an important parameter for modulating device characteristics. Here, we have investigated the effect of variation in gate metal work function range (3.9 eV to 5.2 eV) in DG-TFET device. It affects the especially electrical parameter. The gate's work function controls the height of the potential barrier which controls the energy of electrons and hence make the BTBT suitable [38, 39]. Drain current is directly proportional to transmission probability of carriers. Transmission probability exponentially depends on the difference between height of barrier and energy of carriers. If we increase the work function of both the gates then energy barrier's potential increases. Due to increasing of barrier, the probability of tunneling decreases exponentially. The ON-state current decreases with increasing the channel height and off-current increases to an extent value of work function, so switching ratio drops [40]. All the simulations and results have been carried out and acquired with the help of ATLAS device simulator.

Fig. 4.18 shows the energy band profiles of both gate for different work function of DGTFET in (a) ON-state and (b) OFF-state respectively. In ON-state, when sufficient positive voltage is applied to gate terminal, the potential barrier between the V.B. of p+-region and C.B. of intrinsic region gets narrow and allowing a large tunneling current flow because it reduces the screening length and increases the tunneling probability. The high electric field generates at tunneling junction and allow the BTBT. Here, the gate work function increased from 3.9eV to 5.2eV then the barrier width of tunneling junction and screening length starts to increase because of decrease in the Gate Control Ability and higher tunneling barrier. The energy band becomes less steep curved. BTBT decreases then the drain current will also decrease. It is evident from the fig. 4.18 that for a tunneling gate with lower metal work function, to get the steep band bending at the tunnel junction. BTBT occurs near the source and intrinsic region and lowers the tunneling barrier width. In OFF-state, the electron from the p+-region cannot pass through the potential barrier width between the V.B. of p+-region and C.B. of intrinsic region due to thick tunneling width and hence electron tunneling is almost negligible and a modest leakage current (I_{OFF}) is obtained in the devices. BTBT happens near the drain and intrinsic region. But it is clearly observed from fig. 4.18 (b), gate work function has the relatively significant influence on OFF-state current. As the gate work function increases, the barrier width of tunneling reduces slightly. Due to reduce in tunneling barrier, it increases the OFF-state current [41].

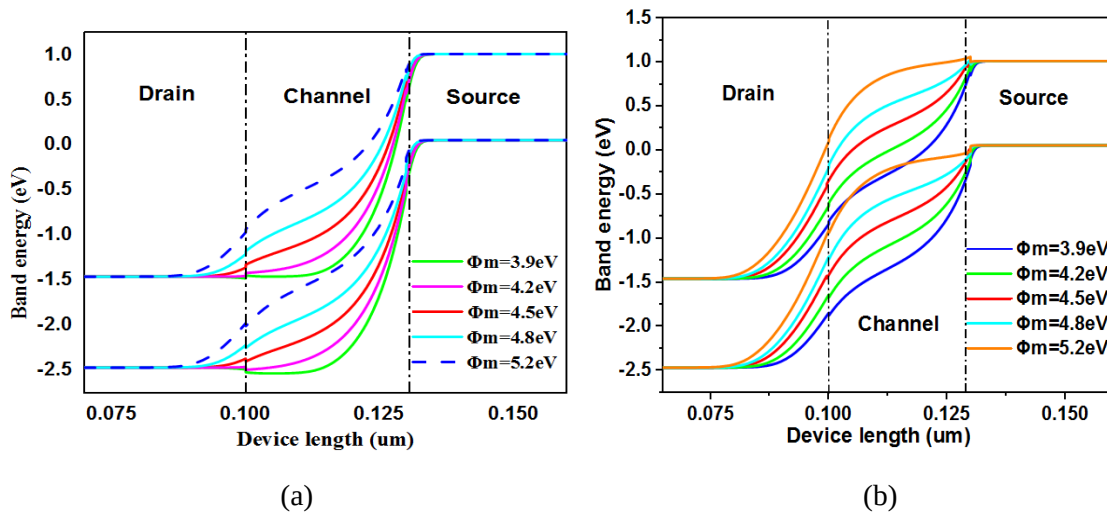


Figure 4.18: Energy band profiles under different both gate work function variation for DGTFET structure is shown in above Fig. (a) ON-state. ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$, $V_{GS}=0V$), cutline 1nm below gate oxide-channel interface.

Transfer Characteristics

The figure 4.19 shows the simulation of I_{DS} - V_{GS} transfer characteristic of DG-TFET for variation in both gate metal work function. From the above figure. 4.19, we have analyzed the impact of variation in both gate work function of metal on DG-TFET device. It is clearly observed that change in gate work function value influences the ON-current of device significantly as shown in device performance in Table 4.6. Here, tunneling gate work function (Φ_m) is changed from 3.9eV to 5.2eV. The ON-state current decreases when work function of gate increases due to the rise in flat band voltage. It reduces the band bending at tunneling junction. Meanwhile, the transfer characteristic curve shifts to the right and threshold voltage increases. The reason for this is that energy barrier at source and channel tunnel junction bends less steep. Furthermore, the OFF-state current increases rapidly when the gate work function is more than 4.2eV. It is clear from the fig. 4.18 and fig. 4.19 that the work function of metal equal to 3.9eV, the band bending is more as compared to other value of work function of gate and achieve a high I_{ON} in the simulation. This is happened due to generating more electric field on tunnel junction as comparison to other work function of gate. Consequently, the BTBT phenomenon is found to be more and the ON-current is also found to be high at 3.9eV. The optimal device design parameters can be obtained by analyzing the effects of variation in work function of metal on the device performance, which can help to reduce the process complexity.

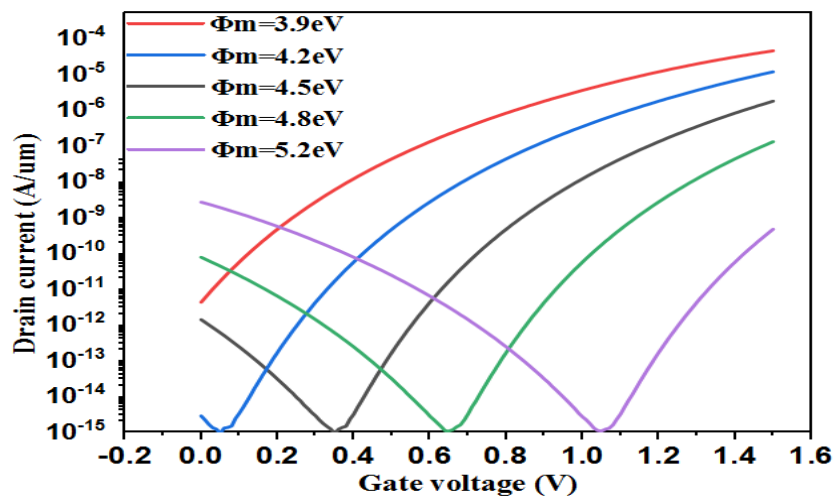


Figure 4.19: Simulation of I_{DS} - V_{GS} Transfer characteristics for different both gate work function with gate voltage varying from 0V-1.5V at $V_{DS}=1.5\text{V}$, $T_{ox}= 3\text{nm}$ in log scale.

Table 4.6: Device Performance Parameters

Parameter(eV)	SS (mV/dec)	$V_T(V)$	$I_{ON}(A/um)$	$I_{OFF}(A/um)$	I_{ON}/I_{OFF}
Gate work function =3.9	80.98	0.57	4.39×10^{-5}	4.20×10^{-12}	1.05×10^7
Gate work function =4.2	45.87	0.87	1.14×10^{-5}	2.81×10^{-15}	4.05×10^9
Gate work function =4.5	50.47	1.17	1.73×10^{-6}	1.37×10^{-12}	1.26×10^6
Gate work function =4.8	52.30	1.47	1.28×10^{-7}	7.56×10^{-11}	1.69×10^3
Gate work function = 5.2	53.54	-	2.62×10^{-9}	2.62×10^{-9}	1

Performance analysis of electrical parameters

Fig. 4.20 display the variation in both gate metal work function on the electrical parameters like (a) ON-current, (b) OFF-current and (c) I_{ON}/I_{OFF} current-ratio respectively for DG-TFET devices. Table 4.5 shows the electrical device performance parameters under the variation in gate work function. It is evident from fig. 4.20 and device performance table 4.5 that I_{ON} is found maximum for gate work function = 3.9eV and is more as compared to gate work function = 5.2eV, but simultaneously there is significant increase in I_{OFF} at the same time. The value achieved by the I_{ON}/I_{OFF} ratio, which is 10^6 orders less than as compared to gate work function = 4.5eV and I_{ON}/I_{OFF} ratio peaks at 4.2eV and its value has found to be 4.05×10^9 . After 4.2eV, I_{ON}/I_{OFF} ratio dropped as gate metal work function increased which affecting the BTBT at the tunnel junction, resulting in a fall in the ratio. Therefore, the value of gate work function must be optimized to get the largest I_{ON}/I_{OFF} ratio at lowest I_{OFF} . In OFF-state, I_{OFF} initially decreases with increasing gate work function value up to 4.2eV due to reduction in gate current but it rises for higher gate work function values due to higher BTBT tunneling between the drain and intrinsic region. After comparing the values, we have found that the gate work function of 4.2eV is optimum value for DG-TFET device.

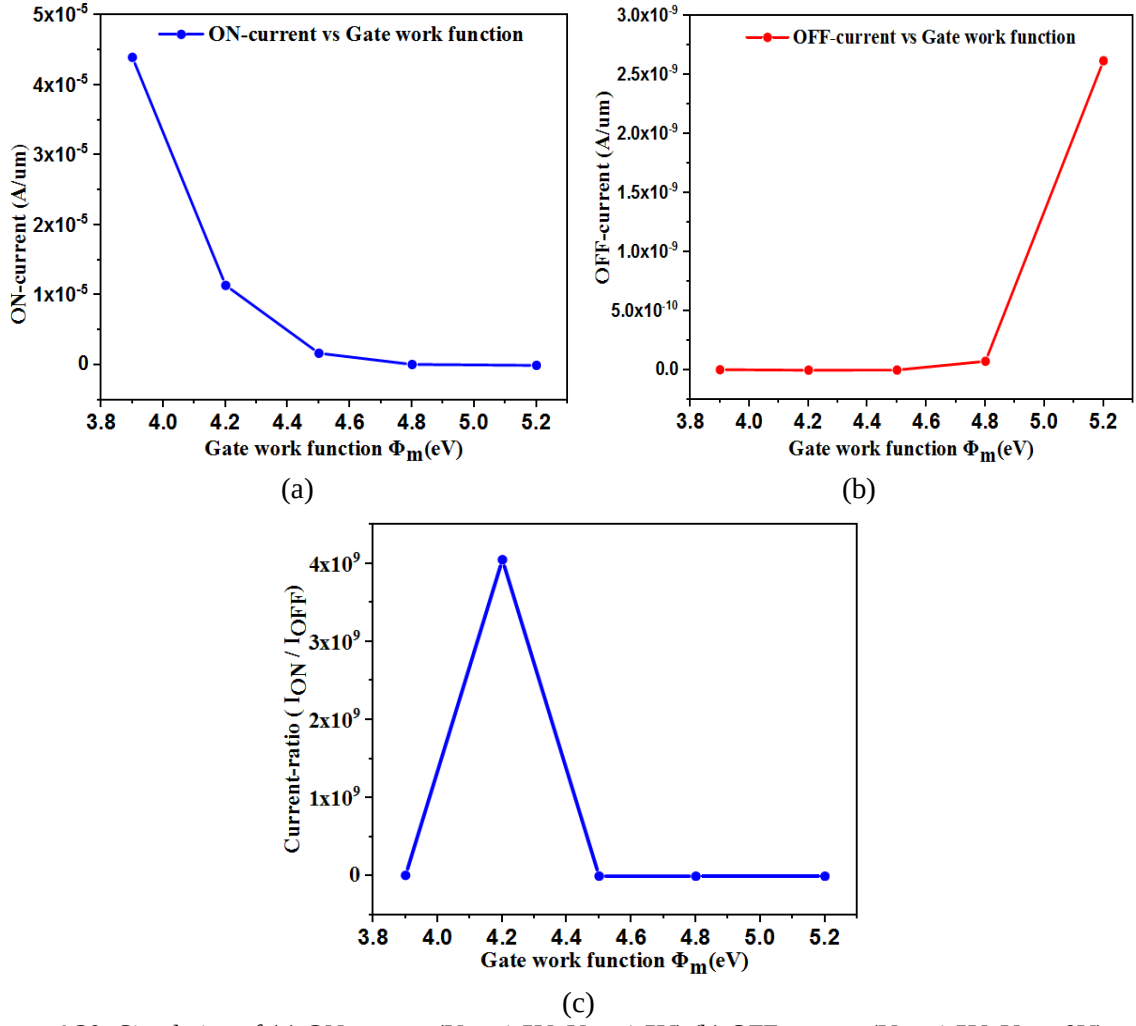


Figure 4.20: Simulation of (a) ON-current ($V_{\text{DS}}=1.5\text{V}$, $V_{\text{GS}}=1.5\text{V}$) (b) OFF-current ($V_{\text{DS}}=1.5\text{V}$, $V_{\text{GS}}=0\text{V}$) (C) $I_{\text{ON}}/I_{\text{OFF}}$ (Current-ratio) variation in both gate work function of metal value varying from (3.9eV to 5.2eV) at $V_{\text{DS}}=1.5\text{V}$.

Here, we analyze the threshold voltage and subthreshold swing of the device plotted in fig. 4.21. Another responsible electrical performance parameters such as SS and V_t are critical in determining the switching power and exact behaviour of any device or circuits. Subthreshold swing and threshold should be possible as low for fast switching and low power applications. The work function of the gate controls the threshold voltage. It is observed from fig. 4.21 that the threshold voltage shows a linear relationship with increasing in the gate metal work function of the varying range 3.9eV to 5.2eV, as in DGTFET. The transfer characteristic shows a similar change with gate work function means that V_t continues to increase by increasing the gate work function. It happens due to increasing the flat band voltage. As a result, threshold voltage shows lowest value at gate work function=3.9eV and maximum value at 5.2eV. So, we can say that threshold voltage is much more strongly depend on the gate metal work function. It is also

observed that subthreshold swing decreases linearly with increasing the gate work function up to 4.2eV and it deteriorates after 4.2eV. Gate work function of value 3.9eV provides the large swing (80.98 mV/decade) whereas, gate work function of value 4.2eV provides the small swing (45.87 mV/decade). Thus, from the Table 4.5 results, it is concluded that gate work function = 4.2 eV is the optimum value of work function of gate metal for which the analyzed device acquired optimum performance parameters.

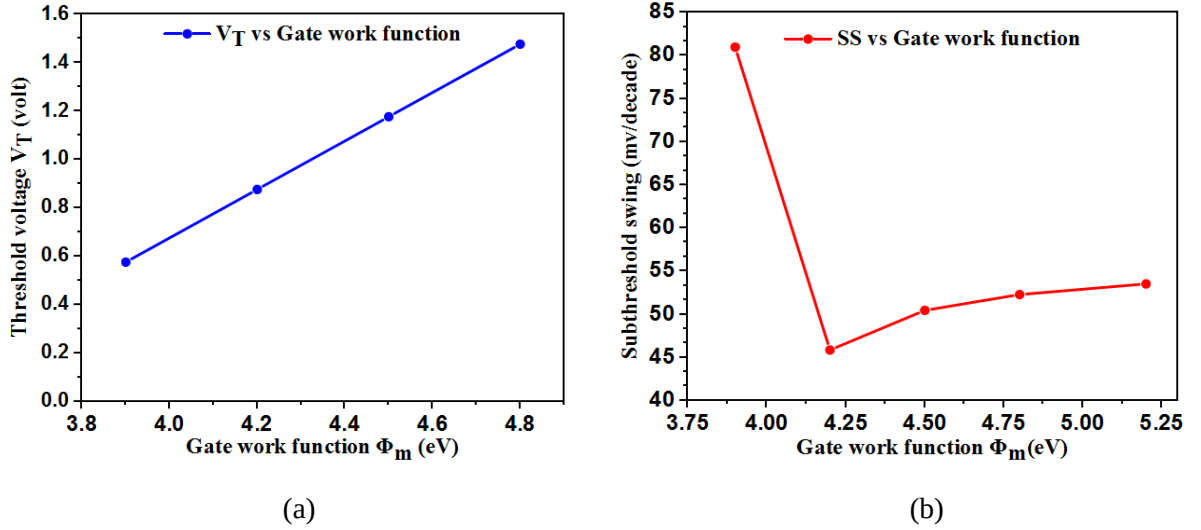


Figure 4.21: (a) Threshold voltage (V_T) (b) Subthreshold swing (SS) graphs show the variation with both gate work function of metal value at $V_{DS}=1.5V$.

4.3.2 Temperature effect on DG-TFET

DG-TFET is an alternative device for ultra-steep CMOS technology, having very excellent device performance including a steeper sub-threshold slope ($< 60\text{mV/decade}$ at $T=300\text{K}$ means room temperature) and very small static leakage current produces. Here, we investigate the impact of temperature on DG-TFET. The results of TCAD simulation shows that leakage current is nearly independent of temperature's increment in the OFF-state. Most of semiconductors exhibit temperature dependence energy band reduction which affects the device performance as temperatures change [42, 43]. In this work, Drain, intrinsic and source region are made of silicon materials, so we analyzed the impact of temperature on the device characteristics. The Silvaco TCAD tool is used to analyze the device characteristics. The ON-current (I_{ON}) in n-type DGTFET depends on the transmission probability of electrons moving from the p+-region of source to the conduction band of channel. The transmission probability can be modelled by WKB approximation technique [44].

$$T_t \approx \exp\left(-\frac{4\lambda\sqrt{2m^*}}{3q(\Delta\Phi+E_g)\hbar} \frac{E_g^{3/2}}{1}\right) \quad (4.7)$$

In the above equation (4.7), m^* is the effective mass, E_g is the bandgap of material. q is the electronic charge, λ is the screening tunneling length and it mostly depends on the device architecture, $\Delta\Phi$ is the energy difference between the C.B. in p+-region and V.B. in the intrinsic region. Normally, we used low band gap materials in conventional DG-TFET architecture for improving the ON-current.

Transfer Characteristics

The figure 4.22 shows the simulated I_{DS} - V_{GS} Transfer characteristics of DG-TFET performing in the temperature range of 298K to 448K. We know that the band gap of silicon is dependent on the temperature. Fig 4.22 exhibits the impact of temperature on the drain-current and it is observed that for a wide range of temperatures, a very modest leakage current ($\approx 1.36 \times 10^{-12}$ to 3.19×10^{-12} A/um) and ON-current vary from ($\approx 1.72 \times 10^{-6}$ to 3.25×10^{-6} A/um) are obtained. The results of the TCAD simulation show that OFF-state current is nearly independent of temperature increment. So, DGTfET can be easily used in large (298K to 448K) temperature range. The drain current is affected by band gap and temperature both.

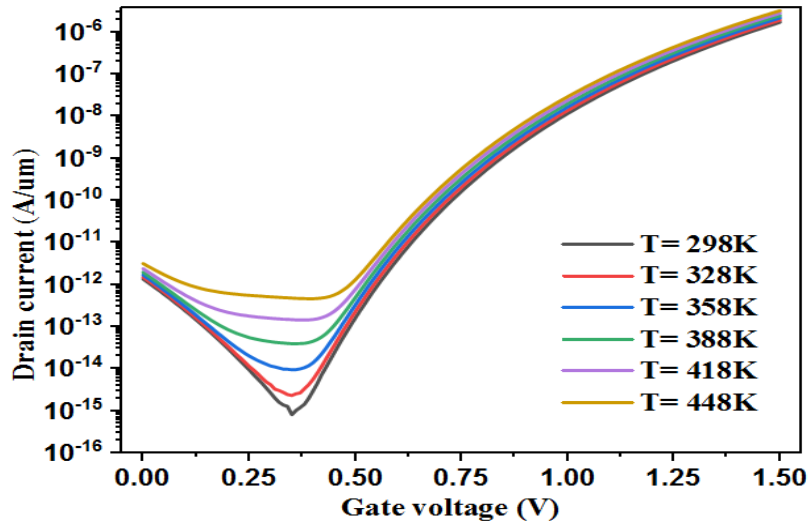


Figure 4.22: Simulation of I_{DS} - V_{GS} Transfer characteristics for temperature range 298K- 448K with gate voltage varying from 0V-1.5V at $V_{DS}=1.5V$, $T_{ox}= 3nm$ in log scale.

Temperature is one of the vital parameters which influences the semiconductor's band-gap. The device's performance is influenced by this physical parameter indirectly and

hence, we focused our research on device performance with temperature variation. Band gap of material is a function of temperature as shown below equation-

$$E_g(T) = E_g(0) - \frac{\alpha T^2}{\beta + T} \quad (4.8)$$

where at $T = 0K$, $E_g(0)$ is the bandgap of the material, α and β are fitting constant parameters. As the temperature rises, the material's bandgap reduces. The band gap of material becomes different for different temperature. The switching operation of DGTFET is usually controlled in two conditions: (a) In ON-state ($V_{GS} > 0V$), when sufficiently positive voltage is applied to the gate terminal, the potential barrier between the V.B. of p+-region and C.B. of intrinsic region gets narrow and allowing a large tunneling current flow because it reduces the screening length and increases the tunneling probability. BTBT occurs at the source-channel junction. (b) In OFF-state ($V_{GS} = 0V$), the electron from the p+-region cannot pass through the potential barrier width between the V.B. of p+-region and C.B. of intrinsic region due to thick tunneling width and hence electron tunneling is almost negligible and only a modest leakage current (I_{OFF}) is obtained from the thermionic emission [45].

One of most important parameters for analyzing the BTBT mechanism in TFET devices is screening tunneling length (λ). The probability of an electron tunneling from the V.B. of p+-region to C.B. of intrinsic region is a weak dependent on temperature. For a wide range of operating temperature (298K to 448K), relatively small fluctuation is obtained in λ as shown in fig. 4.23. and this is exponentially related to transmission probability of tunneling function by Equation (4.7).

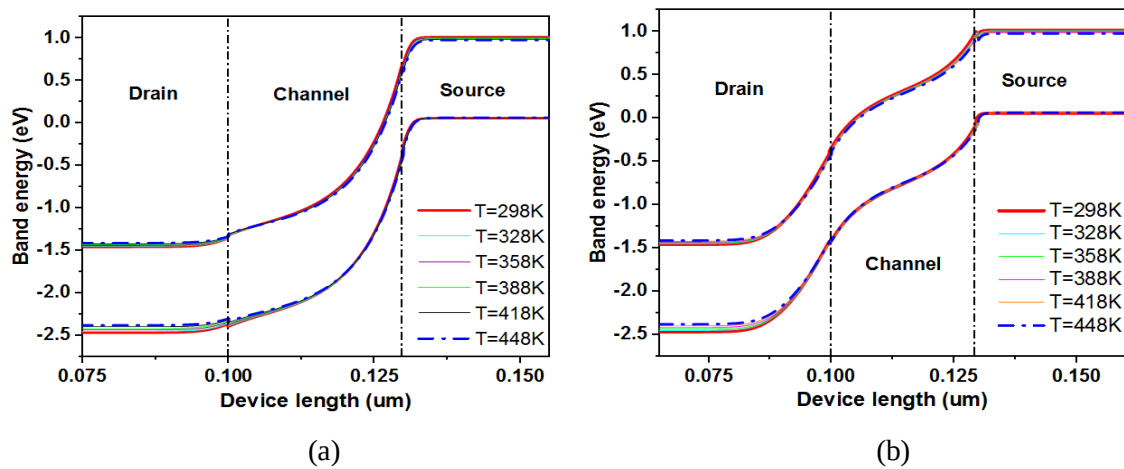


Figure 4.23: Energy band profiles at different temperature for DGTFET structure as shown in Fig. (a) ON-state. ($V_{DS}=1.5V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$, $V_{GS}=0V$), cutline 1nm below gate oxide-channel interface.

Performance analysis of electrical parameters

Fig. 4.24 exhibits the variation of ON-current (I_{ON}) and OFF-current (I_{OFF}) for the range of temperature 298K-448K. Due to increasing of temperature, On-current and OFF-current increases, because of more BTBT. From above figures 4.24, it is observed that the leakage current is nearly constant for temperature 298K-448K while off-current increase but remains extremely low. Thus, DG-TFET can be operated for a large temperature more than 298K. TFET device are predicted to exhibits weak dependence of I-V characteristics on temperature. This is mainly happened due to band gap of material and screening tunneling length decreases only slightly with increasing temperature.

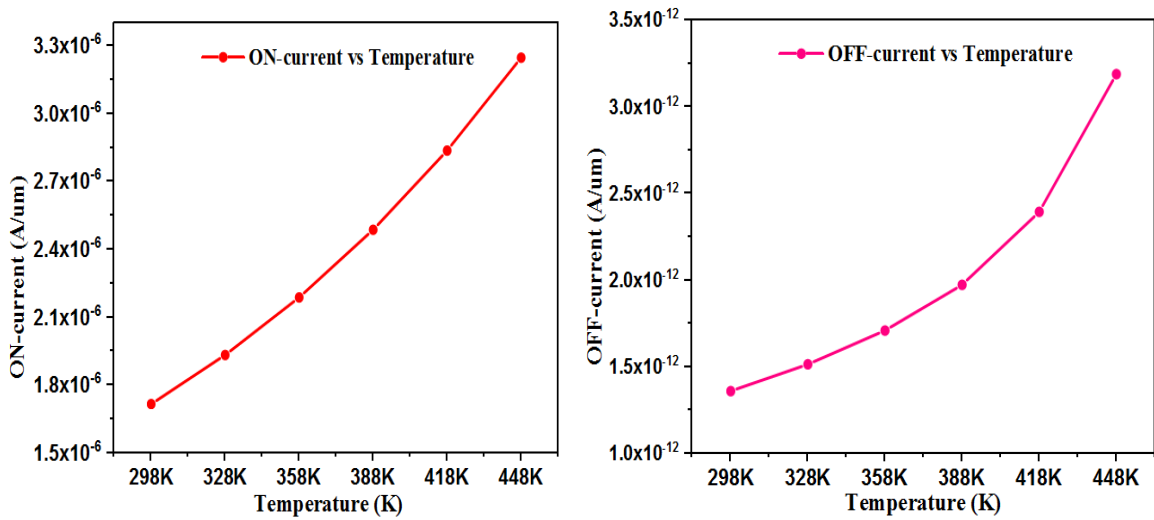


Figure 4.24: Simulation of (a) ON-current (b) OFF-current variation with temperature range 298K- 448K at $V_{DS}=1.5V$.

For better switching performance of devices I_{ON} should be maximum, a low I_{OFF} and the I_{ON}/I_{OFF} ratio should be higher. The current-ratio and its dependence on temperature is shown in fig. 4.25 (a) for temperature range 298K- 448K respectively. We have observed that current-ratio first slightly increases for temperature range 298K-358K, after that current-ratio start to decrease for temperature range 358K- 448K. Current-ratio ($\approx 1.26 \times 10^6$ to 1.02×10^6) is obtained for this temperature range. The current-ratio of optimum value occurs when temperature is $T=358K$ at $V_{GS}=1.5V$. Table 4.7 shows device performance parameters namely subthreshold slope (SS), threshold voltage v_T , I_{ON} , I_{OFF} , I_{ON}/I_{OFF} ratio under the different temperature configurations. The OFF-current in DG-TFET is slightly rises with temperature but it is a less temperature dependent. The slight variation of I_{ON} and I_{OFF} currents with temperature causes minor

dependency of current-ratio on temperature. Qualitatively, the impact of temperature on the DG-TFET can be ignored because of weak dependency.

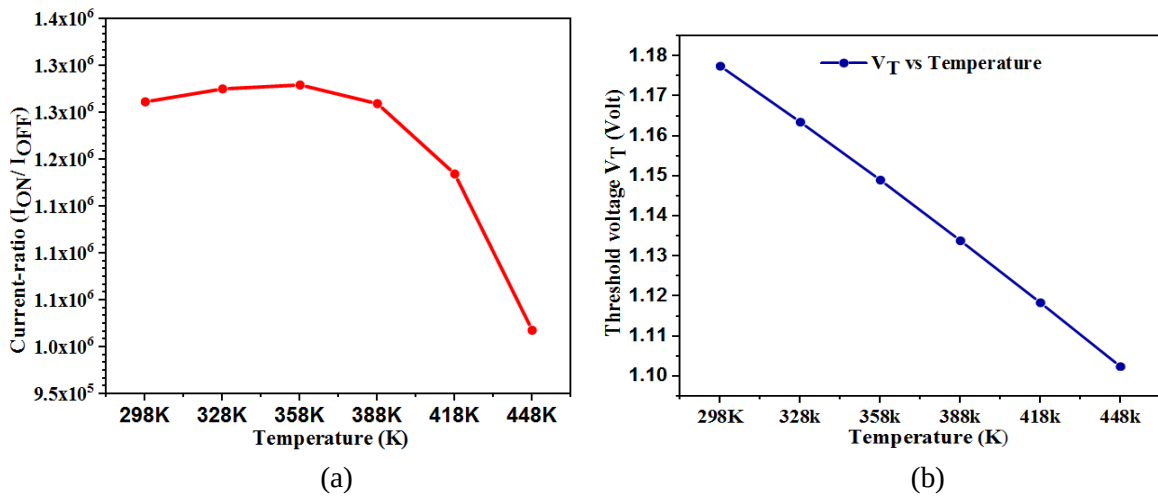


Figure 4.25: (a) I_{ON}/I_{OFF} (Current-ratio) (b) Threshold voltage (V_T) variation with temperature range 298K- 448K at $V_{DS}=1.5V$.

One of the important parameters of a DG-TFET device is its threshold voltage. We have used the constant current method to define the threshold voltage in TFET devices. The gate voltage at which the drain current becomes 1×10^{-7} A/ μ m is taken as the threshold voltage at $V_{DS}=V_{DD}$. Fig.4.25 (b) The threshold voltage shows a slight linearly decreasing relationship with increasing temperature and thus shows a positive temperature coefficient due to change in the Fermi level and decrease in bandgap. We have observed that the threshold voltage changes around 700-mV, when temperature varying from 298K to 448K. Threshold voltage shift in DG-TFET device is relatively small in temperature fluctuations. The threshold voltage for the DG-TFET has been analyzed for different temperature.

Table 4.7: Device Performance Parameters

Parameter (Kelvin)	SS (mV/dec)	$V_T(V)$	$I_{ON}(A/\mu m)$	$I_{OFF}(A/\mu m)$	I_{ON}/I_{OFF}
Temp.= 298K	48.91	1.17	1.72×10^{-6}	1.36×10^{-12}	1.26×10^6
Temp.= 328K	58.93	1.16	1.93×10^{-6}	1.52×10^{-12}	1.27×10^6
Temp.= 358K	65.32	1.15	2.19×10^{-6}	1.71×10^{-12}	1.28×10^6
Temp.= 388K	71.50	1.13	2.49×10^{-6}	1.97×10^{-12}	1.26×10^6
Temp.= 418K	77.60	1.12	2.84×10^{-6}	2.39×10^{-12}	1.18×10^6
Temp.= 448K	83.96	1.10	3.25×10^{-6}	3.19×10^{-12}	1.02×10^6

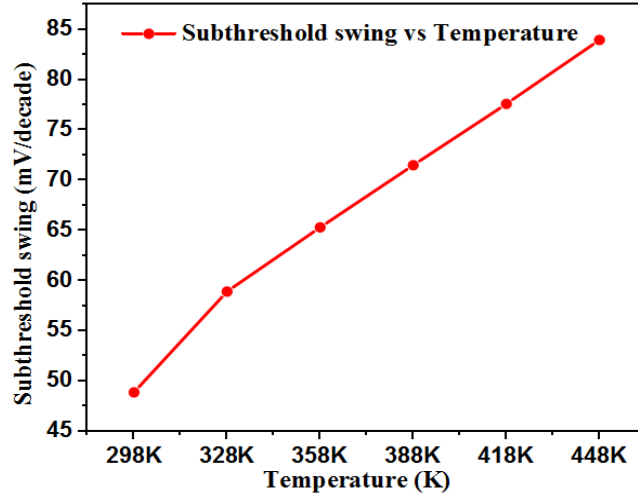


Figure 4.26: Subthreshold swing (SS) variation with temperature range 298K- 448K at $V_{DS}=1.5V$.

The change in gate voltage needed to increase the drain current by a factor of ten is known as Subthreshold swing. It's calculated in millivolts per decade. Small subthreshold swing is good for designing the TFET devices. It is clearly shown in fig. 4.26. that subthreshold swing increases as temperature increase. Subthreshold swing is an important parameter for computing the switching characteristics of a DGTFET devices. It is observed from fig. 4.26, that the Subthreshold swing of the device is less than 60 mV/decade for temperatures at 328K and subthreshold swing varies around 48.91 to 83.96 mV/decade. SS is a weak function of E_G and band gap is also function of temperature but TFET shows less dependence on the temperature. At $V_{DD}=0.35V$, it is also observed from fig. 4.26 that slight variation of the temperature, it effects the subthreshold swing or subthreshold slope which degrade the switching characteristics. Fig. 4.22 shows that subthreshold swing of DG-TFET for fixed value of V_{GS} is approximately constant as temperature increases. The improved subthreshold slope or swing will help in the development of low power switching circuits in the near future.

4.3.3 Impact of scaling in supply voltage

TFETs have been made as a promising device choice for low-power electronic circuit design in the future. V_{DD} scaling is one of the most important techniques for reducing the static and dynamic power dissipation in DG-TFET circuits. So TFETs are called as Green Transistors, because of highly energy efficient, steep sub-threshold swing and potentially applied in ultra-low supply voltage. Low V_{DD} Operation is also desirable for TFETs, as low power circuits are one of the most common applications. Here, we have

investigated on the effect of scaling in the supply voltage by using TCAD simulator. Due to down scaling of drain voltage, TFETs exhibit lower ON-state current, which is the main disadvantage and it is undesirable for TFET device. Low BTBT rate, which depends exponentially on the low lateral electric field in a conventional DG-TFET, is one of the main causes for low ON-state current in TFETs [46,47]. Therefore, some other techniques are needed in order to improve the I_{ON} . A best opportunity for improving the ON-current loss during V_{DD} reduction is to decrease the effective oxide thickness, using high-k dielectric oxide, decrease the silicon body thickness, use material with lower effective mass and smaller the bandgap, use high-k spacer material, high source doping concentration in comparison of drain doping, III-V group of combined semiconductor material offer higher tunneling rates, used to boost the ON-current. The above technique is necessary to keep a high ON-current and low OFF current. Lower value of V_{DD} gives the additional advantages of improving sub-threshold swing, On-current and I_{ON}/I_{OFF} ratio. It is promising candidate for future low power applications. Therefore, main focus for TFET device is to improve the ON current at ultra-low voltage and high-speed operation so that it can fully replace the conventional MOSFET technology in future VLSI circuits [48].

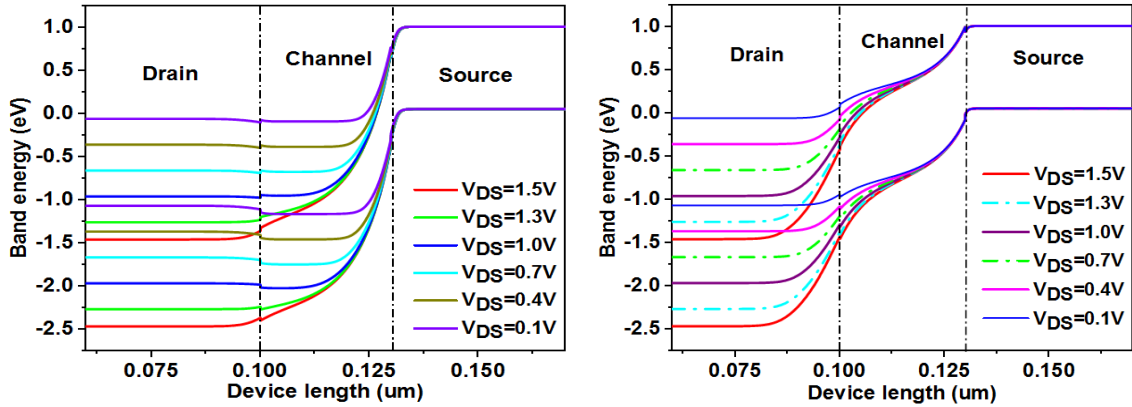


Figure 4.27: Energy band profiles at different drain voltages variation for DGTFET structure as shown in Fig. (a) ON-state. ($V_{DS}=1.5V$ to $0.1V$, $V_{GS}=1.5V$) (b) OFF-state. ($V_{DS}=1.5V$ to $0.1V$, $V_{GS}=0V$), cutline 1nm below gate oxide-channel interface.

Fig. 4.27 shows the energy band profiles at drain voltage variation of DGTFET in (a) ON-state and (b) OFF-state respectively. The drain voltage has the ability to decrease the tunneling energy barrier width with increase in V_{DS} . In ON-state, the positive supply is applied to the gate terminal, the V.B. of p+-region is aligned with the C.B. of intrinsic region and the electrons from source region can pass through the thin energy barrier to

channel region. Tunneling current flows in the device. The high electric field generates at source and channel tunneling junction at ($V_{GS}=1.5V$, $V_{DS}=1.5V$) and it provides the steep band bending as compared to other drain voltages. But the drain voltage gate is decreased from 1.5V to 0.1V then $\Delta\Phi$ starts to decrease. Where $\Delta\Phi$ is the energy range over which tunneling happens at tunneling junction. The energy band becomes less steep curved. BTBT decreases then the drain current also decreases. In OFF-state, the electron from p+-region cannot pass through the barrier width to intrinsic region because of thick tunneling width and decrease in the energy overlap window. So, under the off state there is almost no current flows in the device [49]. But it is clearly observed from fig. 4.27 (b), low drain voltage has a greater impact on OFF-state current. As the drain voltages increases from 0.1v to 1.5V and $V_{GS}=1.5V$, the $\Delta\Phi$ starts to increase gradually. Due to reduce in the energy overlap window it would cause a decrease in OFF-state current.

Transfer Characteristics

The figure 4.28 shows the simulation of I_{DS} - V_{GS} transfer characteristic of DGTFTFET curve for reduction in V_{DD} . From the above figure. 4.28, we have analyzed the impact of scaling in supply voltage on DG-TFET device. It is clearly observed that the reduction in supply voltage from 1.5V to 0.1V affects the I_{ON} of device significantly as reducing in the drain voltage values. The $I_D(V_G)$ curves of TCAD simulations for a DGTFTFET show that for the best case ($V_{DS}=1.5V$), the ON-current at a given voltage of $V_{GS}=1.5V$ is maximum current 1.73×10^{-6} A/um and BTBT take place easily. The tunnelling current is significantly worse at lower drain voltages but the OFF-current remains same as increase in drain voltage from 0.1V to 1.5V. However, the ambipolar current is also more notable at high V_{DS} . Due to low OFF-current the power dissipation is also low in TFETs. When we apply gate voltage which is high enough to create an inversion layer, the drain voltage starts the tunnelling process and electrons are attached to the drain terminal, producing a net flow of current. Fig. 4.28 also show that tunnelling current rises with drain voltages, however this effect is gradually saturating beyond the gate voltage equal to 0.8V (as seen for $V_{DS}= 1.5V$ to 0.4V) due to the short channel effect. In TFETs, the vertical electric field produced by the V_{GS} competes with parallel electric field formed by the V_{DS} . If drain voltage is increased gradually, depletion of the inversion layer happens nearly at the drain junction where electrons are attached. As a result, the tunnelling occurs at the intrinsic/n+ region rather than the p+-region/intrinsic region, and to generate the inversion layer near the source region it will compulsorily

supply a higher gate voltage, which is obviously not appropriate for low-power applications. It can be observed that in the fig. 4.28 shows a low OFF-current as well as low ON-current when drain voltage is equal to 0.1V. Fig. 4.28 shows that for drain voltage more than 0.4V is the optimum value of supply voltage which is suitable for low power application to analyzed the device. Remaining electrical device performance parameters namely subthreshold swing (SS), threshold voltage v_T , I_{ON} , I_{OFF} , I_{ON}/I_{OFF} ratio at different drain voltage configurations will be analyzed in the future for ultra-low-power integrated devices.

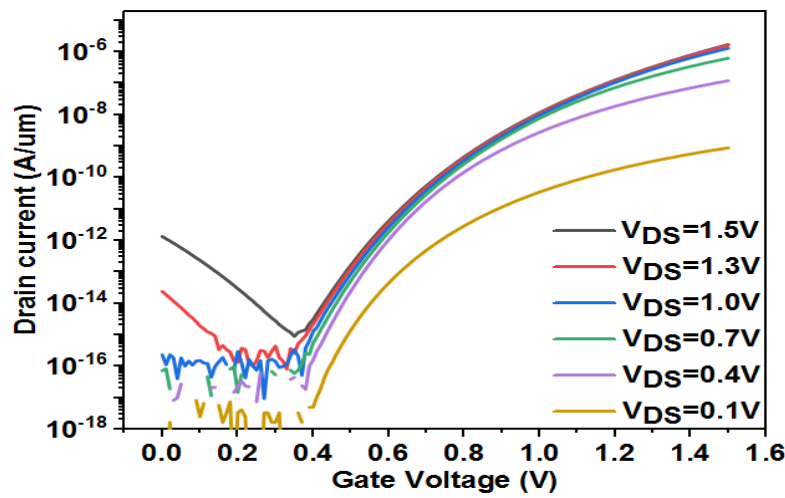


Figure 4.28: Simulation of I_{DS} - V_{GS} Transfer characteristics for different drain voltages with gate voltage varying from 0V-1.5V, $T_{ox} = 3\text{nm}$ in log scale.

4.3.3.1 DGTFET Ambipolar leakage current

Ambipolar current flow is an unwanted phenomenon that occurs in TFETs due to structural similarities between an n-TFET and p-TFET. Ambipolar currents develop when a device conducts for both strong positive and negative supply while maintaining the constant drain voltage. The source/channel tunnelling and drain-channel tunnelling junction are two types of tunnelling junctions exist in a TFET. When the TFET is symmetric, which implies the same doping concentration is utilised on the source and drain region but of the opposite type, the TFET exhibits ambipolar behaviour. It's important to note that at the same V_{DS} , ambipolar behaviour implies that the similar TFET exhibits n-type conduct with electrons playing a significant role in current transport mechanism and p-type behaviour with holes playing a big contribution in current transport.

In a n-type DGTFET configuration when V_{GS} is zero, the conduction band of intrinsic region lies above the valance band of the p+-region. As a result, no BTBT tunneling occurs in the device and TFET is turned OFF. When applying $V_{GS} > 0V$, $V_{DS} > 0V$ then BTBT occurs on the source/channel junction due to bands get aligned and gate voltage controls the tunneling current in the device. During this condition, TFET acts as ON-state. When $V_{GS} < 0V$, $V_{DS} > 0V$ then tunneling junction occurs at intrinsic-drain region instead of the p+-region/intrinsic region and bands in the drain of n+-region and intrinsic region get aligned. As a result, TFET shows the BTBT from channel to drain and the conduction current is known as ambipolar current. Ambipolar current flows from drain to source in TFET device. Furthermore, a p-type TFET will be operated in the ambipolar region when the V_{GS} is negative and BTBT conducts on the source/channel junction, and when the V_{GS} is positive and BTBT happens on the drain/channel region, while maintaining the V_{DS} negative value constant [50].

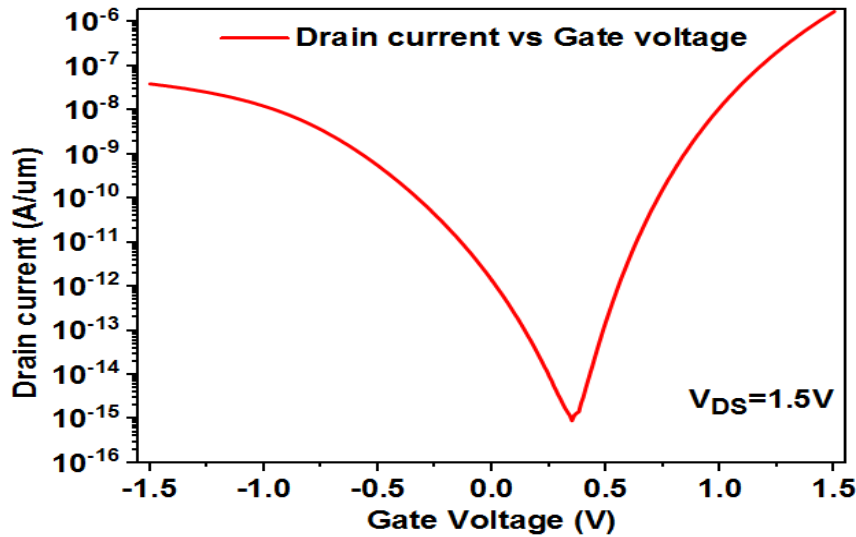


Figure 4.29: Simulation of I_{DS} - V_{GS} Transfer characteristics in the DGTFET for different doping on the source and the drain region with varying the gate voltage from (-1.5V to 1.5V) at $V_{DS}=1.5V$, $T_{ox}= 3nm$ in log scale.

Fig. 4.29 shows the I_{DS} - V_{GS} transfer characteristics of an n-type DGTFET that has the different doping concentration for the source $N_A= 1 \times 10^{20}$ atoms/ cm^3 , $N_D= 5 \times 10^{18}$ atoms/ cm^3 for the drain at $V_{DS}= 1.5V$ and V_{GS} varying from -1.5V to 1.5V. At $V_{GS}= -1.5V$, the ambipolar current is high and approximately equal to 0.1uA/um. Ambipolarity is a serious challenge in TFETs and recent years a device-level optimization techniques have been developed to limit ambipolar leakage currents.

In a TFET, Ambipolar conduction is not desirable for several reasons:

In a TFET, the ambipolar current might promote a greater OFF-state current. In TFETs using materials with a lower bandgap, the increase in I_{OFF} because of the ambipolar current is particularly noticeable. The thermal leakage current in these devices can be dominated by the ambipolar current in the OFF-state, which occurs because of the BTBT on the drain/channel junction. This outcome is in the form of an OFF-state current, which is represented by the ambipolar current rather than the thermal leakage current. As a result, the TFET must be designed in such a way that the ambipolar current is reduced. Therefore, ambipolar current should be avoided as much as possible in TFET devices. Reduce drain doping concentration, create a drain-gate underlap, use a heterojunction, or insert any type of asymmetry into the device to achieve this. A low drain doping can be used to minimize the ambipolarity and OFF-current. Due to drain-gate underlap region, the gate electric field at the drain is greatly decreased and BTBT at the drain/channel is minimized. In a TFET device, these methods are used to minimise the ambipolar current [1].

4.3.3.2 TFET p-i-n forward leakage current

The figure 4.30 shows the simulation of I_{DS} - V_{DS} output characteristic of n-type DG-TFET curve for varying the drain voltage range (-1.5V to 1.5V) at $V_{GS} = 1.5V$ and $-1.5V$. We have analyzed the impact of large negative V_{DS} . It is clear observed that significant leakage current transport instead of unidirectional current conduction. When positive V_{DS} is applied, n-type TFET shows desired transistor behaviour with high ON-current. TFET shows the considerable p-i-n forward leakage current flow with negative V_{DS} because of forward biasing of p-i-n device. It can be also seen that as V_{DS} become more negative, the leakage current increases rapidly. In this condition, TFET behaves like as a standard p-n diode and carriers moving from drain to source due to drift of carriers instead of BTBT mechanism. At $V_{DS} = -1.5V$, n-type TFET shows a leakage current of 4×10^{-3} A/um in the reverse direction flows (from source to drain). A negative supply V_{DS} (-0.5V to -0.7V) produces considerable leakage current (lower when compared to I_{ON}) and it is rarely controlled by gate potential. From the fig. 4.30, it can be seen that TFET represents similar magnitude of p-i-n forward current by changing gate voltages from -1.5 to 1.5V. This means that as V_{DS} gets more negative due to the forward biasing of p-i-n architecture, the gate terminal loses control of channel and tunneling

current is highly controlled by only V_{DS} . It can be also observed that magnitude of p-i-n forward leakage is more than the ambipolar current. Hence, p-i-n forward leakage is a severe problem that raises the power consumption of TFET circuit and degrades the reliability [50].

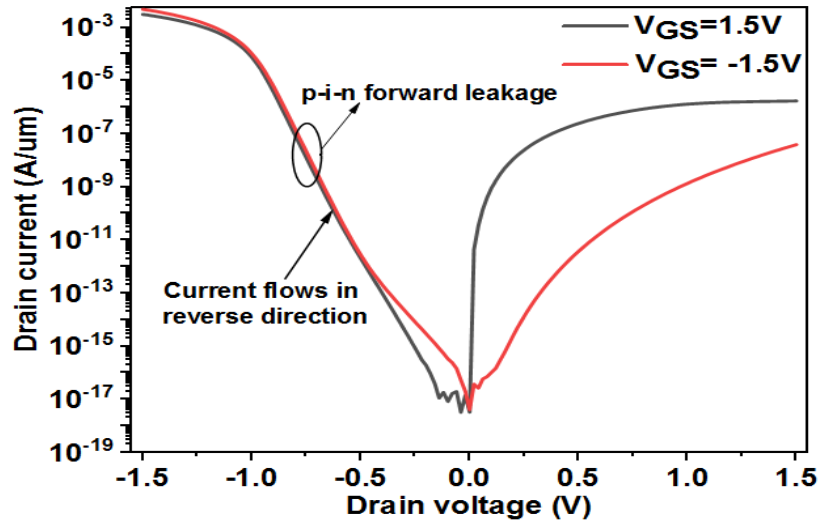


Figure 4.30: Simulation of I_{DS} - V_{DS} output characteristics in the DGTFET for different doping on the source and the drain region with varying the drain voltage from (-1.5V to 1.5V) at $V_{GS}=1.5V$ and -1.5V, $T_{ox}= 3nm$ in log scale.

Chapter 5

In this chapter we will briefly discuss conclusion and future work related to the analyzed structure.

5.1 Conclusion

We have investigated the electrical performance parameter of conventional DG-TFET with low- k dielectric gate oxide in the process-voltage-temperature (PVT) analysis. In the analysis of PVT, we have analyzed the effect of bottom gate misalignment towards the source and drain side, effect of interface fixed charges at Si-SiO₂ interface, impact of variation in both gate oxide thickness, impact of variation in upper gate oxide thickness, effect of variation in gate metal work function, effect of scaling in supply voltages and effect of temperature variation in the conventional DG-TFET device which especially affects the electrical parameter. In the process, At the first, we have analyzed the effect of bottom gate misaligned towards the source and drain side. A comparative analysis has been discussed for different parameters such as ON-current, OFF-current, threshold voltage, subthreshold swing and I_{ON}/I_{OFF} ratio etc. When bottom gate is misaligned towards source then the device performance of ON-current will slightly decrease because control of gate on the channel slightly reduces at the tunnel junction. But simultaneously, there is significant decrease in I_{OFF} and threshold voltage is almost constant. As the bottom gate is misaligned towards drain then the device performance degrades and provides worst results at BGM towards 15nm drain side. Best performance is observed when top and bottom gate will be perfectly aligned. Furthermore, we analyze the effect of interface fixed charges exist at the Si-SiO₂ interface on the device performance. It has been found that for increasing positive interface charges reduce the tunneling width and more BTBT occurs at tunnel junction due to high lateral electric field and it increases the ON-current but simultaneously it also reduces the OFF-current as well as threshold voltage and increases the current-ratio. But in the presence of negative interface, charges at Si-SiO₂ have been found that it decreases the ON-current but simultaneously increases the OFF-current as well as threshold voltage and decreases the current-ratio. Hence, no variation takes place in tunneling current and threshold voltage for zero interface charge. We have also investigated the effect of variation in both gate dielectric thickness range (3nm to 5nm) in DG-TFET device. It has been observed that when oxide thickness is increased from 3nm to 5nm, ON-current decreases due to loses in the gate controllability

over the tunnel junction but simultaneously OFF-current decreases while current-ratio increases and threshold voltage almost increases linearly. Furthermore, we have analyzed the variation in upper gate oxide thickness from 3nm to 5nm in DG-TFET. It has been observed that when top oxide thickness is increased from 3nm to 5nm, ON-current decreases due to losses in the gate controllability over the tunnel junction but simultaneously OFF-current decreases and threshold voltage almost increases linearly. Moreover, I have also investigated the impact of variation in gate work function of metal on DG-TFET device. I have found that ON-state current decreases when work function of gate increases due to the rise in flat band voltage which means energy barrier's potential increases and probability of tunneling decreases exponentially. OFF-state current increases rapidly when the gate work function is more than 4.2eV. It has been found that the threshold voltage shows a linear relationship with increase in the gate metal work function as in DG-TFET. It happens due to increase in the flat band voltage and current-ratio decreases after 4.2eV. After comparing the values, it has been found that the gate metal work function of 4.2eV is optimum value for DG-TFET device. We have analyzed the impact of scaling in supply voltage for reducing the static and dynamic power dissipation on DGTFET device. It has been also observed that the reduction in supply voltage affects the I_{ON} of device significantly but the OFF-current remains same as decrease in drain voltages. It happens due to Low lateral electric field which is created at the tunnel junction at low drain voltages. It can be observed that drain voltage more than 0.4V is the optimum value of supply voltage which is suitable for low power application to analyzed the device. We have also analyzed the different regions of operation by varying both gate to source voltage (V_{GS}) and drain to source (V_{DS}) from negative to positive voltages in the DG-TFET device. We have found that the ambipolar current is high and approximately equal to 0.1uA/um. Ambipolarity is a serious challenge and undesirable in TFETs. The ambipolar current might promote a greater OFF-state current. We have analyzed the impact of large negative V_{DS} . It has been observed that when V_{DS} become more negative, the p-i-n forward leakage current flows rapidly and its value equal to 4×10^{-3} A/um in the reverse direction which flows from source to drain. Because the gate terminal loses control of channel and therefore the tunneling current is highly controlled by V_{DS} only. It can be also observed that magnitude of p-i-n forward leakage is more than the ambipolar current. Hence, p-i-n forward leakage is a severe problem that raises the power consumption and degrades the performance of the TFET device. At last, we have investigated the effect of temperature

with variation of range from 298K to 448K on the TFET device. It has been found that a very modest leakage current ($\approx 1.36 \times 10^{-12}$ to 3.19×10^{-12} A/ μ m) and ON-current vary from ($\approx 1.72 \times 10^{-6}$ to 3.25×10^{-6} A/ μ m) is obtained. But OFF-current increases extremely low due to weak temperature effect. Current-ratio decreases for temperature range 298K- 448K and its obtained value is ($\approx 1.26 \times 10^6$ to 1.02×10^6). We have also observed that threshold voltage decreases with increase in the temperature. It has been found that the Subthreshold swing of the device is less than 60 mV/decade for temperatures at 328K and subthreshold swing varies around 48.91 to 83.96 mV/decade for temp. range 298K to 448K. We have analyzed the process-voltage-temperature (PVT) in DG-TFET and observed that all these types of parameter variation will affect the electrical performance parameter in overall TFET device behaviour.

5.2 Future work and scope

The initial results are very promising which make it an attractive alternative for lower technology nodes. Furthermore, it is exciting to see its applicability in circuit design. This can be only possible when either we make an analytical spice model or capture device behaviour using Verilog A model. So, for future work, we can analyze the effect of variation in source doping, drain doping, channel doping, Ge material used in intrinsic region, High-k gate dielectric oxide, body thickness and channel length on the TFET based device. To improve the DC parameters, all of these parameters can be optimised as performance boosters and achieve superior device characteristics. ON-current must be improved without compromising the off-current to achieve the TFET target parameters.

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